Sharp tests of instrument validity for general outcome distributions

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Abstract

I combine novel identification and inference results to obtain the first test of instrument validity that can accommodate outcomes with unrestricted, potentially infinite, and multi-dimensional support, along with a large class of shape restrictions. The main result shows that, even if the outcome is continuous, the sharp testable implications of the exclusion restriction can be reduced to a finite set of linear inequality constraints on the model for selection into treatment. Hence, if the instrument and treatment are discrete, an infinite system of equations is reduced to an equivalent finite one. The resulting system involves non-differentiable functions of the data, for which existing inference procedures are not applicable. I therefore develop a procedure for joint inference on these parameters, which delivers a formal test of instrument validity. I apply the proposed test to the Oregon Health Experiment, where I reject the hypothesis that the intervention affected insurance coverage only on the extensive margin. Instead, I find evidence that individuals were incentivized to retain insurance for at least six months. Leveraging a shape constraint that models this empirical finding, I obtain estimates of the effect of insurance on health and hospital utilization that reflect the intensive margin effect of the intervention, and are consistent with previous studies.

JEL classification: C12, C21, C25, C26, C31, C36.

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1 Introduction

Instrumental variables (IV) is one of the most used methods in empirical economics. Following seminal work by Imbens and Angrist (1994), a common approach to IV is to leverage non-parametric shape restrictions (such as no-defiers or monotonicity) to identify causal parameters without restricting treatment effect heterogeneity. Furthermore, the assumptions that underpin IV models for heterogeneous treatment effects are equivalent to the hypothesis that a set of variables can fully account for the effect of an exogenous treatment in the context of mediation analysis. Existing tests for the validity of IV models for heterogeneous treatment effects either apply to and rely on specific shape restrictions, or require that the outcome be discrete or discretized. When the outcome is continuous, tests that use discretization neither exhaust the model's testable implications nor use the entire empirical content of the data.

In this paper, I combine novel identification and inference results to propose the first test of instrument validity that can accommodate general outcome distributions and a large class of non-parametric shape restrictions. The test is sharp: it employs a set of conditions that is logically equivalent to the IV assumptions. The observable implications involve sub-density overlaps, a class of parameters for which no valid joint inference methods exist. Thus, I develop a novel method for inference on multiple density-overlaps based on non-parametric classification. Density and sub-density overlaps arise in econometric characterizations beyond IV validity, as well as in other disciplines such as medicine, political science, and biology. The proposed inferential method may therefore be of independent interest.

Finally, I propose a variant of the main test based on a novel micro-foundation for IV models with binary instruments and shape restrictions that prevent switches between specific pairs of treatments. This variant allows for simpler inference, and identifies the specific behaviors that exist in the population but contradict the model's assumptions. Additionally, in contexts other than IV, it can test nonparametric assumptions about how an exogenous treatment affects the preferences and choices of a population of decision makers. In an application to the Oregon Health Experiment (OHE hereafter), I illustrate how this framework can be used to obtain valid, credible conclusions from IV methods that allow for heterogeneous treatment effects.

Identification: I consider a potential outcomes model with a discrete instrument Z, a discrete treatment X, and an outcome Y with unrestricted, and potentially multi-dimensional, support. The model is defined by potential treatments $X = X(Z)$ and potential outcomes $Y = Y(X, Z)$.

The key assumptions are: *i*) Z is independent from potential outcomes as if it was randomly assigned, *ii*) Z affects X without violating some pre-specified restrictions (e.g. no-defiers), and *iii*) an exclusion restriction on Z, which reduces potential outcomes to $Y = Y(X)$. The goal is to obtain a sharp set observable implications for which a fully non-parametric test is feasible.

The model is not falsified if potential treatment and outcome functions exist that rationalize the observed conditional treatment probabilities:

$$\mathbb{P}[X = x|Z = z] = \mathbb{P}[X(z) = x|Z = z], \tag{1}$$

and conditional outcome probabilities:

$$\mathbb{P}[Y \in B_Y, X = x|Z = z] = \mathbb{P}[Y(x) \in B_Y, X(z) = x|Z = z] \tag{2}$$

for every x and z in the support of X and Z, and every measurable subset B_Y of the support of Y.

X and Z have finite support, thus, selection into treatment (equation 1) is described by a finite set of equations. But, if Y has infinite support, there are infinitely many measurable subsets B_Y and thus, infinitely many instances of equation 2. The novel insight that enables the test is that the sharp observable implications of the exclusion restriction combined with conditional outcome probabilities are equivalent to a finite set of linear restrictions on the selection part of the model. Thus, I reduce an infinite set of equations to a finite one.

The set of linear inequalities that expresses the sharp observable implications of the exclusion restriction involves a finite collection of sub-density overlaps. The overlap of n functions is the area that lies strictly bellow all of them, or, in other words, the integral of their point-wise minimum. In this paper, subdensities are the functions obtained by restricting the joint densities of the outcome and treatment conditional on the instrument to a unique treatment value. Thus, the target parameters are integrals of the point-wise minimum of the functions $f_{Y,X|Z=z}(y, x)$ across all possible values of z and some fixed x.

Inference: Estimation and inference on sub-density overlaps is a non-standard problem for which no previous, valid, fully non-parametric method exists. The non-standard nature of the problem arises from the region of the support of the outcome in which two or more subdensities are equal. This region is referred to as the contact set. The main inferential challenge follows from the fact that the minimizer is not unique in this region.

Ruling the contact set out is problematic. First, I argue that it might naturally arise whenever compliance (i.e. the effect of Z on X) is correlated with the outcome (for example,

if the instrument does not affect the treatment choices of the healthiest subjects in a given population). Second, it also arises whenever the instrument has no effect on treatment adoption (i.e. instrument irrelevance), which, in the context of testing, might not be an *ex-ante* plausible assumption. Finally, the assumption that the contact set has zero measure is fundamentally untestable. Thus, a test robust to this possibility is more credible.

Furthermore, the contact set is the reason why discretization based tests are not sharp. Discretization is equivalent to integration with a coarse grid, and it is precisely around the points where subdensities cross that coarse grids overestimate or underestimate the integral of the minimum. Hence, discretization is only sharp if the contact set is known or estimated with high precision. Thus, to obtain a sharp test, the problems associated with the contact set cannot be avoided.

The contact set is problematic because non-differentiability of the minimum impedes using the delta method, while the non-parametric nature of the problem implies there are no guarantees that extensions of the delta method for directionally differentiable functions such as Fang and Santos (2019) work.

Furthermore, if the minimum of two densities is not unique in some non-negligible subset of the support of the outcome, the asymptotic behavior of plug-in estimators is non standard. Any finite-sample sub-density estimator will have a statistical noise component that shifts the estimator away from the true value of the target function. If two sub-densities are estimated from samples that are approximately similar in size, the magnitude of the noise component of both estimators will be of the same order, and its ratio will remain constant even as the sample size grows. Hence, for any given sample size, the identity of in-sample minimizers inside the contact set is determined by statistical noise. As a consequence, the limit of the estimated minimizer, and thus of a plug-in overlap estimator, might be a random variable rather than constant.

To circumvent these problems, I note that inference on sub-density overlaps is equivalent to inference on the loss of an asymmetric classifier: the goal is not to learn the identity of the maximum/minimum across the support of Y, but the aggregate frequency with which the observed values of X and Y come from the sub-density associated with the least likely instrument value. The main advantage of this approach is that, precisely in the set in which sub-densities are equal, any classification decision attains the exact same loss. In other words, the estimated classifier might be random (even in the limit), but its loss, which corresponds to the target parameter, is deterministic.

I propose a consistent, asymptotically normal, and asymptotically unbiased estimator for the overlap of two or more sub-densities. To derive its asymptotic properties, I combine the classification framework with uniform rates of convergence for kernel density estimators

(Giné and Guillou (2002)) and a margin condition that controls the magnitude of classification errors in the proximity of the contact set. The estimator uses sample splitting. One fraction of the data is used to train a non-parametric classifier based on kernel density estimators, and the rest to evaluate its loss. Sample splitting ensures that the empirical loss of the trained classifier is unbiased for its theoretical loss.

In Monte-Carlo simulations, I show that the overlap estimators have negligible bias, and that my proposed confidence intervals achieve coverage close to the nominal level in large samples. Given this valid method for inference on sub-density overlaps, testing instrument validity reduces to verifying that a linear system of equations admits at least one solution. This is a well studied problem for which inference procedures exist. Using the methodology from Cox and Shi (2023) I formally show that the resulting test has asymptotically correct size. A second set of Monte-Carlo simulations supports the theoretical properties of the test.

Generalized no-defiers and empirical application: One of the most used approaches to identification in IV models with heterogeneous treatment effects is to assume that some treatment switches cannot occur as the instrument changes (e.g. the no defiers assumption). I call restrictions in this class *generalized no-defiers assumptions*. For models with binary instruments that impose *generalized no-defiers assumptions*, I propose a specialized, simpler test based on a microeconomic interpretation of such restrictions.

Generalized no-defiers restrictions specify that, as the instrument increases, some treatments are weakly promoted (at least in relative terms) with respect to others. Thus, if the assumption holds, adoption of the promoted treatments should weakly increase. I formalize the notion of *promoted treatments* in terms of restrictions on how the instrument affects the preferences of units in the population, and characterize how adoption of the promoted treatments *weakly increases* using a generalization of first order stochastic dominance (GFOSD) to arbitrary binary relations.

Additionally, the sharp testable implications of any *generalized no-defiers restriction* combined with the exclusion restriction are still a set of inequalities with a GFOSD interpretation. The difference is that the exclusion restriction imposes an upper bound on the probabilities associated with some of the promoted treatments, which results in a *tightening* of the GFOSD inequalities relative to the inequalities associated with the *generalized no-defiers restriction* alone.

These results provide a microfoundation for *generalized no-defiers assumptions*, an interpretable set of sharp testable implications, and a framework to formulate plausible restrictions that are consistent with the observed data.

I illustrate the proposed framework in an empirical application to the Oregon Health

Experiment, a state-level lottery that randomized eligibility to the Medicaid program among low income households (Finkelstein et al. (2012)). Specifically, I test and reject the assumption that the intervention only affected insurance on the extensive margin proposed by Rose and Shem-Tov (2024). Instead, I find that the treatment also affected the duration of insurance for households that would otherwise have acquired insurance for less than six months. That is, the intervention also had an effect on the intensive margin. Leveraging a shape restriction that models my empirical findings, I estimate the effect of increased insurance periods on hospital utilization and health outcomes. My empirical results are consistent with previous results by Finkelstein et al. (2012) and Rose and Shem-Tov (2024).

Related literature: This paper contributes to the literature on testing the validity of instrumental variables models for heterogeneous treatment effects. The first to consider instrument validity for heterogeneous treatment effects where Balke and Pearl (1997), but without a formal inference procedure and only for binary outcomes. Subsequently, Kitagawa (2015), Huber and Mellace (2015), and Mourifié and Wan (2017) developed tests for the binary instrument and binary treatment case under a monotonicity assumption. Kédagni and Mourifié (2020) proposed a joint test for the exclusion restriction and random assignment but their results are only sharp for a binary outcome and do not allow for shape restrictions. Subsequent work has derived sharp testable implications for IV designs on a case-by-case basis (Bai et al., 2024c), or proposed general tests that are not sharp (Sun, 2023), Frandsen et al. (2023). More recently, **?** and Kaido and Ponomarev (2025) provided the first sharp characterizations for general IV models, but their proposed inference procedures require that the outcome be either discrete, or discretized at the cost of losing sharpness.

The observable implications I propose apply to a range of IV methods for heterogeneous treatment effects. Imbens and Angrist (1994) showed how to use non-parametric restrictions on instrument-response heterogeneity to identify the LATE parameter. Subsequent work has extended their approach to a wide range of settings and other parameters of interest, such as: re-interpretations of monotonicity (Heckman and Pinto (2018), Navjeevan and Pinto (2022)); alternative restrictions on how the instrument affects treatment take-up (Richardson and Robins (2010), De Chaisemartin (2017), Dahl et al. (2023), Goff (2024)); and generalizations to settings with multiple instruments and treatmens (Lee and Salanié (2020), Goff (2020), Mogstad et al. (2021), van't Hoff et al. (2023), Kazemi and Olsen (2024), Fusejima (2024), Bai et al. (2024a)). These methodological advances have been applied in diverse empirical contexts, including education and labor market studies (Kline and Walters (2016), Kirkeboen et al. (2016), Pinto (2021), Mountjoy (2022)). The testing procedures I propose apply to all the aforementioned papers.

The GFOSD characterization and its relation to non-parametric random utility models contributes to the literature that investigates the empirical content of response type restrictions (Goff (2024) and Bai et al. (2024b)) and their microeconomic foundations (Vytlacil (2002), Heckman and Pinto (2018), Navjeevan and Pinto (2022)).

To obtain the GFOSD inequalities I show that finding a distribution over potential treatments and potential outcomes that satisfies the sharp instrument validity conditions amounts to solving a well known transportation problem over a directed graph. This technique, and its equivalence to the maximum flow/minimum cut problems, was first used to characterize incomplete econometric models by Ekeland et al. (2010) and Galichon and Henry (2011). My approach differs in two ways: first, instead of considering transportation problems from the space of observed variables to the domain of latent unobservables, my transportation problem is defined between observed treatment probabilities conditional on different instrument values. Second, to incorporate the exclusion restriction, I extend their minimum-cut characterization method to a broader class of capacity constraints.

My optimal transport approach is related to contemporaneous and independent work by Kaido and Ponomarev (2025) who study the falsification of shape constraints and exclusion restrictions using graph theory. The approaches are not nested. The main advantage of the approach from Kaido and Ponomarev (2025) is its generality since it is sharp for the multiple instruments case and applies to other econometric models beyond IV. The main advantage of my approach is that it does not rely on partitioning the support of the outcome (discretization), obtaining a finite characterization even when the outcome has infinite support. A second advantage is that my approach can accommodate a wider class of restrictions on treatment adoption, for example, the *more compliers that defiers* assumption from De Chaisemartin (2017).

All the testable implications I provide are equivalent to the requirement that a particular linear system of inequalities admits at least one solution. Inference on the feasibility of linear systems and linear programs is an active area of research (Bai et al. (2022), Fang et al. (2023), Cho and Russell (2024), Goff and Mbakop (2025)) that is tightly related to inference methods for models characterized by moment inequalities such as Cox and Shi (2023) or Andrews et al. (2023).

To test IV validity using inferential methods for linear systems I first require a method for non-parametric joint inference on sub-density overlaps. This is a particular instance of the more general problem of estimating the integral of non-differentiable transformations of a collection of functions or densities. Two closely related papers are Semenova (2025) and Levis et al. (2025) who consider inference on the integral of the point-wise minimum/maximum of smooth functions, and its applications to IV models. Neither of these

papers studies testing instrument validity. For inference, Semenova (2025) assumes that the minimum is unique almost-everywhere, which eliminates the non-regularity issue. Levis et al. (2025) propose a similar solution, and also consider the non-negligible contact set case. Their solution is to restore regularity using a continuous approximation to the minimum function. While the soft-min approach would be valid in the context of this paper, the continuous approximation to the minimum shifts the target parameter. Consequently, the resulting inequalities would be a set of necessary conditions for IV validity, but not the sharp testable implications

To the best of my knowledge, only two papers have considered exact inference on parameters defined by the integral of the minimum/maximum of a collection of functions that is robust to non-negligible contact sets. Anderson et al. (2012) consider a plug-in approach for inference on a single overlap parameter. Unfortunately, their variance estimator is not compatible with existing inference methods for linear systems. Luedtke and Van Der Laan (2016) propose using online estimation to conduct inference on the mean outcome under a non-unique optimal policy. While similar non-regularity issues arise in both problems, in sub-density overlap estimation, realized and potential outcomes enter the target parameter through their distribution rather than directly. Thus, the online estimation solution is not applicable without substantial modifications.

The key argument sustaining my inference procedure is that, in the contact set, the loss of a non-parametric classifier is invariant, and hence, an estimator based on the empirical loss of a classifier is robust to the erratic behavior of the minimum inside the contact set. This insight reduces the inference problem to controlling the behavior of the classifier out of the contact set. To do so, I combine sample splitting, a margin assumption that controls the misclassification rate near the boundary of the contact set (Tsybakov (2004), Audibert and Tsybakov (2007)), and classical results on the rates of uniform convergence for kernel-density estimators by Giné and Guillou (2002).

The alternative estimator I propose to enable K-fold cross fitting relies on estimating a set that is known to contain the contact set with high probability. This idea was first proposed in the context of a test for first order stochastic dominance by Linton et al. (2010). In the context of inference on overlaps, it was also used by Anderson et al. (2012).

My method for inference on density and sub-density overlaps might be of independent interest. First, other IV identification methods such as Kitagawa (2010) or Kédagni et al. (2024) obtain identification results that involve or are related to the vector of sub-density overlaps from this paper, but do not have inference methods that allow for continuous outcomes. Second, similar parameters appear in other econometric settings such as Borusyak (2015) or Meango et al. (2025). Finally, density overlaps are a relevant non-parametric mea-

sure of similarity between populations in other disciplines such as political science, ecology, and medicine (see Dey et al. (2025) for a recent review).

The remainder of this paper is organized as follows: In section 2 I introduce the class of models and restrictions considered. In section 3 I derive the sharp testable implications of the exclusion restriction and provide the finite sharp characterization for IV models. In section 4 I specialize the characterization from section 2 to instrument-response-type restrictions and link them to FOSD and random utility models using optimal transport. In section 5 I describe my method for joint estimation and inference on sub-density overlaps and prove its asymptotic properties. Then, I discuss how the estimators can be used to construct a sharp test for instrument validity. In section 6 I illustrate the performance of both the estimators and the test. In section **??**, I use my inference method and the GFOSD characterization to study the effect of the oregon health experiment on the duration of insurance and its effect on health and hospital utilization. In section **??** I conclude.

2 IV models for heterogeneous treatment effects

The model has three observable random variables: an outcome Y, a treatment X and an instrument Z with joint distribution $F_{Y,X,Z}$. Both X and Z have discrete finite supports denoted $\mathcal{X} = \{x_0, x_1, ..., x_{L-1}\}$ and $\mathcal{Z} = \{z_0, z_1, ..., z_{K-1}\}$ respectively. The support of Y which I denote as $\mathcal{Y}$ has unrestricted cardinality, finite dimension $d \geq 1$, and a measure μ_Y over $\mathcal{Y}$ exists.

I adopt the potential outcomes framework. The joint distribution of (Y, X, Z) is determined by the potential treatment and potential outcome functions $X(z) : \mathcal{Z} \mapsto \mathcal{X}$ and $Y(x, z) : \mathcal{X} \times \mathcal{Z} \mapsto \mathcal{Y}$. These functions are random variables, and their joint distribution along with the observed marginal distribution of Z fully determine the model.

Formally, for any measurable subset of the support of the outcome $B_Y \subset \mathcal{Y}$, and any fixed treatment and instrument values $x_l \in \mathcal{X}$ and $z_k \in \mathcal{Z}$, the joint distribution of (Y, X, Z) satisfies:

$$\mathbb{P}[Y \in B_Y, X = x_l, Z = z_k] = \mathbb{P}[Y(x_l, z_k) \in B_Y, X(z_k) = x_l, Z = z_k] \tag{3}$$

I assume that all other relevant covariates are exogenous. Since they play no role in the characterizations I provide, I assume that all the identification arguments are conditional on such covariates and omit them from the formal proofs and results.

Unless otherwise noted, I will present and discuss all the results using the IV notation

and terminology. All results would also hold for the sharp null of full mediation from **?** after appropriately relabeling the values of X as mediators and the values of Z as treatments.

I denote the joint density of X and Y conditional on $Z = z_k$ induced by $F_{Y,X,Z}$ as $f_{Y,X|Z=z_k}(y, x)$.

2.1 Instrument validity and response types

For Z to be a valid instrument, two assumptions are typically invoked, instrument independence, and an exclusion restriction:

Assumption 2.1 (Instrument validity)**.**

1. ***Instrument independence****: The potential outcomes and potential treatment functions $Y(\cdot, \cdot)$ and $X(\cdot)$ are independent from the instrument Z.*
2. ***Exclusion restriction****: $Y(x_l, z_k) = Y(x_l)$, for every $x_l \in \mathcal{X}$ and $z_k \in \mathcal{Z}$*

Put in words, the instrument must be independent from potential outcomes and treatments as if it was randomly assigned, and it must have no effect on outcomes after accounting for treatment.

Assumption 2.1 implies that the observed joint distribution of (X, Y, Z) is fully determined by the observed distribution of Z, and the unobserved distribution of the potential outcomes and potential treatments functions. This motivates the following definitions:

Definition 2.1.

1. *An instrument-response type $t^Z \in \mathcal{T}_Z$ is a function $t^Z : \mathcal{Z} \longrightarrow \mathcal{X}$ that maps instrument values to treatment values.*
2. *A treatment-response type $t^X \in \mathcal{T}_X$ is a function $t^X : \mathcal{X} \longrightarrow \mathcal{Y}$ that maps treatment values to outcome values.*
3. *A response type t is a pair $t = (t_z, t_x) \in \mathcal{T} = \mathcal{T}_Z \times \mathcal{T}_X$.*

An important feature of response types is that by construction they satisfy the exclusion restriction. For any $x_l \in \mathcal{X}$, a response type $t = (t^Z, t^X)$ assigns a unique outcome value $t^X(x_l)$ when $X = x_l$ regardless of the value that the instrument takes. This is why, under assumption 2.1, the model is fully determined by response types and the marginal distribution of Z.

Since Y is μ_y-measurable, and $\mathcal{X}$ is finite, the set $\mathcal{T}$ is measurable with respect to the product measure induced by μ_Y and the count measure on $\mathcal{X}$. I denote the resulting measure space as $(\mathcal{T}, \mathcal{B}_T)$.

Definition 2.2.

1. *A distribution over response types is a probability measure* $g : \mathcal{B}_T \longrightarrow [0,1]$ *over the measure space* $(\mathcal{T}, \mathcal{B}_T)$.

2. *A distribution over instrument-response types is a function* $p : \mathcal{T}_Z \longrightarrow [0,1]$ *such that* $p(t^Z) \geq 0$ *for all* $t^Z \in \mathcal{T}_Z$, *and* $\sum_{t^Z \in \mathcal{T}_Z} p(t^Z) = 1$.

Remark 2.1. *Any distribution over response types* g *defines a unique distribution over instrument-response types* f_g *according to the following equation:*

$$p_g(t'^Z) := g\Big(\big\{t = (t^Z, t^X) \in \mathcal{T} : t^Z = t'^Z, t^X \in \mathcal{Y}^L\big\}\Big), \qquad \forall\ t'^Z \in \mathcal{T}_Z, \tag{4}$$

$$= g\Big(\big\{t = (t^Z, t^X) \in \mathcal{T} : t^Z = t'^Z\big\}\Big), \qquad \forall\ t'^Z \in \mathcal{T}_Z. \tag{5}$$

If assumption 2.1 holds, then an unobserved true distribution over response types g_0 must exist that generates the observed distribution of (Y, X, Z).

Definition 2.3 (Consistency)**.** *A distribution over response types* $g \in \Delta(\mathcal{T})$ *is consistent with observed conditional outcome and treatment probabilities if:*

$$\mathbb{P}[Y \in B_Y, X = x_l | Z = z_k] = g_0\Big(\big\{(t^Z, t^X) \in \mathcal{T} : (t^Z)_k = x_l, (t^X)_l \in B_Y\big\}\Big), \tag{6}$$

$$\forall x_l \in \mathcal{X}, z_k \in \mathcal{Z},\ \text{and}\ B_Y \in \mathcal{B}_Y.$$

In the previous equation, the observed conditional probability on the left hand side must correspond to the mass that g_0 assigns to response types whose potential treatment is equal to x_l when the instrument takes the value z_k and realize outcomes in B_Y when treatment is equal to x_l.

Let $\Delta(\mathcal{T})$ be the set of all distributions over response types, assumption 2.1 is falsified if no $g \in \Delta(\mathcal{T})$ exists that satisfies equation 6.

2.2 Shape restrictions

The class of shape restrictions I consider is general. It covers any restriction that either rules out specific instrument response types, or imposes inequality restrictions on their prevalence and relative frequency. Formally stated, I allow for any restriction than can be expressed as a set of linear constraints on the set of admissible distributions over instrument-response types.

The most prominent example of a restriction that rules out specific response types is monotonicity when both X and Z are binary, and switcing from 0 to 1 is assumed to promote

treatment addoption, thus, defiers[1] are ruled out from the population. Subsequent work has explored relaxations and extensions of this assumption to settings with multiple instruments and multiple, ordered and unordered, treatments (Heckman and Pinto (2018),Goff (2020), Mogstad et al. (2021), van't Hoff et al. (2023), Lee and Salanié (2020), Bai et al. (2024c)), these generalizations typically rely on ruling out particular instrument-response types.

Alternatively, researchers may impose inequality restrictions on the prevalence or relative frequency of certain instrument-response types—for example, requiring that the proportion of compliers exceeds that of defiers (De Chaisemartin (2017)), or imposing upper bounds on the share of defiers for sensitivity analysis as Huber (2014), Noack (2021) or Yap (2025)).

To formalize the shape restrictions allowed, note that the total number of possible instrument-response types is $|\mathcal{T}_Z| := |\mathcal{X}|^{|\mathcal{Z}|} = (L)^K$. Therefore, a distribution p over $\mathcal{T}_Z$ can be represented as a vector $\boldsymbol{p}$ in $\mathbb{R}_+^{|\mathcal{T}_Z|}$, where each coordinate corresponds to the probability that $p \in \Delta(\mathcal{T}_Z)$ assigns to a different $t \in \mathcal{T}_Z$.[2]

I consider any shape restriction that takes the form $A_Z\boldsymbol{p} \leq \boldsymbol{r}$ where A_Z is an M by $|\mathcal{T}_Z|$ matrix with known coefficients for some fixed $M \geq 1$, and $\boldsymbol{r}$ is an M dimensional vector with entries that are either known or point identified from the data. I use the subscript Z to emphasize that the matrix A_Z encodes restrictions that pertain to treatment assignment as a function of the instrument.

Assumption 2.2. *Let $M \geq$ be a fixed integer, A_Z a known fixed matrix $A_Z \in \mathbb{R}^{M \times |\mathcal{T}_Z|}$, and $\boldsymbol{r}_Z \in \mathbb{R}^M$ a point identified vector that depends on the joint distribution of (Y, X, Z).*

A distribution over response types $g_0 \in \Delta(\mathcal{T})$ exists such that if $\boldsymbol{p}_{g_0}$ denotes the vector representation of p_{g_0}, the following system of inequalities holds:

$$A_Z\boldsymbol{p}_{g_0} \leq \boldsymbol{r}_Z. \tag{7}$$

Note that strict equality restrictions can be incorporated by representing them as pairs of inequality constraints. In particular, if the k-th entry of $\boldsymbol{p}$ corresponds to the instrument-response type $t^Z \in \mathcal{T}_Z$, the restriction that t^Z occurs with zero probability can be imposed by including the canonical basis vectors e_k and $-e_k$ as rows of A_Z, and setting the corresponding entries of $\boldsymbol{r}_Z$ to zero.[3] Thus, assumption 2.2 can accommodate restrictions

[1] i.e. units that switch from being treated to being untreated as the instrument changes from 0 to 1.

[2] To formally define $\boldsymbol{p}$ is suffices to note that the set of response types is finite and thus can be well ordered. Then, the $n-th$ coordinate of $\boldsymbol{p}$ represents the probability that the distribution $p \in \Delta(\mathcal{T}_Z)$ assigns to the $n-th$ response type. The specific ordering of response types is not relevant as long as it is well defined.

[3] e_k denotes the vector that takes the value 0 in every entry except the k-th which is equal to 1.

that rule-out specific response types. In appendix D I provide examples of models and restrictions encompassed by assumption 2.2.

To conclude the discussion of shape restrictions, note that if no distribution over instrument-response types $p \in \Delta(\mathcal{T}_Z)$ exists such that $\boldsymbol{p}$ satisfies equation 7 and is consistent with observed conditional treatment probabilities, then, the restrictions on instrument-response heterogeneity imposed by assumption 2.2 are falsified.

Proposition 2.1 (**Sharp observable implications of assumptions 2.1 and 2.2**). *A distribution over instrument-response types $p \in \Delta(\mathcal{T}_Z)$ is consistent with assumption 2.2 and observed conditional treatment probabilitites, if and only if it satisfies the following system of linear equalities and inequalities:*

$$A_Z \boldsymbol{p} \leq \boldsymbol{r}_Z, \tag{8}$$

$$A_X \boldsymbol{p} = \boldsymbol{r}_X. \tag{9}$$

Where A_Z and $\boldsymbol{r}_Z$ are defined as in assumption 2.2 and the system $A_X \boldsymbol{p} = \boldsymbol{r}_X$ stacks the following set of equations:

$$\mathbb{P}[X_i = x_l | Z_i = z_k] = \sum_{t^Z \in \mathcal{T}_Z : t^Z(z_k) = x_l} p(t^Z), \qquad \forall z_l \in \mathcal{Z}, x_k \in \mathcal{X}. \tag{10}$$

The proof of proposition 2.1 is straightforward and follows from the previous discussion.

3 Sharp testable implications

In this section I show how to express the restrictions implied by the observed conditional distribution of the outcome combined with the exclusion restriction as a finite set of linear restrictions on instrument-response probabilities. This result, combined with proposition 2.1 transforms the potentially infinite set of linear inequalities that describe the original model into an equivalent finite one. Here, I consider binary instruments for which I prove the sharpness of my characterization. In appendix **??** I discuss non-trivial extensions to the multiple instruments case.

In the binary instrument case, the sharp testable implications are just a collection of upper bounds on the share of units that realize the exact same treatment at every instrument value. To define these bounds, I first introduce additional notation.

For any $x_l \in \mathcal{X}$, I refer to units that realize treatment x_l at all possible instrument values as x_l-always takers. The population frequency of always-x_l-takers is given by the quantity $\mathbb{P}[X(z_0) = X(z_1) = x_l]$.

Let $p\left(AT_{x_l}\right)$ denote the probability that a distribution over response types $p \in \Delta(\mathcal{T}_Z)$ assigns to always-x_l takers. Formally:

$$p\left(AT_{x_l}\right) := \sum_{t^Z \in \mathcal{T}_Z} p(t^Z) \boldsymbol{I}_{\{t^Z(z_0)=t^Z(z_1)=x_l\}}, \tag{11}$$

where $\boldsymbol{I}_{\{t(z_0)=t(z_1)\}}$ is an indicator that response type t^Z realizes treatment x_l at both values of Z. Note that this equation is linear in $\boldsymbol{p}$ with known, deterministic coefficients.

Let $f_{Y,X|Z=z_k}$ denote the joint distribution of Y and X conditional on $Z = z_k$. Then, for any fixed $x_l \in \mathcal{X}$, define the function $\Psi_{x_l} : \mathcal{Y} \longrightarrow \mathbb{R}_+$ as follows:

$$\psi_{x_l}(y) = \underset{k=0,1}{min} \left\{ \left[f_{Y,X|Z=z_k}(y, x_l) \right] \right\}, \tag{12}$$

and the quantity:

$$\Psi_{x_l} = \int_{\mathcal{Y}} \psi_{x_l}(y) d\mu_Y. \tag{13}$$

Let t_j^Z denote the j-th instrument-response type according to some fixed, arbitrary, pre-specified order over instrument-response types. Let A_Y be a $L \times |\mathcal{T}_Z|$ matrix with entires $(A_Y)_{l,j} = \boldsymbol{I}_{\{t_j^Z(z_0)=t_j^Z(z_1)=x_l\}}$ and let $\boldsymbol{\Psi}$ denote the vector that has Ψ_{x_l} in its $l+1$-th coordinate. As the next theorem shows, the sharp observable implications of the exclusion restriction are equivalent to $A_Y \boldsymbol{p} \leq \boldsymbol{\Psi}$.

Theorem 3.1. *[Sharp observable implications of binary instruments]*
The following statements are equivalent:

1. *A distribution over response types $g \in \Delta(\mathcal{T})$ that satisfies, assumption 2.1 (instrument validity), assumption 2.2 (shape restrictions) and definition 2.3 (consistency with observed probabilities) exists.*

2. *A distribution over instrument-response types $p \in \Delta(\mathcal{T}_Z)$ exists such that p satisfies:*

$$A_Z \boldsymbol{p} \leq \boldsymbol{r}_Z, \tag{14}$$

$$A_X \boldsymbol{p} = \boldsymbol{r}_X, \tag{15}$$

$$A_Y \boldsymbol{p} \leq \boldsymbol{\Psi}. \tag{16}$$

Where A_Z, A_X, r_Z and r_X are finite dimensional matrices as defined in assumption 2.2 and proposition 2.1.

The formal proof of theorem 3.1 is given in appendix A. Here, I provide a graphical sketch of the proof and compare it to existing falsification procedures.

To see why the inequality $p(AT_{x_l}) \leq \Psi_{x_l}$ must hold for every $x_l \in \mathcal{X}$ note that always-x_l takers are a potentially strict subset of all the units that realize treatment x_l both when $Z = z_0$ and when $Z = z_1$. Therefore, if the exclusion restriction holds, the group of always-x_l takers that realize outcomes in any measurable set $B_Y \subset \mathcal{B}_Y$ must also be a (potentially strict) subset of the units that realize outcomes in B_Y when $X = x_l$ and $Z = Z_k$ for $z_k \in \{z_0, z_1\}$. Thus, the fraction of always takers with outcomes in B_Y must be weakly dominated by $\psi_{x_l}(y)$ in B_Y (see figure 1 for a graphical representation). Setting $B_Y = \mathcal{Y}$ yields the desired result.

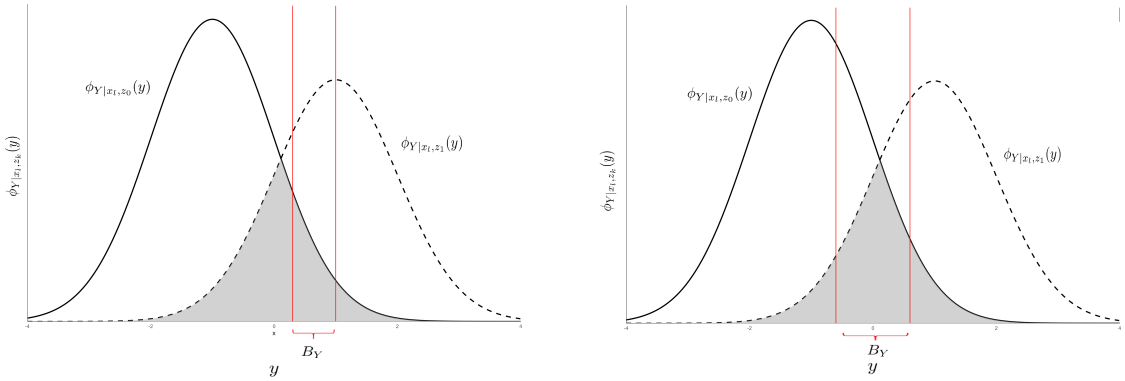
To see the sufficiency of the restrictions, suppose that some $p \in \mathcal{T}_Z$ is consistent with observed conditional treatment probabilities and satisfies $p(AT_{x_l}) \leq \Psi_{x_l}$ for every $x_l \in \mathcal{X}$. To construct a distribution over response types consistent with the observed joint distribution of (Y, X, Z), the unobserved marginal distribution of potential outcomes for always-x_l takers when $X = x_l$ can be set to $\left[\frac{p(AT_{x_l})}{\Psi_{x_l}}\right]\psi_{x_l}(y)$ so that always-x_l takers account for a fraction $\frac{p(AT_{x_l})}{\Psi_{x_l}}$ of the mass represented by Ψ_{x_l}. This construction guarantees that the unconditional sub-density associated to the outcomes of always-x_l-takers is dominated by $f_{Y,X|Z=z_0}(y, x_l)$ and $f_{Y,X|Z=z_1}(y, x_l)$. Moreover, this construction ensures that the distribution of the outcome for always takers is invariant to instrument changes, which ensures the exclusion restriction is satisfied.

It remains to distribute the fraction $1 - \frac{p(AT_{x_l})}{\Psi_{x_l}}$ of the area Ψ_{x_l} as well as any mass strictly above at least one sub-density across response types that realize x_l at just one instrument value (i.e. x_l-compliers and x_l-defiers). Assigning potential outcomes to these units is simple because –since they realize treatment x_l at just one instrument value– the exclusion restriction does not bind their potential outcomes. Figure 2a provides a visual illustration of this procedure. The formal construction is described in appendix A.

A benchmark case is when assumption 2.2 imposes that treatment x_l can only increase (or decrease) as the instrument changes. This case corresponds to the classical monotone binary instrument and treatment model from Imbens and Angrist (1994),Kitagawa (2015) and Mourifié and Wan (2017). It also corresponds to maximal elements in the ordered or unordered case as in Sun (2023).

Suppose that addoption of x_l weakly increases as Z changes from z_0 to z_1. Note that in this special case, the function $f_{Y,X|Z=z_1}(y, x_l)$ weakly dominates the function $f_{Y.X|Z=z_0}(y, x_l)$, therefore, $\psi_{x_l}(y)$ is equal to $f_{Y,X|Z=z_0}(y, x_l)$. Furthermore, the fraction of always x_l takers is identified and coincides with $\mathbf{\Psi}_{x_l}$ and $\mathbb{P}[X = x_l|Z = z_0]$. This

Figure 1: Upper bounds on always takers.



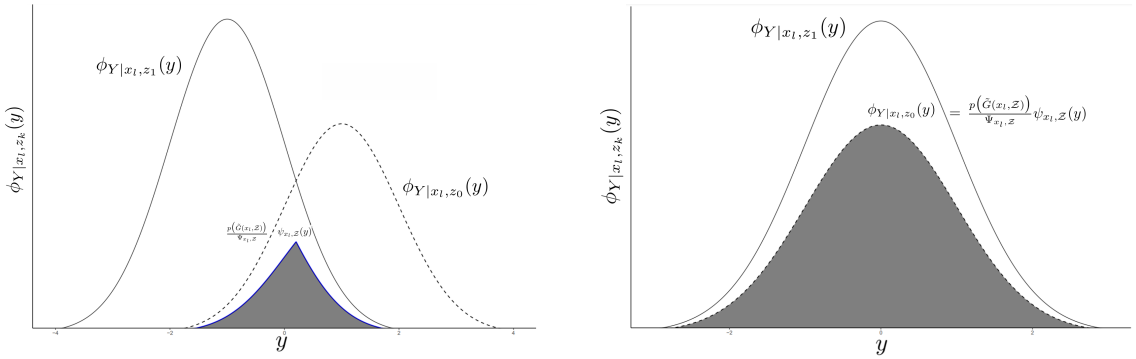
(a) (b)

Notes: *In both sub-figures the solid and dashed black lines respectively represent hypothetical sub-densities* $f_{Y,X|Z=z_0}(y, x_l)$ *and* $f_{Y,X|Z=z_1}(y, x_l)$*. The constant* Ψ_{x_l} *is defined as the integral over* $\mathcal{Y}$ *of the point-wise minimum of these two sub-densities, thus, its magnitude corresponds to the gray colored area. The two vertical red lines delimit a measurable interval* $B_Y \in \mathcal{B}_Y$*.*
The exclusion restriction implies that the distribution of outcomes for the group AT_{x_l} *of subjects that realize treatment* x_l *at both instrument values (*z_0 *and* z_1*) must be the same conditional on* $Z_i = z_0$ *or* $Z_i = z_1$*. Moreover, the units in* AT_{x_l} *whose outcomes belong to some measurable* $B_Y \in \mathcal{B}_Y$ *are a (potentially strict) subset of all the units that would realize treatment* x_l *and outcomes in* B_Y*. As a consequence, conditional on* $Z_i = z_k$*, the probability that the outcomes of such group belong to any measurable set* $B_Y \in \mathcal{B}_Y$ *must be lower that the probability that the events* $\{Y_i \in B_Y\}$ *and* $\{X_i = x_l\}$ *are jointly realized. For instance, in subfigure 1a, the fraction of units that realize treatment* x_l *and outcomes in* B_Y *conditional on* $Z = z_0$ *strictly coincides with the intersection of the gray area and the rectangle delimited by the red lines. On the other hand, conditional on* $Z = z_1$*, there is an additional group of units (whose mass corresponds to the area delimited by the red lines, the dashed black line, and the contour of the gray area) who realize outcomes in* B_Y*. This additional group cannot belong to* AT_{x_l}*, otherwise, their outcomes would change without their treatment changing, which would be a violation of the exclusion restriction. In subfigure 1b none of the two sub-densities delimits the upper contour of the gray area, and therefore, the integral of the point-wise minimum yields a tighter bound on always takers that realize outcomes in* B_Y *than the minimum of the integrals of* $f_{Y,X|Z=z_0}(y, x_l)$ *and* $f_{Y,X|Z=z_1}(y, x_l)$ *over* B_Y*. This argument is formalized in appendix* **??**.

particular case is illustrated in figure 2b.

Figure 2b illustrates why, while seemingly simpler, my approach is equivalent to Kitagawa (2015) in the binary monotone case. The integral of $\psi_{x_l}(y)$ over $\mathcal{Y}$ constrains the frequency of always takers, this statement is true even if monotonicity does not hold. The test of Kitagawa (2015) takes the function $f_{Y,X|Z=z_0}(y, x_l)$ and asks if, indeed, it coincides with ψ_{x_l} almost everywhere. Instead, I first compute the integral of $\psi_{x_l}(y)$ without invoking monotonicity. Then, I note that if violations of monotonicity occur with positive probability, $\psi_{x_l}(y)$ will be strictly smaller than $f_{Y,X|Z=z_0}(y, x_l)$ in some positive measure set. Thus, the two integrals are equal if and

Figure 2: Sufficiency of the necessary conditions



(a) General case (b) Monotone IV

Notes: *In both sub-figures the solid and dashed black lines represent hypothetical sub-densities* $\mathbf{f}_{Y,X|Z=z_0}(y,x_l)$ *and* $f_{Y,X|Z=z_1}(y,x_l)$ *respectively. The function* $\psi_{x_l}(y)$ *is defined as the point-wise minimum* $\psi_{x_l}(y) := min\{f_{Y,X|Z=z_0}(y), f_{Y,X|Z=z_1}(y)\}$*, and the constant* Ψ_{x_l} *as its integral over* $\mathcal{Y}$*, thus, its magnitude corresponds to the area that lies strictly below both sub-densities. In this figure, the gray area equals the quantity* $p(AT_{x_l})$*. The blue solid line represents the sub-density* $\frac{p(AT_{x_l})}{\Psi_{x_l}}\psi_{x_l}(y)$ *which is proportional to the distribution of potential outcomes for always-*x_l*-takers. Then, the distribution of potential outcomes for* x_l*-compliers and* x_l*-defiers can be defined to be proportional to the sub-density* $f_{Y,X|Z=z_k}(y,x_l) - \frac{p(AT_{x_l})}{\Psi_{x_l}}\psi_{x_l}(y)$ *for* $k = 0, 1$ *respectively.*
Sub-figure (a) corresponds to the general case without monotonicity. It also illustrates a situation in which a valid distribution over instrument response types $p \in \Delta(\mathcal{T}_Z)$ *is such that* $p(AT_{x_l}) < \Psi_{x_l}$*. The monotone case is illustrated in sub-figure (b). In this case, monotonicity enforces that* $p(AT_{x_l}) = \Psi_{x_l}$*, furthermore, the distribution of potential outcomes for always-*x_l*-takers is identified and is proportional to* $\psi_{x_l}(y)$*.*

only if random assignment, exclusion, and monotonicity are not violated.

For the general, non-monotone case, **?** note that the population frequency of x_l-defiers (units that realize $X_i = x_l$ only when $Z_i = z_0$) is bounded below by the supremum over measurable sets $B_Y \subset \mathcal{Y}$ of the difference $\mathbb{P}[Y_i \in B_Y, X_i = x_l | Z_i = z_0] - \mathbb{P}[Y_i \in B_Y, X_i = x_l | Z_i = z_0]$. This supremum is attained in the set $\left\{y \in \mathcal{Y} : f_{Y,X|Z=z_0}(y,x_l) \geq f_{Y,X|Z=z_1}(y,x_l)\right\}$ and its value coincides with the quantity $\mathbb{P}[X_i = x_l, Z_i = z_0] - \Psi_{x_l}$. Since at $Z = z_0$ only x_l-defiers and x_l-always takers can realize x_l, the upper bound on always takers I derive implies and is implied by the lower bound on compliers from **?**. The difference is that instead of taking a supremum over a potentially infinite collection of measurable sets, I obtain an equivalent bound by integrating the point-wise minimum of two sub-densities.

4 Generalized no-defiers restrictions

Now I specialize theorem 3.1 to binary instruments combined with restrictions that rule out specific treatment switches from the population. Examples of restrictions in this class include ordered and unordered monotonicity (Imbens and Angrist (1994), Angrist and Imbens (1995), Heckman and Pinto (2018)). I refer to these type of restriction as *generalized no-defiers restrictions*. The specialized characterization is a set of sharp inequalities with a clear microeconomic interpretation.

The essence of generalized no-defiers restrictions is that as the instrument changes, some treatments are promoted relative to others. I formalize this notion as follows: If alternative x_1 is preferred over x_0 absent any exogenous incentive, then, it must also be preferred over x_0 when the instrument promotes x_1. Hence, no units that switch from x_1 to x_0 exist in the population.

Furthermore, if generalized no-defiers restriction assume that some treatments are promoted relative to others by the instrument, then, the promoted treatments should be more prevalent at $Z = z_1$ than at $Z = z_0$. GFOSD extends this idea from the monotone, ordered case, to arbitrary generalized no-defiers restrictions.

In the ordered monotone case, treatment values are ordered (e.g. by intensity), and $Z = z_1$ is assumed to promote adoption of higher ranked treatments. The sharp observable implications of this restriction are known to be first order stochastic dominance. That is, for any given value of the treatment x_l, treatments ranked at least as high as x_l should be more likely when $Z = z_1$.

Generalized no-defiers restrictions accommodate a large class of assumptions on how Z affects realizations of X other than ordered monotonicity. Examples are, the effect on the extensive margin assumption from Rose and Shem-Tov (2024), versions of the unordered monotonicity assumption by Heckman and Pinto (2018), or other restrictions that allow for richer heterogeneity such as the examples I provide in appendix D. For these restrictions it is not obvious neither which treatments are being promoted, nor exactly how the promoted treatments should be more prevalent when $Z = z_1$.

I answer these questions by showing that any generalized no-defiers restriction is uniquely associated with a non-parametric restriction on the potential preferences of units in the population, that such unique restriction is defined by a unique binary relation over $\mathcal{X}$, and that the sharp testable implications of the restriction are GFOSD

with respect to that unique associated binary relation.

Furthermore, I show that the sharp testable implications of any *generalized no-defiers restriction* combined with the exclusion restriction from assumption 2.1 is also a set of inequalities with a GFOSD interpretation. The reason is that the exclusion restriction on its own does not rule-out any instrument-response type. But, combined with the observed distribution of the outcome, it imposes upper bounds on the prevalence of always-takers, which in turn imposes upper bounds on the total share of units that can realize certain treatments.

The GFOSD characterization is a set of inequalities rather than a system of linear equations. Having a characterization in terms of inequalities has two advantages, first, it eliminates nuisance parameters which allows for simpler inference, second, every inequality is linked to a set of behaviors that the imposed shape restriction (on its own or combined with the exclusion restriction) rules out. Hence, if an inequality is violated , it reveals a set of behaviors that exist in the population but are inconsistent with the model's assumptions.

4.1 Sharp testable implications

First, I formally define *generalized no-defiers restrictions* and show that any such restriction can be represented as a binary relation.

Assumption 4.1 (Generalized no-defiers)**.**
Let $R \subset \mathcal{T}_Z = \mathcal{X} \times \mathcal{X}$ be a fixed and known collection of instrument-response types. The underlying distribution over response types $g_0 \in \Delta(\mathcal{T})$ is such that $p_{g_0}(t^Z) = 0$ for all $t^Z \in R$.

Definition 4.1 (Binary relation associated to $R \subset \mathcal{T}_Z$)**.**
Let $R \subset \mathcal{T}_Z$ be a collection of response types. Define the binary relation $\geq^R \subset \mathcal{X} \times \mathcal{X}$ as follows:

$$x_l \geq^R x_{l'} \iff (x_l, x_{l'}) \in R \tag{17}$$

Put in words, a *generalized no-defiers restrictions* simply imposes that instrument-response types contained in the set R occur with zero probability. Moreover, x_l is related to $x_{l'}$ according to $\geq^R$ if switches from x_l to $x_{l'}$ when the instrument goes

from z_0 to z_1 are ruled out by the *generalized no-defiers restrictions* associated with R.

Suppose that a natural ordering over treatments exist (e.g. the number of months a household had insurance in a given period), and suppose that an exogenous intervention promotes the adoption of weakly higher treatments (e.g. having insurance for longer periods). Then, no unit should realize a strictly lower treatment as a consequence of the intervention, or, put formally, no unit should realize a a treatment in the lower contour of its initial one.

The generalization of the previous idea is that, for any given set of treatments S, there is a unique second set of treatments $L_{\geq^R}(S)$ such that $X(z_0) \in S$ rules out the possibility that $X(z_1) \in L_{\geq^R}(S)$. That is, any unit that realizes a treatment in S when $Z = z_0$, cannot realize a treatment in $L_{\geq^R}(S)$ when $Z = z_1$. For a *generalized no-defiers restrictions* associated with a binary relation R, the set $L_{\geq^R}(S)$ is the *common lower contour* of the set S according to $\geq^R$.

Definition 4.2 (Common lower contour of $\geq^R$)**.**
Let $\geq^R$ be a binary relation over $\mathcal{X}$ associated with the generalized no-defiers restriction R. For any $S \subset \mathcal{X}$, the common lower contour of S according to $\geq^R$ is:

$$L_{\geq^R}(S) = \{x_k \in \mathcal{X} : s \geq^R x_k, \forall s \in S\} \tag{18}$$

Having defined common lower contours, I state the sharp observable implications of *generalized no-defiers restrictyions* and binary instruments. The sharp testable implications can be summarized as follows: For every $S \subset \mathcal{X}$, no unit can have its potential treatment $X(z_0)$ in S and $X(z_1)$ in $L_{\geq^{R(S)}}$.

Theorem 4.1.
A distribution over response types $p \in \Delta(\mathcal{T}_Z)$ that satisfies assumption 2.1 (instrument validity), assumption 4.1 (instrument-response type restriction), and definition 2.3 (consistency) exists, if and only if:

1. ***Shape restriction+Independence:*** *The following inequality holds for every $S \subset \mathcal{X}$.*

$$\mathbb{P}[X_i \in S | Z_i = z_0] \leq \mathbb{P}[X_i \in L_{\geq^R}(S)^C | Z_i = z_1] \tag{19}$$

2. ***Shape restriction+Independence+Exclusion restriction:*** *For any* $S \subset \mathcal{X}$ *and any partition* (Λ_S, Λ'_S) *of* $L_{\geq^R}(S)^C$ *satisfying (i)* $\emptyset \neq \Lambda'_S \subset S$*, (ii)* $L_{\geq^R}(x_{l'})^C \setminus \{x_{l'}\} \subset \Lambda_S$ *for every* $x_{l'} \in \Lambda'_S$*, and (iii)* $L_{\geq^R}(S \setminus \Lambda'_S)^C \subset \Lambda_S$*:*

$$\mathbb{P}[X_i \in S | Z = z_0] \leq \mathbb{P}[X_i \in \Lambda_S | Z = z_1] + \sum_{x_{l'} \in \Lambda'_S} \Psi_{x_{l'}} \tag{20}$$

Where Ψ_{x_l} *is defined in equation 13.*

The formal proof of theorem 4.1 is given in appendix B. The key idea is to represent the instrument validity conditions as a flow problem: If the observed conditional treatment probability $\mathbb{P}[X_i = x_l | Z_i = z_0]$ represents the share of the population that realizes treatment x_l at instrument value z_0, then, finding a valid distribution over response types is equivalent to finding a valid set of flows that, for every $x_l \in \mathcal{X}$, takes the share $\mathbb{P}[X = x_l | Z = z_0]$ of the population that starts with treatment x_l and distributes it across potential counterfactual treatments in a way that matches the observed treatment probabilities conditional on $Z = z_1$ without allocating mass to the flows, or treatment switches that the *generalized no-defiers restriction* forbids.

This flow representation of the falsification problem is equivalent to a pair of well known combinatorial problems: the maximum flow and minimum cut problems.[4] The minimum cut formulation is particularly useful to derive the sharp observable implications of incomplete models in terms of inequalities that only depend on observable quantities as first noted by Ekeland et al. (2010) and Galichon and Henry (2011).

Another interpretation for the first part of theorem 4.1 (i.e. falsification of *generalized no-defiers restrictions* without an outcome and an exclusion restriction), is that mutually exclusive events cannot occur in the population with probability greater than one (the events $\{X_i(0) \in S\}$ and $X_i(1) \in L_{\geq^R(S)}$ are mutually exclusive). Kaido and Ponomarev (2025) recently generalized a logically equivalent idea beyond the binary instrument case, but their inference procedure only covers discrete outcomes.

To incorporate an outcome along with the exclusion restriction, Kaido and Ponomarev (2025) expand the graph representation of the model to the space of treatment-outcome pairs (i.e. $(x, y) \in \mathcal{X} \times \mathcal{Y}$). For discrete outcomes, this expanded space

[4]The equivalence is also proven in appendix B. The argument also applies to more general restrictions such as those considered in assumption 2.2. The only difference is that more general restrictions cannot necessarily be stated as capacity constraints on the edges of the graph. This would complicate using the equivalence with minimum-cut problem to obtain observable inequalities that do not depend on unobservables.

substantially increases the total number of inequalities to verify, and for continuous outcomes results in an infinite number of inequalities making statistical inference impractical. In contrast, with my approach, the number of inequalities remains the same when an outcome and an exclusion restriction are added.

The second part of theorem 4.1 incorporate the exclusion restriction which, as I show in theorem 3.1, binds the prevalence of always-x_l takers. In some cases, these bounds on always takers *tighten* the GFOSD inequalities by further limiting the maximum of units that could potentially realize treatments in certain upper contours.

The fact that the exclusion restriction *tightens* the GFOSD inequalities by replacing some conditional treatment probabilities $\mathbb{P}[X_i = x_{l'}|Z = z_1]$ with the weakly smaller quantities $\Psi_{x_{l'}}$ allows to distinguish violations of the exclusion restriction from violations of instrument-response restrictions. In particular, one of the following three cases must occur:

1. All inequalities are satisfied. Hence, neither the exclusion restriction nor the response-type restriction is falsified.

2. Only inequalities from part 2 are violated. Hence, the response type restriction is not falsified on its own, but only when combined with the exclusion restriction.

3. At least one inequality from part 1 is violated. Hence, the response type restriction is falsified

Additionally, every inequality is linked to exactly one set S. Thus, if the inequalities associated with S are not satisfied, it must be that the assumed restriction on how the instrument affects the potential treatments (or preferences over treatments) of units that realize treatments in S when $Z_i = z_0$ is falsified, while restrictions associated to other treatment switches might not.

To illustrate its usefulness, I use theorem 4.1 to characterize the binary monotone instrument IV model from Angrist and Imbens (1995).

Corollary 4.1 (**Sharp observable implications of binary monotone instruments**). *Suppose that* $\mathcal{X} = \{x_0, x_1, ..., x_{L-1}\}$, *with* $x_l < x_{l'}$ *for every* $l < l'$. *A distribution over response types exists that satisfies assumptions 2.1 and:*

$$\mathbb{P}[X(z_0) \leq X(z_1)] = 0 \tag{21}$$

If and only if:

$$\mathbb{P}[X \geq x_l | Z = z_0] \leq \Psi_{x_l} + \mathbb{P}[X > x_{l+1} | Z = z_1] \qquad \forall x_l \in \mathcal{X} \setminus \{x_{L-1}\} \tag{22}$$

In the ordered monotone case, the set $L_{\geq}(x_l)^C$ corresponds to the standard upper contour of x_l (i.e. $\{x_l, x_{l+1}, ..., x_L\}$), thus, the sharp inequalities reduce to FOSD but *tightened* by the exclusion restriction in the sense that the probability $\mathbb{P}[X = x_l | Z = z_1]$ associated to the minimal element of every $L_{\geq}(x_l)^C$ (i.e. x_l) is replaced by the weakly smaller quantity Ψ_{x_l}.[5]

4.2 A submonotone interpretation

I conclude this section by formally deriving *generalized no-defiers restrictions* from non-parametric restrictions on how the instrument affects the preferences or utility of units in the population. In particular, I assume that selection into treatment follows a utility maximization model, and, rather than restricting potential treatment choices, I restrict potential preferences over the set of treatments $\mathcal{X}$.

Assumption 4.2 ($\geq^*$-Submonotonicity)**.** *Let $\geq^*$ be an irreflexive[6] binary relation over $\mathcal{X}$. Suppose that:*

1. *Units are endowed with strict preference profiles over $\mathcal{X}$ that depend on Z_i denoted $\succ_i^{(z_k)}$ for $k = 0, 1$ where the function $\succ_i^{(z_k)}$ maps $\mathcal{Z}$ to strict preference profiles over $\mathcal{X}$. In other words, units have potential preferences.*

2. *Units are rational in the sense that $X_i(z_k) = x_l$ implies $x_l \succ_i^{(z)} x_{l'}$ almost surely for all $l, l' \in \{0, 1, ..., L-1\}$ such that $l \neq l'$.*

3. *There are no preference reversals against $\geq^*$ when the instrument changes from z_0 to z_1. That is, if $x_l \geq^* x_{l'}$ and $x_l \succ_i^{(z_0)} x_{l'}$, then, $x_l \succ_i^{(z_1)} x_{l'}$ almost surely.*

The first two points of assumption 4.2 simply posit that response types arise from latent utility maximization. The third point captures the the notion that if $x_l \geq^* x_{l'}$,

[5] Hence, it is easy to see that the conditions from Sun (2023) do not incorporate the effect that the exclusion restriction has on upper contours associated with non minimal/maximal elements. Consequently, the observable implications from Sun (2023) are not sharp for monotone instruments, and why they are nested by the ones proposed in this paper.

[6] The extension to reflexive binary relations is conceptually simple. The reasoning remains the same but requires special treatment of response type restrictions that rule out "always takers" of one particular treatment. In the interest of brevity I address this case in appendix B.2.

then alternative x_l is weakly incentivized or promoted by the instrument. Note that this promotion of x_l is not necesarilly relative to all other treatments, but relative to $x_{l'}$.

I refer to assumption 4.2 submonotonicity because it encodes a notion weaker than monotonicity. It rules out two-way flows between certain pairs of treatments, without necessarily involving all alternatives or treatments. In this sense, any submonotonicity assumption can always be extended to a full monotonicity condition with respect to a total order (or the linear extension of a total order) by incorporating additional restrictions. For instance, both ordered and unordered monotonicity correspond to special cases of submonotonicity where the binary relation is transitive. The notion of preventing "two-way flows" from Navjeevan and Pinto (2022) corresponds to assuming that the binary relation in question is asymmetric.

Theorem 4.2. *Let $\geq^*$ be an irreflexive binary relation over $\mathcal{X}$ and suppose that $R \subset \mathcal{T}$ is such that $\geq^R = \geq^*$. Then:*

1. *If $\geq^*$-Submonotonicity holds, then, $p_{g_0}(t^Z) = 0$ for every $t^Z \notin R$.*
2. *R is the smallest cardinality subset of $\mathcal{T}_Z$ such that if $p_{g_0}(t) = 0$ for all $t \notin R$, then, $\geq^*$-Submonotonicity cannot be falsified.*

The assumption that $\geq^*$ is irreflexive is not necessary, but it substantially simplifies exposition. The proof of theorem 4.2 is straightforward and can be found in appendix B.2 where the irreflexive case is also covered.

Theorem 4.2 shows that $\geq^R$-submonotonicity is sufficient to guarantee that instrument-response types inside of R occur with zero probability. Moreover, R is the minimal (in the sense of set inclusion) restriction over instrument-response types, such that $\geq^R$-submonotonicity is not falsified. Thus, the set of inequalities in 19 comprises the sharp observable implications of $\geq^R$-submonotonicity.

In Appendix E, I discuss in depth the interpretation of different assumptions within the submonotonicity class and provide multiple motivating examples. These examples include extensions of unordered monotonicity (Heckman and Pinto (2018)); assumptions that draw from both ordered and unordered monotonicity; specific cases of targeting and encouragement designs as well as some possible relaxations and generalizations (Lee and Salanié (2020), Bai et al. (2022)); and designs in which the instrument encourages variety and experimentation, or its effect depends on *status-quo* ($Z = z_0$) choices.

In section **??** I use this framework to asses the *compliers in the extensive margin* assumption of Rose and Shem-Tov (2024) in the Oregon Health Experiment, and formulate alternative restrictions that are not rejected by the data.

5 Inference

In this section I present the inference framework that yields a sharp test of IV validity. In subsection 5.1 I discuss why robustness to the contact set is desirable, and the issues that arise from it. In subsection 5.2 I introduce a classification-based estimator and argue how the classification approach yields an estimator with a tractable characterization. In subsection 5.3 I use this characterization to analyze the asymptotic properties of the estimator, and state formal asymptotic inference results. Finally, in subsection 5.4 I combine the proposed overlap estimator with the inference method for linear systems of Cox and Shi (2023) to obtain a valid, sharp test of IV validity.

Throughout this entire section, I asume that the data is an *i.i.d.* random sample $\{(Y_i, X_i, Z_i)\}_{i=1}^n$ drawn from the distribution $F_{Y,X,Z}$. In what follows, I assume that the outcome is a real-valued vector of finite dimension d so that $\mathcal{Y} = \mathbb{R}^d$.

In this section I consider the two instruments case, but the results of this section extend naturally to estimation of the overlap of any finite number of densities or subdensities.

5.1 Contact set, non-differentiability, and plug-in estimators

The main challenge for inference on the vector of sub-density overlaps arises from the region of the support of the outcome in which two or more subdensities are equal. Let $\mathcal{C}_{x_l} = \{y \in \mathbb{R}^d : f_{Y,X|Z=z_0}(y, x_l) = f_{Y,X|Z=z_1}(y, x_l)\})$ denote this region to which I refer as the contact set. When the contact set has positive measure, non-differentiability of the minimum implies that plug-in estimators are biased and might not have a limiting distribution. These problems are exacerbated if the goal is to have a fully non-parametric method.

Before elaborating on the technical challenges, I argue that, in the context of testing IV validity, robustness to the possibility that the contact set has positive measure is desirable. The first and simplest reason is that an IV validity test should be robust to the possibility that the instrument has no effect on realized treatments,

in other words, it should allow for instrument irrelevance. If the instrument has no effect on realized treatments and the exclusion restriction holds, sub-densities will not change with the instrument and therefore the contact set is equal to the entire support of the outcome.

A second, related but subtler reason has to do with the quasi-experimental nature of IV models. Consider a randomized intervention that seeks to induce individuals in a population to acquire medical insurance with the goal of estimating its effect on health outcomes. If the intervention had a 100% compliance rate, the setup would correspond to a perfectly randomized experiment; it is precisely because of non-compliance that an IV method is needed.

But non-compliance is likely to be correlated with the outcome. The healthiest subjects in the population might not get insurance even if it is promoted by the instrument. If healthier subjects do not respond to the instrument, their treatment and outcomes do not change. Thus, the joint distribution of the outcome and treatment is the same at different instrument values which results in a non-negligible contact set.

Finally, unless the relevant subdensities do not cross, in finite samplesthe possibility that the contact set has positive measure cannot be ruled out. The reason is that, for any given sample size, around the true crossing points there is a positive measure set in which the difference between densities is small and statistically indistinguishable from zero. Hence, equality on a non-negligible set cannot be rejected. If the existence of a positive measure contact set is fundamentally untestable, robustness to this possibility becomes a desirable property for both overlap estimators and instrument validity tests.

To illustrate the challenges that arise from non-differentiability, recall that the target parameter is a vector whose $l+1$-th entry, for $l \in \{0, ..., L-1\}$ is:

$$\Psi_{x_l} = \int_{\mathbb{R}^d} \min\{f_{Y,X|Z=z_0}(y, x_l), f_{Y,X|Z=z_1}(y, x_l)\}dy \tag{23}$$

A plug-in estimator for Ψ_{x_l} replaces the conditional densities $f_{Y,X|Z=z_k}(y, x_l)$ for $k = 0, 1$ in equation 23 with non-parametric density estimators $\hat{f}_{Y,X|Z=z_k}(y, x_l)$ and integrates the function $\hat{\psi}_{x_l}(y) := \min\{\hat{f}_{Y,X|Z=z_0}(\tilde{y}, x_l), \hat{f}_{Y,X|Z=z_1}(\tilde{y}, x_l)\}$.

These density estimators admit a decomposition into a signal component $f_{Y,X|Z=z_k}(y, x_l)$, and a noise component $\hat{f}_{Y,X|Z=z_k}(y, x_l) - f_{Y,X|Z=z_k}(y, x_l)$. For any $y \in \mathcal{C}_{x_l}$, the signal

components are the same for $Z = z_0$ and $Z = z_1$. Thus, for any given sample size, the estimated minimizer at y is entirely determined by statistical noise.

If the minimum is determined by statistical noise regardless of the sample size, then, for any fixed $y \in \mathcal{C}_{x_l}$, the minimizer can switch back and forth from z_0 to z_1 as the sample size increases. Consequently, the identity of the estimated minimizer at y need not converge in probability to any fixed limit.

The asymptotic distribution of a plug-in estimator depends on the random process $\{\hat{\psi}_{x_l}(y) : y \in \mathbb{R}^d\}$ which in turn depends on the estimated minimizers. The randomness in the minimizer's identity makes characterizing the variance-covariance of this process and understanding how it affects plug-in estimators technically challenging.

Moreover, aplying the mimimum operator in the contact set creates a bias problem. Instead of averaging out statistical noise, the minimum consistently picks negative noise which results in a downward biased estimator.

Finally observe that non-parametric density estimators, and more specifically kernel density estimators, do not satisfy the tightness assumption required for the *m out of n* boostrap method proposed by Fang and Santos (2019). Hence, even if the minimum is a directionally differentiable function, there are no guarantees that this bootstrap approach is valid.

To circumvent these technical challenges, I recast sub-density overlaps as the loss of an asymmetric classifier. This alternative representation of the target parameters has the advantage of eliminating most of the randomness that arises from the contact set.

5.2 A classification-based estimator

Given the functions $f_{Y,X|Z=z_0}(y, x_l)$ and $f_{Y,X|Z=z_1}(y, x_l)$, calculating Ψ_{x_l} amounts to identifying the regions in which each of these two functions is less or equal than the other, and then computing their integrals over such sets. Thus, the core problem is to identify the points in $\mathbb{R}^d$ that belong to the set $\{y \in \mathbb{R}^n : f_{Y,X|Z=z_1}(y, x_l) \geq f_{Y,X|Z=z_0}(y, x_l)\}$. Identifying this set is a classification problem.

Consider the random vector (Y, X, Z). The classification problem consists of predicting Z given Y and X. A classifier is a function $\eta : \mathcal{Y} \times \mathcal{X} \longrightarrow \{0, 1\}$ that maps pairs (y, x) to an instrument label $k = 0, 1$. A classification mistake occurs on the event $\{\eta(Y, X) \neq k, Z = z_k\}$ for $k = 0, 1$.

Given a classifier η, the probability that it incurs into a classification mistake is $\mathbb{E}[\mathbf{1}_{\{\eta(Y,X)\neq k, Z=z_k\}}] = \mathbb{P}[\eta(Y,X) \neq k, Z = z_k]$. This is a measure of the quality of the classifier η.

If one value of Z is more prevalent than other in the population, a classifier might attain a low loss by always returning the most likely value of Z. This can be avoided by penalizing classification mistakes differently when $Z = z_0$ than when $Z = z_1$. One simple alternative is to consider the sum of conditional missclassification probabilities $\mathbb{P}[\eta(Y,X) \neq k, Z = z_0] + \mathbb{P}[\eta(Y,X) \neq k, Z = z_1]$. As the next theorem shows, for the oracle classifier η^*, this quantity coincides with Ψ_{x_l}

Proposition 5.1. *Define the classifier* $\eta^*(y, x_l) := \mathbf{1}\big\{f_{Y,X|Z=z_1}(y, x_l) \geq f_{Y,X|Z=z_0}(y, x_l)\big\}$ *and the loss function* $L^*(\eta^*, x_l)$ *as:*

$$L^*(\eta^*, x_l) := \sum_{k=0,1} \mathbb{P}\big[\eta^*(Y,X) \neq k, Z = z_k, X = x_l \big| Z = z_k\big]. \tag{24}$$

Then:

$$\Psi_{x_l} = L^*(\eta^*, x_l). \tag{25}$$

Furthermore, for any classifier $\eta'(y, x_l)$ *such that* $\eta'(y, x_l) = \eta^*(y, x_l)$ *for every* $y \in \mathbb{R}^n \setminus \mathcal{C}_{k_l}$*:*

$$L^*(\eta^*, x_l) = L^*(\eta', x_l). \tag{26}$$

Proposition 5.1 shows that the overlap parameter is equivalent to the loss of the classifier $L^*(\eta^*, x_l)$. Furthermore, it shows that any classifier that only differs from $L^*(\eta^*, x_l)$ in the contact set attains the same loss as $L^*(\eta^*, x_l)$, resulting in an alternative valid expression for Ψ_{x_l}. Thus, at the population level, the classification approach is invariant to the behavior of classifiers in the contact set. This invariance property makes analyzing the asymptotic behavior of a classification-based estimator tractable.

I propose the following estimation procedure: first, split the sample in two subsamples S_1 and S_2 with n_1 and n_2 observations respectively, where the ration $n1/n2$ tends to some $\tau \in (0, 1)$ as the sample size tends to infinity. In the first split, estimate the classifier η^*. Then, evaluated the empirical loss of the estimated classifier in the

second split. The resulting empirical loss is an estimator for Ψ_{x_l}.

To formalize the estimator, let $K(\cdot)$ be a kernel function, and h_n a bandwidth sequence with $K_n(y - Y_i) := K\big(\frac{y-Y_i}{h_n}\big)$. The sub-density estimators are:

$$f_{Y,X|Z=z_k,n}(y, x_l) := P_{k,n_1}^{-1}\Big[\frac{1}{n_1 h_n^d}\sum_{i=1}^{n_1}\mathbf{1}\{X_i = x_l, Z_i = z_k\}K_n\big(y - Y_i\big)\Big], \tag{27}$$

where the sum is over all units in the first split, $n_{k,1} := \sum_{i=1}^{n_1}\mathbf{1}\{Z_i = z_k\}$ and $P_{k,n_1} := n_{k,1}/n_1$.

Then, define $\Delta_{n_1}(y, x_l) := f_{Y,X|Z=z_1,n}(y, x_l) - f_{Y,X|Z=z_0,n}(y, x_l)$ and $\eta^*_{n_1}(y, x_l) := \mathbf{1}\{\Delta_{n_1}(y, x_l) \geq 0\}$. The empirical loss of the estimated classifier $\eta^*_{n_1}(y, x_l)$ evaluated in the second split is the empirical analogue of the last expression in (25):

$$\begin{aligned} L_{n_2}(\eta^*_{n_1}, x_l) := \frac{1}{n_2}\sum_{i=1}^{n_2}\Big(&\frac{\mathbf{1}\{\eta^*_{n_1}(Y_i, X_i) = 1, Z_i = z_0, X_i = x_l\}}{P_{0,n_2}} \\ &+ \frac{\mathbf{1}\{\eta^*_{n_1}(Y_i, X_i) = 0, Z_i = z_1, X_i = x_l\}}{P_{1,n_2}}\Big), \end{aligned} \tag{28}$$

where the sum is over all observations that belong to the second split and P_{k,n_2} denotes the sample probability that Z is equal to z_k in the second split. The estimator is $L_{n_2}(\eta^*_{n_1}, x_l)$.

Note that, by construction, the estimated classifier $\eta^*_{n_1}$ yields a higher theoretical loss than the oracle classifier η^*. Thus, the estimator is upward biased. But, as I show in the next sub-section, this bias is asymptotically zero.[7]

Sample splitting plays a crucial role in deriving the asymptotic properties of the estimator. First, it eliminates the finite sample, downward bias that arises from the contact set. The reason is that, for any value of y inside the contact set, the expected empirical loss in the second sample, is invariant to classification decisions made in the first sample. Furthermore, sample splitting implies that the difference between the theoretical loss of the oracle classifier η^* and its estimator $\eta^*_{n_1}$ is uncorrelated with the difference between the empirical and theoretical loos of $\eta^*_{n_1}$. This last insight simplifies the analysis of the asymptotic variance of the estimator.

Two natural questions are whether if using the proposed estimators for the overlap and the variance of its estimator, a full sample approach would be valid, or if K-fold

[7] In simulations I find that this bias is negligible even in small and medium samples.

cross-fitting could recover full sample asymptotics. I investigate both alternatives in a set of monte-carlo simulations presented in appendix 7. I find that both alternatives perform well, as long as the contact set is negligible. But, if the contact set is non-negligible, the full-sample estimator is biased, and that confidence intervals for both of these methods have coverage below the nominal level. The in the simulations, realized coverage ranges from 23% to 85% which is substantially below the 95% target.

Furthermore, in appendix C.6 I analyze K-fold cross fitting estimators. In particular, I propose modifying the sample splitting estimator so that, in a set that is guaranteed to contain the contact set with probability one as the sample size tends to infinity, the classifier takes a pre-specified, deterministic decision. This ensures that the classifier has an fixed, non-stochastic limit, and thus, the overlap estimator admits an asymptotically linear expression. I derive necessary conditions under which this modified estimator is valid. The key condition is a stronger margin condition. As the simulations in appendix 7, in practice, sample splitting might outperform K-folds cross-fitting.

The IV validity conditions involve the vector of overlaps $\mathbf{\Psi}$ along with conditional treatment probabilities $\mathbb{P}[X = x_l | Z = z_k]$. Thus, in the evaluation split I also compute estimates of conditional treatment probabilities:

$$P_{x_l|z_k,n_2} = \frac{\sum_{i=1}^{n_2} \mathbf{1}\{X_i = x_l, Z_i = z_k\}}{\sum_{i=1}^{n_2} \mathbf{1}\{Z_i = z_k\}}. \tag{29}$$

Conditional treatment probabilities are linearly deficient, to avoid degenerate asymptotic distributions, I restrict the target parameter to all but one observed treatment values. Let $\mathcal{X}_k = \{l : \mathbb{P}[X = x_l | Z = z_k] > 0\}$ denote the set of treatment indices that occur with strictly positive probability conditional on $Z = z_k$[8]. Define $\boldsymbol{P}_{\boldsymbol{\mathcal{X}_k,n}}$ as the vector that stacks all the estimators $P_{x_0|z_k,n_2}$ for every $l \in \mathcal{X}_k$ except the one associated with the highest treatment index in $\mathcal{X}_k$. Let $\boldsymbol{P}_{\boldsymbol{\mathcal{X}_k}}$ be the vector that stacks the associated target conditional treatment probabilities.[9]

[8] In some empirical applications, this set might be known. In some others this set might not be obvious ex-ante. However, with a sufficiently large sample, the identity of all the treatment values that occur with positive probability can be identified.

[9] I restrict the conditional probability estimators to treatment values in $\mathcal{X}_k$ to ensure that none of the estimators has a degenerate limit distribution.

5.3 Asymptotic properties

To analyze the asymptotic properties of the estimator, I decompose its scaled difference with respect to the target parameter into two terms. One that corresponds to the theoretical difference between the loss of target and the trained classifiers, and another that corresponds to the difference between the empirical and theoretical loss of the trained classifier.

$$\sqrt{n_2}\Big[L_{n_2}(\eta^*_{n_1}, x_l) - \Psi_{x_l}\Big] = \sqrt{n_2}\Big[L_{n_2}(\eta^*_{n_1}, x_l) - L(\eta^*_{n_1}, x_l)\Big] + \sqrt{n_2}\Big[L(\eta^*_{n_1}, x_l) - L(\eta^*, x_l)\Big] \tag{30}$$

To obtain valid inference, I note that the asymptotic behavior of the first term depends on neither the distribution of the classification decision at particular values of $y \in \mathbb{R}^d$, nor the correlation between classification decisions at different values of $y \in \mathbb{R}^d$. It only depends on aggregate missclassification rates, which substantially simplifies analyzing and computing its variance. Combining this insight with a set of sufficient conditions that make the second term $o_p(1)$ I prove that the estimator is asymptotically normal and propose a variance estimator.

To make the second term in equation 30 $o_p(1)$ I first use the fact that the population level loss is invariant to the behavior of $\eta^*_{n_1}$ in the contact set (see proposition 5.1).

It remains to analyze the loss of $\eta^*_{n_1}$ out of the contact set. The key insight is that, under mild regularity conditions satisfied by most standard kernel functions, the rate of uniform convergence for KDEs is $r_n = \sqrt{\frac{log(h_n^{-1})}{nh_n^d}}$ (Giné and Guillou (2002)). Therefore, in the region of $\mathbb{R}^d$ in which the absolute difference $|\Delta_{x_l}(y)| := |f_{Y,X|Z=z_1}(y, x_l) - f_{Y,X|Z=z_0}(y, x_l)|$ is greater than r_n, classification mistakes occur with very low probability. Consequently, while the additional loss in which $\eta^*_{n_1}$ might incur relative to η^* in this region can be large, the expected additional loss is negligible.

The only remaining region is the set in which $0 < |\Delta_{x_l}(y)| \leq r_n$. Since r_n tends to zero, both the measure of this set and the magnitude of the extra loss from suboptimal classification decisions in this region tend to zero as the sample grows. The declining loss and measure of the region, however, are not enough to guarantee that the extra loss is negligible at the rate $n^{-1/2}$. Thus, I make an additional assumption that forces the region $\{y \in \mathbb{R}^d : 0 < |\Delta_{x_l}(y)| \leq r_n\}$ to shrink at a sufficiently fast rate, or, in other words, I impose a margin condition (see assumption 5.1.3).

Now I formalize the assumptions I make on the functions $f_{Y,X|Z=z_k}$, the kernel function, and bandwidth sequence:.

Assumption 5.1.

1. ***Non-zero instrument probabilities:*** *A constant* $C \in (0,1)$ *exists such that:*

$$\mathbb{P}[Z = z_k] > C \ \ \forall z_k \in \mathcal{Z}. \tag{31}$$

2. ***Bounded continuous Hessian :*** *For every* $x_l \in \mathcal{X}$ *and* $z_k \in \mathcal{Z}$*, the function* $f_{Y,X|Z=z_k}(y, x_l)$ *is twice continuously differentiable in* y *with Hessian matrix* $\nabla^2 f_{Y|X=x_l,Z=z_k}$*. Furthermore, fixed constants* $M > 0$ *and* $M' > 0$ *exist such that:*

$$\sup_{y \in \mathbb{R}^d} f_{Y,X|Z=z_k}(y, x_l) \leq M, \tag{32}$$

and

$$\sup_{||u||=1} ||(\nabla^2 f_{Y|X=x_l,Z=z_k})u|| \leq M' \tag{33}$$

3. ***Margin condition:*** *For every* $x_l \in \mathcal{X}$ *and* $z_k \in \mathcal{Z}$*, positive constants* B, α*, and* $\delta > 0$ *exist such that* $\forall t \in (0, \delta)$

$$\boldsymbol{\mu}\big\{y \in \mathcal{Y} : 0 < |f_{Y,X|Z=z_1}(y, x_l) - f_{Y,X|Z=z_0}(y, x_l)| \leq t\big\} \leq Bt^{\alpha} \tag{34}$$

Where $\boldsymbol{\mu}$ *denotes the Lebesgue measure over* $\mathbb{R}^d$*.*

Assumption 5.1 imposes mild conditions on the data generating process. 5.1.1 requires that all instrument values occur with positive probability in the population, which is guaranteed to occur in most empirical settings. 5.1.2 requires that the distribution of the outcome be bounded, and twice continuously differentiable with bounded Hessian. I combine this assumption with standard properties of kernel density estimators (KDE) to argue that the contribution of density estimation bias is negligible for classification mistakes. The margin condition from 5.1.3 is typically imposed in the non-parametric classification literature to obtain fast rates of convergence for estimated classifiers (Tsybakov (2004)).[10] In the one-dimensional outcome

[10]It is also used in optimal policy and DML (add citations!!!!).

case, to satisfy the margin condition, values of α as low as 1/4 suffice, depending on the bandwidth choice.

Assumption 5.2.

1. ***Kernel regularity:*** *The kernel function $K : \mathcal{Y} \mapsto \mathbb{R}$ is positive, symmetric around zero, bounded, and square integrable. Furthermore, it satisfies assumption (K_1) from Giné and Guillou (2002).*[11]

2. ***Bandwidth sequence restrictions:*** *The bandwidth sequence h_n satisfies:*

   (a) *$h_n = n^{-\gamma}$ for some $\gamma > \frac{1}{d+4}$.*

   (b) *$nh_n^d / ln(n) \longrightarrow \infty$.*

   (c) *The constant α from assumption 5.1.3 and γ satisfy: $\alpha > \frac{d\gamma}{1-d\gamma}$.*

Assumption 5.2 is a set of common requirements for kernel functions and bandwidth sequences. 5.2.1 is required to control bias and to use the results from Giné and Guillou (2002) to bound the probability that KDEs take extreme values and thus incur in classification mistakes. Typical kernel choices such as gaussian, triangular, uniform or Epanechnikov satisfy this assumption. Part (a) of 5.2.2 simply requires some degree of undersmoothing so that KDEs bias is negligible, part (b) requires that the bandwith decreases at a rate slower than $n^{-1/d}$. Part (c) is necessary to invoke the margin condition from assumption 5.1.3 and guarantee that the bias caused by classification mistakes in the proximity of the contact set is asymptotically negligible.

Assumption 5.3.

1. ***Strictly positive-overlaps:*** *A fixed constant $\epsilon > 0$ exist such that, for every $x_l \in \mathcal{X}$:*

$$\Psi_{x_l} > \epsilon. \tag{35}$$

[11] The assumption requires that K lies in the linear span of functions $\{g : \mathcal{Y} \mapsto \mathbb{R}_+\}$ such that the set $\{(s,u) : g(s) \geq u\}$ can be represented as a finite number of Boolean operations among sets of the form $\{(s,u) : p(s,u) \geq \phi(u)\}$ where p is a polynomial and ϕ an arbitrary real function. See Giné and Guillou (2002) for a discussion of this assumption.

2. ***Non-trivial classification problem*** *A constant* $\epsilon > 0$ *exist such that for every* $x_l \in \mathcal{X}$ *and every* $z_k \in \mathcal{Z}$*:*

$$\operatorname*{plim}_{n\to\infty} \int_{D_k} f_{Y,X|Z=z_k}(y)dy > \epsilon \tag{36}$$

Where $D_k = \mathbb{R}^d \setminus \{y \in \mathbb{R}^d : \eta^*_{n_1}(y, x_l) \neq k\}$.

Assumption 5.3 is only necessary for testing IV validity, it is not required for inference on sub-density overlaps. The condition $\Psi_{x_l} > 0$ from assumption 5.3.1 ensures the limit distribution of the overlap estimator is non-degenerate asymptotically. This condition is testable using the proposed inference framework on the overlap parameters alone.

Assumption 5.3.2 rules out the possibility that, as the sample grows, the classifier estimated in the first split always selects the same instrument value. This condition guarantees that the overlap estimator is not perfectly correlated with conditional treatment probability estimators.

Assumption 5.3.2 can only be violated in two cases, the first is when the contact set is equal to $\mathbb{R}^d$ and yet, even as the sample size tends to infinity, the estimated classifier picks the same fixed instrument value z_k for every $y \in \mathbb{R}^d$. Since this first case appears implausible in practice, I rule it out. The second case is when one sub-density strictly dominates the other almost everywhere. This second condition is testable and, if verified, both the classification and overlap estimation problems become trivial with a sufficiently large sample.[12]

Under these assumptions, I obtain valid joint inference on conditional treatment probabilities and sub-density overlaps.

Theorem 5.1. *Suppose assumptions 5.1, 5.1.2, and 5.3 hold.*

Define the vectors $\boldsymbol{\theta}_n^T = (\boldsymbol{P}^T_{\boldsymbol{\mathcal{X}_0,n}}, \boldsymbol{P}^T_{\boldsymbol{\mathcal{X}_1,n}}, \boldsymbol{\Psi}_n^T)$ *and* $\boldsymbol{\theta}_{\mathbf{0}}^T = (\boldsymbol{P}^T_{\boldsymbol{\mathcal{X}_0}}, \boldsymbol{P}^T_{\boldsymbol{\mathcal{X}_1}}, \boldsymbol{\Psi}^T)$ *and let* m *denote their dimension. Let* $\boldsymbol{\Omega}_n$ *denote the symmetric positive definite matrix* $n_2 Var(\boldsymbol{\theta}_n|S_1)$.

[12]For the alternative estimator based on a semi-deterministic classifier that I introduce in the next section, assumption 5.3.2 can be guaranteed to hold by choosing an appropriate default deterministic classification rule.

Then:

$$\sqrt{n_2}\boldsymbol{\Omega}_n^{-1/2}\big(\boldsymbol{\theta}_n - \boldsymbol{\theta}_0\big) \xrightarrow{d} \mathcal{N}(0, \boldsymbol{I}_m) \tag{37}$$

Furthermore, the formulas from lemma C.8 in appendix C show how to construct an estimator $\boldsymbol{\Sigma}_n$ for the matrix $\boldsymbol{\Omega}_n$. The matrix $\boldsymbol{\Sigma}_n$ satisfies:

$$\big|\big|\boldsymbol{\Sigma}_n - \boldsymbol{\Omega}_n\big|\big| \xrightarrow{p} 0. \tag{38}$$

In lemma C.5 from appendix C I show that sample splitting implies that the expectation of the empirical loss $L_{n_2}(\eta^*_{n_1} m, x_l)$ is equal to the theoretical loss $L(\eta^*_{n_1} m, x_l)$. Hence, since $L(\eta^*_{n_1} m, x_l)$ converges to Ψ_{x_l}, the overlap estimators are asymptotically unbiased.

Note that theorem 5.1 does not yield asymptotic normality for $\boldsymbol{\theta}_n$ but for the normalized sequence $\boldsymbol{\Omega}_n^{-1/2}\boldsymbol{\theta}_n$.The reason is that, even if the overlap estimator Ψ_{x_l,n_1} converges to a fixed constant, it is obtained by using the second split to evaluate the empirical loss of the classifier trained in the first split $\eta^*_{n_1}$. Therefore, its variance conditional on the first split depends on $\eta^*_{n_1}$. If the contact set has positive measure, the estimated classifier $\eta^*_{n_1}$ has no stable limit and thus $\sqrt{n_2}(\Psi_{x_l,n} - \Psi_{x_l})$ has no stable asymptotic distribution.

In particular, the conditional variance of the empirical loss in the second split is given by:

$$n_2 Var\big(\Psi_{x_l,n} - \Psi_{x_l}\big|S_1\big) = \sum_{k=0,1} \frac{P_{\xi,x_l|z_k,n_2}(1 - P_{\xi,x_l|z_k,n_2})}{\mathbb{P}[Z = z_k]} \tag{39}$$

Where $P_{\xi,x_l|z_k,n_2} := \mathbb{P}[\eta^*_{n_1}(Y, X_l, Z; z_k, x_l) \neq k, X = x_l | Z = z_k]$.

If classification decisions in the contact set are dictated by statistical noise, then, $\eta^*_{n_1}$ randomly re-classifies subsets of the contact set from one instrument value to another. Furthermore, this re-classification does not stop as the sample size tends to infinity. Hence, $P_{\xi,x_l|z_k,n_2}$ does not converge to a fixed constant and as a consequence the limit of the scaled variance is not a constant. The same phenomenon occurs with the covariance between overlap estimators associated with different treatment values, and their covariance with conditional treatment probabilities.

Variance instability does not prevent inference on the vector of overlap parameters,

it only implies that it has no fixed limit distribution. That is why normalization by $\boldsymbol{\Omega}_n^{-1/2}$ is required to obtain asymptotic normality.

5.4 Formal sharp tests of IV validity

Now I discuss how to implement a test of instrument validity based on the estimators from subsection 5.3 and the characterizations from theorems 3.1 and 4.1. In particular, I show that the testing problem satisfies the assumptions from Cox and Shi (2023) and thus their method is applicable.

Let $\boldsymbol{\theta_0}$ denote the vector of non-redundant conditional treatment probabilities and sub-density overlaps. Theorem 3.1 shows that the null hypothesis that assumptions 2.1 (IV independence and exclusion) and 2.2 (shape restriction) are jointly satisfied is equivalent to the hypothesis that the linear system:

$$\boldsymbol{H}_0 : A\boldsymbol{p} \leq \boldsymbol{\theta}_0, \quad \boldsymbol{p}_0 \in \Delta(\mathcal{T}), \tag{40}$$

admits at least one solution. Here, A_{Sharp} is a deterministic matrix given by $A_{Sharp}^T = (A_Z^T, A_X^T, A_Y^T)$, the matrices A_Z^T, A_X^T and A_Y^T are as in theorem 3.1, and $\boldsymbol{p}_0$ is a distribution over instrument-response types. That is, $\boldsymbol{p}_0$ is a non-negative vector of dimension L^K such that the sum of its coordinates is equal to one.

In the case of *generalized no-defiers restrictions* from section 4, theorem 4.1 shows that the hypothesis can also be cast as:

$$A_{GFOSD}\boldsymbol{\theta}_0 \leq 0 \tag{41}$$

Where A_{GFOSD} is a matrix with deterministic, known entries, obtained from stacking all the equations from theorem 4.1.

Cox and Shi (2023) show that both the null hypotheses can be recast as

$$\boldsymbol{H}_0 : T := \min_{\boldsymbol{b}: A\boldsymbol{b} \leq 0} (\boldsymbol{\theta} - \boldsymbol{b})^T \boldsymbol{\Omega}^{-1} (\boldsymbol{\theta} - \boldsymbol{b}) = 0, \tag{42}$$

for some fixed, deterministic matrix A with known coefficients and a positive-definite matrix $\boldsymbol{\Omega}$. Furthermore, they propose a testing procedure based on comparing the sample analogue of T to the critical values of a χ-squared distribution with degrees of freedom equal to the rank of the rows of A associated with the inequalities that

bind at the optimal $\boldsymbol{b}$.

Definition 5.1 (Admissible nulls)**.** *Let $\mathcal{F}$ be the set of all distributions over $\mathbb{R}^d \times \mathcal{X} \times \mathcal{Z}$ such that:*

1. *F satisfies assumptions 5.1 and 5.3 for some fixed values of C, M, M', α, B and ϵ.*

2. *The set of instrument-response types $p \in \Delta(\mathcal{T}_Z)$ such that p satisfies equations 14 to 16 from theorem 3.1 with respect to the probabilities induced by $F_{Y,X,Z} \in \mathcal{F}$ is non-empty.*

Theorem 5.2. *Suppose that $\boldsymbol{\theta}_n$ is the estimator from theorem 5.1 with variance estimator $\boldsymbol{\Sigma}_n$ and let A be the matrix from equation (42).*

Define:

$$T_n := \min_{\boldsymbol{b}: A\boldsymbol{b} \leq 0} (\boldsymbol{\theta}_n - \boldsymbol{b})^T \boldsymbol{\Sigma}_{\boldsymbol{n}}^{-1} (\boldsymbol{\theta}_n - \boldsymbol{b}), \tag{43}$$

and let $\hat{r}$ be the rank of the matrix comprised by the rows of A associated with the inequalities that bind at the optimal $\boldsymbol{b}$ in equation 43.

For $q \in [0,1]$, let $\chi^2_{m,1-q}$ be the $1-q$-quantile of a χ-squared distribution with m degrees of freedom. Define $\phi_n^{CC}(q) := \mathbf{1}_{\{T_n > \max\{\chi^2_{\hat{r},1-q}, tol\}\}}$ and some tolerance $tol > 0$. Then:

$$\limsup_{n\to\infty} \sup_{F_{Y,X,Z} \in \mathcal{F}} \mathbb{P}_{F_{Y,X,Z}} \left[\phi_n^{CC}(q) = 1\right] \leqq \tag{44}$$

To prove theorem 5.2 I verify that every sequence F_n in $\mathcal{F}$ is such that the centered estimator $\sqrt{n}(\boldsymbol{\theta}_n - \boldsymbol{\theta})$ and its variance estimator $\boldsymbol{\Sigma}_n^{-1/2}$ satisfy assumption 3(a) from Cox and Shi (2023). The proof is in appendix C. The only requirement is that, as definition 5.1 imposes, assumptions 5.1 and 5.3 must hold for the same fixed, strictly positive constants for all admissible nulls. This requirement ensures that all admissible distributions yield non-degenerate variance-covariance matrices and therefore the test statistic is well defined. The requirement rules out sequences of null hypothesis with conditional treatment probabilities $\mathbb{P}[X = x_l | Z = z_k]$ or overlaps Ψ_{x_l} that are strictly positive for every sample size n, but decrease with n and reach zero in the limit.

6 Simulations

In this section, I conduct a series of Monte Carlo simulations to evaluate the performance of the test of IV validity and the method for inference on subdensity overlaps that I propose in this paper. I find that the theoretical properties established in section 5 are supported by the simulation results, and that both the test and the estimator are robust to choices of tuning parameters such as the kernel function, the bandwidth sequence, and which fraction of the data is used to estimate the kernel-based classifiers. Moreover, I show that, for some data generating processes, my test outperforms existing, discretization-based tests of IV validity.

In subsection 6.1 I show simulations that asses the performance of the method for inference on subdensity overlaps. In subsection 6.2 I asses the performance of the test of IV validity.

Tuning parameters: The classification-based approach to overlap estimation I propose has three main tuning parameters: the kernel function, the bandwidth sequence, and the fraction of the data that is used to estimate the classifier. I denote these tuning parameters as K, h_n and c_p respectively. Furthermore, for $p \in (0,1)$, c_p denotes that a share p of the data is used to estimate the classifier, and thus, a share $1-p$ is used to evaluate its loss.

Throughout all simulations I consider three kernel functions: Uniform, Gaussian and Epanechnikov. For the bandwidth sequence, I consider three different rates: $h_n = s_n n^{-1/3}$, $h_n = s_n n^{-1/4}$ and $h_n = s_n n^{-1/5}$ where the scaling factor s_n is defined so that, when $h_n \propto n^{-1/5}$, the bandwidth coincides with Silverman's rule of thumb[13] (Silverman (2018)). Finally, I consider the splits $c_{1/3}$, $c_{1/2}$ and $c_{2/3}$.

6.1 Joint estimation and inference on subdensity overlaps

Data generating processes: The main motivation for the classification-based estimator I propose is its robustness to the contact set. Moreover, the margin condition from assumption 5.1.3 plays a key role in deriving the asymptotic properties of the estimator. Thus, in addition assessing how different choices of tuning parameters affect the estimator, the simulations also examine how the estimator's performance is affected by the measure of the contact set, and the slopes of the subdensities $f_{Y,X|Z=z_0}$

[13] That is, $s_n = 0.9 \min\left\{\hat{\sigma}, \frac{IQR}{1.34}\right\}$, where $\hat{\sigma}$ is the sample standard deviation

and $f_{Y,X|Z=z_1}$ near the boundary of the contact set.

The data generating processes of this subsection are tailored to study the performance and robustnees of the subdensity overlap estimator without depending or relying on the IV structure for interpretation. I consider simulations with two instrument, and two treatment values (i.e. $\mathcal{Z} = \{z_0, z_1\}$ and $\mathcal{X} = \{x_0, x_1\}$). The unconditional instrument probabilities are $\mathbb{P}[Z = z_0] = 1/3$ and $\mathbb{P}[Z = z_1] = 2/3$, that is, $Z = z_1$ is more likely and thus, subdensities associated with $Z = z_0$ will be more precisely estimated. Since the focus of these simulations is overlap estimation and not instrument validity, I impose independence between the random variable X and the random vector (Y, Z). Hence, treatment probabilities are given by $\mathbb{P}[X = x_l|Z = z_k] = 0.5$, for $k = 0, 1$.

In the simplest case, I assume that $Y|Z = z_k$ follows a normal distribution with standard deviation $\sigma = 1$ and let the mean depend on Z. More specifically, I assume that $Y|Z = z_k \sim \mathcal{N}(\mu_{z_k}, 1)$ with $\mu_{z_1} = 0$ and $\mu_{z_k} = \tau > 0$. Observe that when $\mu_1 = 0$, the subdensities $f_{Y,X|Z=z_0}$ and $f_{Y,X|Z=z_1}$ are identical and thus the contact set correspond to the entire real line. On the other hand, if $\mu_1 > 0$, the subdensities differ everywhere except in one point, and the contact set has zero measure.

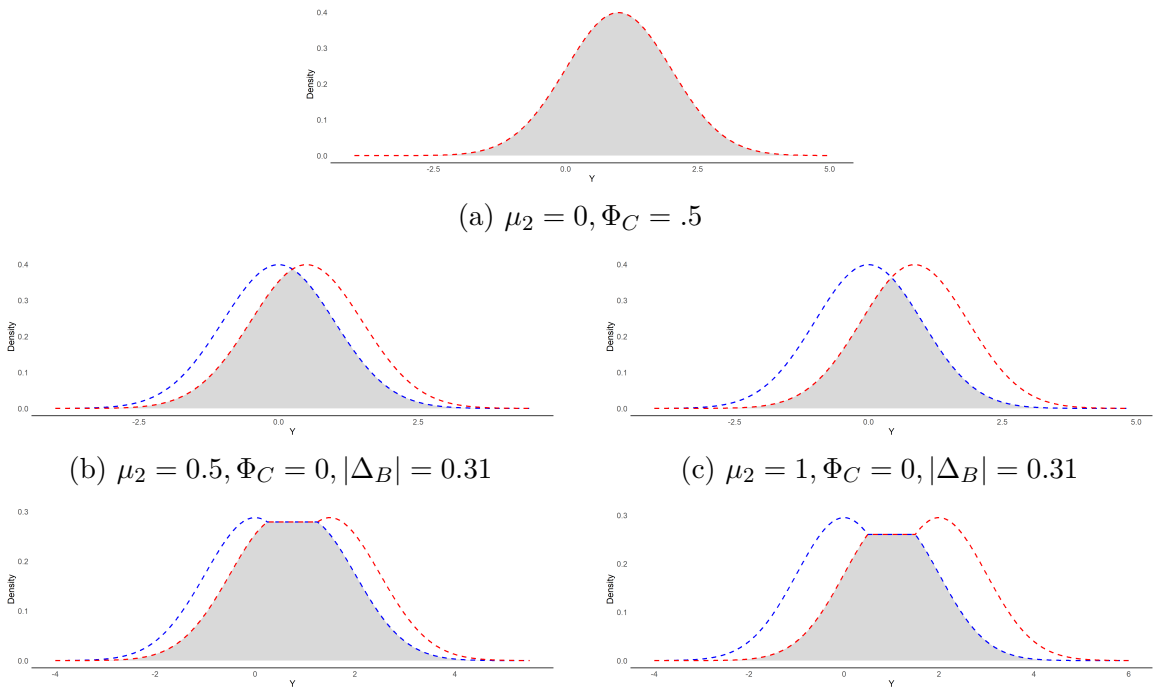
Since μ_1 affects the point in which $f_{Y,X|Z=z_0}$ and $f_{Y,X|Z=z_1}$ cross, it is related to the margin condition. For example, suppose that μ_1 is small, then, the two densities will cross near the point $y = 0$ in which the derivative of the normal density is zero. Consequently, in a neighborhood of the crossing point, the derivatives of $f_{Y,X|Z=z_0}$ and $f_{Y,X|Z=z_1}$ are also close to zero. Thus, separation between subdensities is slow and an estimated classifier will require a large sample to confidently identify the true minimizer in such neighborhood. Furthermore, since $y = 0$ is the global maximizer of the standard normal distribution, realizations of Y near the crossing point occur with non-negligible probability. On the other hand, if μ_1 is large, the densities will cross at a point in which either their derivatives are larger, or their mass is close to zero. In both of these two cases, classification mistakes have a smaller impact on the estimator than when μ_1 is close to zero.

Additionally, I consider DGPs in which the contact set is non-negligible, but different from the entire support of Y. Specifically, I modify the normal densities so that exactly in the crossing point, a region in which both densities are flat and identical appears. I denote the width of this flat contact region by w. Thus, different combinations of μ_1 and w allow me to asses how the performance of the estimator

is affected as the measure of the contact set, and separation near its boundary vary jointly, or alone.

In figure 3 I illustrate the data generating process for different values of μ_1 and w. Below every figure I report the parameters μ_1 and w, as well as the integral of $f_{Y,X|Z=z_k}(y, x_l)$ over the contact set (denoted Φ_C), and the absolute difference of the derivatives of the normal densities at the crossing points, that is $\Delta_C := |f_{Y,X|Z=z_0}(\tilde{y}, x_l) - f_{Y,X|Z=z_1}(\tilde{y}, x_l)|$[14], where $\tilde{y}$ is the crossing point. The data generating process is formally described in appendix 7.

Figure 3: Outcome distributions, contact set, and density separation under different DGPs.



(a) $\mu_2 = 0, \Phi_C = .5$

(b) $\mu_2 = 0.5, \Phi_C = 0, |\Delta_B| = 0.31$

(c) $\mu_2 = 1, \Phi_C = 0, |\Delta_B| = 0.31$

(d) $\mu_2 = 0.5, \Phi_C = .28, |\Delta_B| = 0.16$

(e) $\mu_2 = 1, \Phi_C = .27, |\Delta_B| = 0.31$

Reported statistics: I report three main summaries of the simulations. First, I report the mean of the quantity $\Psi_{x_0,n} - \Psi_{x_0}$, that is the difference between the estimated and true overlap associated with $X = x_0$. Second, I use the overlap and variance estimators I propose in section 5 to construct confidence intervals at the 95% confidence level for a single overlap. Finally, the instrument validity test requires joint

[14]Since X does not affect the distribution of Y, I omit the dependence of Δ_C on x_l.

inference on the vector $\boldsymbol{\theta}$ of subdensity overlaps and conditional treatment probabilities, hence, I report the observed coverage frequency of ellipsoidal confidence regions at the 95% confidence level for $\boldsymbol{\theta}$ constructed using the joint estimator and variance from section 5.

To construct ellipsoidal confidence regions, I consider the set:

$$E_{0.95} = \{\tilde{\boldsymbol{\theta}} : (\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_n)^T \boldsymbol{\Sigma}_n^{-1} (\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_n) \leq \chi^2_{3d,0.95}\} \tag{45}$$

Where $\xi^2_{3L,0.95}$ denotes the .95 quantile of a χ squared random variable with $3L$ degrees of freedom[15]. I count and report in which percent of the simulations the true vector $\boldsymbol{\theta}$ is contained in $E_{0.95}$.

Simulation results: Table 1 reports the results from 500 simulations for sample sizes $n = 500$, $n = 1000$, and $n = 2000$. Each column corresponds to a different combination of the bandwidth rate h_n and the sample-splitting regime c_p. The rows correspond to different combinations of μ_1 and w along with one of the three reported statistics. Here, I report the results for uniform Kernels, the results for uniform and Epanechnikov kernels are quantitatively and qualitatively similar and can be found in appendix (7).

The simulations show that the bias of the classification-based estimator is small and the confidence intervals yield coverage rates close to the nominal level.

The sample splitting regime $c_{1/3}$ under-performs relative to the other alternatives I consider. This result is consistent with the fact that my inference method does not require inference on the estimated classifier, only to evaluate its expected loss, and that sample splitting implies that the loss of the estimated classifier evaluated in the second sample-split, is unbiased for its theoretical loss. Thus, a larger estimation sample improves the accuracy of the classifier without biasing its empirical loss.

In the simulations, bias and confidence interval coverage do not change in a clear direction as the bandwidth varies. One explanation is that, while under-smoothing (i.e. bandwidths that shrink at a rate faster than $n^{-1/5}$) reduces the bias of the first stage kernels and, consequently, of the classifier, it also requires a stronger margin condition. Thus, the gains from improving the classifier far away from the contact set, are offset by an increase in the measure of the region around the contact set

[15]Recall that L is the cardinality of the support of X. The vector $\boldsymbol{\theta}$ comprises d subdensity overlaps, and $2d$ conditional treatment probabilities.

where classification is noisy.

Table 1: Bias, overlap coverage, and joint overlaps and CTPs coverage for sample splitting estimators.

			(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
			$h_n = n^{-1/3}$			$h_n = n^{-1/4}$			$h_n = n^{-1/5}$		
$n = 500$											
$\mathbb{E}n_{z_0} = 167$, $\mathbb{E}n_{z_1} = 333$			$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$
$w = 0$	$\mu_1 = 0$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.94	0.95	0.95	0.95	0.94	0.93	0.94	0.93
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.93	0.93	0.93	0.92	0.93	0.92	0.89	0.91	0.92
	$\mu_1 = 0.5$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.04	0.03	0.02	0.03	0.02	0.02	0.03	0.02	0.01
		$\Psi_{x_0} \in C_{0.95}$	0.85	0.91	0.92	0.87	0.91	0.93	0.88	0.92	0.93
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.73	0.82	0.87	0.74	0.83	0.89	0.73	0.82	0.90
	$\mu_1 = \tau$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.93	0.94	0.94	0.94	0.93	0.93	0.94	0.93	0.93
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.93	0.92	0.90	0.92	0.92	0.90	0.92	0.91	0.91
$w = 1$	$\mu_1 = 0.5$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.03	0.02	0.01	0.02	0.01	0.01	0.02	0.01	0.01
		$\Psi_{x_0} \in C_{0.95}$	0.90	0.90	0.93	0.88	0.90	0.93	0.90	0.92	0.94
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.81	0.88	0.90	0.80	0.87	0.91	0.79	0.87	0.89
	$\mu_1 = \tau$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.93	0.95	0.93	0.94	0.94	0.93	0.94	0.94	0.93
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.92	0.91	0.91	0.92	0.91	0.92	0.92	0.91	0.92
$n = 1000$											
$\mathbb{E}n_{z_0} = 333$, $\mathbb{E}n_{z_1} = 666$			$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$
$w = 0$	$\mu_1 = 0$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.93	0.96	0.95	0.94	0.96	0.95	0.94	0.96	0.95
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.94	0.94	0.93	0.94	0.95	0.94	0.93	0.94	0.93
	$\mu_1 = 0.5$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.03	0.02	0.02	0.02	0.02	0.01	0.02	0.01	0.01
		$\Psi_{x_0} \in C_{0.95}$	0.84	0.89	0.92	0.86	0.91	0.92	0.88	0.92	0.92
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.72	0.85	0.89	0.77	0.88	0.91	0.79	0.89	0.92
	$\mu_1 = \tau$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.95	0.95	0.95	0.95	0.94	0.95	0.96	0.94
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.93	0.94	0.92	0.93	0.94	0.93	0.92	0.94	0.93
$w = 1$	$\mu_1 = 0.5$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.02	0.01	0.01	0.02	0.01	0.01	0.01	0.01	0.01
		$\Psi_{x_0} \in C_{0.95}$	0.89	0.92	0.93	0.90	0.92	0.93	0.89	0.92	0.93
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.80	0.88	0.90	0.81	0.89	0.90	0.81	0.87	0.91
	$\mu_1 = \tau$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.96	0.95	0.94	0.96	0.93	0.95	0.94	0.94
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.94	0.95	0.94	0.94	0.96	0.95	0.95	0.96	0.95
$n = 2000$											
$\mathbb{E}n_{z_0,x_l} = 666$, $\mathbb{E}n_{z_1,x_l} = 1333$			$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$
$w = 0$	$\mu_1 = 0$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.96	0.95	0.95	0.95	0.95	0.94	0.94	0.95
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.94	0.96	0.95	0.95	0.96	0.95	0.94	0.94	0.95
	$\mu_1 = 0.5$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.02	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.87	0.92	0.93	0.92	0.93	0.94	0.92	0.93	0.94
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.79	0.89	0.92	0.86	0.91	0.92	0.87	0.91	0.92
	$\mu_1 = \tau$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.95	0.94	0.96	0.96	0.94	0.96	0.95	0.95
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.95	0.93	0.94	0.95	0.94	0.94	0.94	0.94	0.94
$w = 1$	$\mu_1 = 0.5$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.01	0.01	0.01	0.01	0.01	0.00	0.01	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.91	0.93	0.94	0.92	0.94	0.94	0.92	0.93	0.93
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.85	0.91	0.93	0.87	0.92	0.93	0.86	0.92	0.94
	$\mu_1 = \tau$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.96	0.95	0.95	0.95	0.95	0.95	0.94	0.93
		$\boldsymbol{\theta}_0 \in E_{0.95}$	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.93

6.2 Testing IV validity

Now I present a set of simulations designed to asses the performance of the test of IV validity I propose. Specifically, I investigate the size and power of the test, and compare it to the discretization-based test proposed by Kwon and Roth (2024). The simulations show that the test has correct size control, power against violations of IV validity, and that it can outperform discretization based tests.

The comparison with discretization-based tests allows me to illustrate two risks that discretizing entails and my test avoids: under rejection due to loss of information, and over-rejection due the bias that arises from approximating a smooth integral with a finite grid. Both risks are caused by the fact that discretization-based tests use conditions that either imply or are implied by the sharp testable implications, but are not logically equivalent to them.

Discretization risks: To discuss discretization, I first define it. Let $\mathcal{P}_{Disc} = \{P_0, ..., P_{N_{Disc}-1}\}$ denote a finite partition of the support of the outcome $\mathcal{Y}$ where each of the sets P_0 is called a *discretization bin*[16]. Then, a *discretized* outcome Y_{Disc} can be constructed according to the following rule: $Y_{Disc} := \sum_{m=0}^{N_{Disc}-1} m\mathbf{1}\{Y \in P_m\}$. Observe that since Y_{Disc} is a deterministic function of Y, it is a random variable with support $\{0, ..., N_{Disc} - 1\}$.

The first problem that discretizing might cause is that any feature of the original distribution of (Y, X, Z) that is inconsistent with the IV validity assumptions, but occurs within the same discretization bin, cannot be detected by discretization.

For example, suppose that the outcome of interest is a health index, that the treatment is a binary indicator that individuals have insurance, and that the instrument reduces the cost of insurance and thus, it weakly promotes insurance acquisition. Furthermore, having a cheaper alternative allows units that chose it to save. Suppose that the healthier subjects are more likely to spend those savings into improving their diet and exercising, while the rest of the population would spend it on goods that have no effect on their health. In this hypothetical scenario, violations of the exclusion restriction can only occur and be detected at high values of Y, that is, near the right tail of its distribution.

Now suppose that, as is common practice (Kwon and Roth (2024)), the outcome is partitioned into N discretization bins constructed from the quantiles of the uncon-

[16]Note that in practice, when $\mathcal{Y} = \mathbb{R}$, the sets P_0 typically are intervals.

ditional distribution of the health index. Then, unless the rigth tail of the support of Y is spread across different discretization bins, the discretized outcome will not reflect the violations of the exclusion that occur in the right tail, while the true distribution of Y might. Thus, the model can only be falsified by a sufficiently fine partition or by a test that does not discretizes.

Furthermore, while fine discretizations might detect violations of IV validity at the population level, in practice, they impose both computational and statistical costs. The computational cost comes from how the system of equations that describe the discretized IV model increases with the cardinality of the discretizing partition. The statistical cost comes from the fact that discretization requires estimating the conditional probability that every single bin is realized. Large cardinality partitions imply that the average number of observations per bin is small, and thus, bin probabilities are noisily estimated.

Discretizing can not only lead to tests that fail to detect violations of IV validity, it can also lead to tests that reject, even when the null hypothesis holds at the population level. To see this, note that discretization is equivalent to using a finite integration grid to estimate the overlap parameter Ψ_{x_l}. Unless the contact set is known, using a discrete grid will necessarily either over-estimate, or under-estimate Ψ_{x_l}. If Ψ_{x_l} is over-estimated, discretization leads to a test that uses a set of sufficient, but not necesary conditions for IV validity, and thus, under-rejects. If Ψ_{x_l} is under-estimated, discretization leads to a test that uses necessary but not sufficient conditions for IV validity, and thus it over-rejects.

Neither testing necessary, nor sufficient conditions can avoid the problem of failing to detect some violations of IV validity. Thus, a testing procedure that over-estimates Ψ_{x_l} can simulatenously and erroneously over and under reject the null of IV validity depending on the true data generating process. In practical applications, there is no way for researchers to know in which of the two scenarios they are.

Data generating processes (Undetectable violations of IV validity): The first data generating process I consider corresponds to a case in which the exclusion restriction is only violated at the right hand tail of the distribution of the outcome.

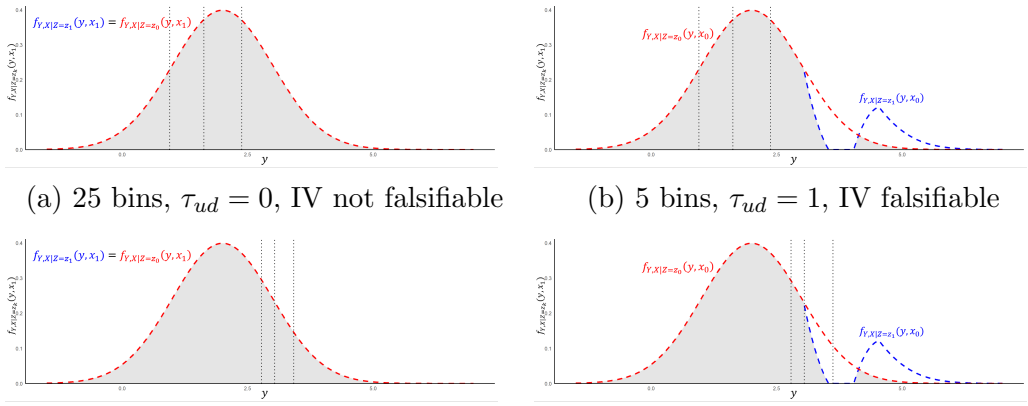
I consider a binary instrument ($\mathcal{Z} = \{z_0, z_1\}$), three different treatment levels ($\mathcal{X} = \{x_0, x_1, x_2\}$), and violations of the exclusion restriction that only occur when $X = x_2$ and near the rightmost tail of the subdensity $f_{Y,X|Z=z_k}(y, x_2)$.

To simplify the simulations, I impose standard ordered monotonicity and let in-

strument assignment be independent from the realized potential treatments. Thus, absent violations of the exclusion restriction, the densities $f_{Y,X|Z=z_0}(y,x)$ and $f_{Y,X|Z=z_1}(y,x)$ are identical. When exclusion is violated, they only differ when $X = x_2$ and near the righmost tail of $f_{Y,X|Z=z_0}(y,x_2)$. This difference implies that the overlap parameter is strictly smaller than $\mathbb{P}[X = x_2|Z = z_0]$ (which, under monotonicity, corresponds to the share of always-x_2-takers) and therefore, the model is falsified.

I parametrize violations of exclusion with a scalar $\tau_{ud} \in (0, \infty)$. When $\tau_{ud} = 0$ the IV model is not falsifiable, and when $\tau_{ud} > 0$ it is. The data generating process is formally defined in appendix 7. The following figure illustrates the functions $f_{Y,X|Z=z_k}$ for both $k = 0, 1$, evaluated at $X = x_2$, and for two different values of τ_{ud}. Additionally, the figures depict the partitions that result from discretizing using 5 and 25 bins constructed from the quantiles of the unconditional outcome distribution.

Figure 4: Data generating process: Undetectable IV validity violations



(a) 25 bins, $\tau_{ud} = 0$, IV not falsifiable

(b) 5 bins, $\tau_{ud} = 1$, IV falsifiable

(c) 5 bins, $\tau_{ud} = 0$, IV not falsifiable

(d) 25 bins, $\tau_{ud} = 0$, IV falsifiable

Notes: *These figures depict the functions $f_{Y,X|Z=z_k}(y, x_l)$ fixing $x_l = x_2$, for $k = 0, 1$, for the data generating process in which violations of the exclusion restriction only occur at the rightmost tail of the distribution. The instrument is binary $\mathcal{Z} = \{z_0, z_1\}$, and the treatment takes three values $\mathcal{X} = \{x_0, x_1, x_2\}$. The outcome Y is continuous with support $\mathbb{R}$. The scalar τ_{ud} parametrizes violations of IV validity, $\tau_{ud} = 0$ implies that the IV assumptions are not falsified.*

The figures in the first column (subfigures (a) and (c)) show the baseline case in which the two densities are equal and thus, there are no violations of the exclusion restriction, and outcomes only depend on realized treatment. These two figures depict the subdensity that results from fixing $X = x_2$, that is, $f_{Y,X|Z=z_k}(y, x_2)$ for every $y \in \mathbb{R}$. In this baseline case, $Y|X, Z$ follows a normal distribution and X affects its mean linearly. Thus, the equivalent figures for $X = x_0$ and $X = x_1$ would be similar, just translated by the constant treatment effect.

The figures in the second column (subfigures (b) and (d)) depict a case with violations of the exclusion restriction. In particular, at treatment value $X = x_2$, some of the mass in the right tail of the normal distribution is shifted further right, but only if $Z = z_1$. This results in an overlap area that is strictly smaller than in the benchmark case, which in turn leads to independence, exclusion, and monotonicity being jointly falsified. In this case, the equivalent figures for $X = x_0$ and $X = x_1$ are not affected, and thus, would coincide with the baseline case, that is, they would depict two identical functions, proportional to the distribution of a normal random variable.

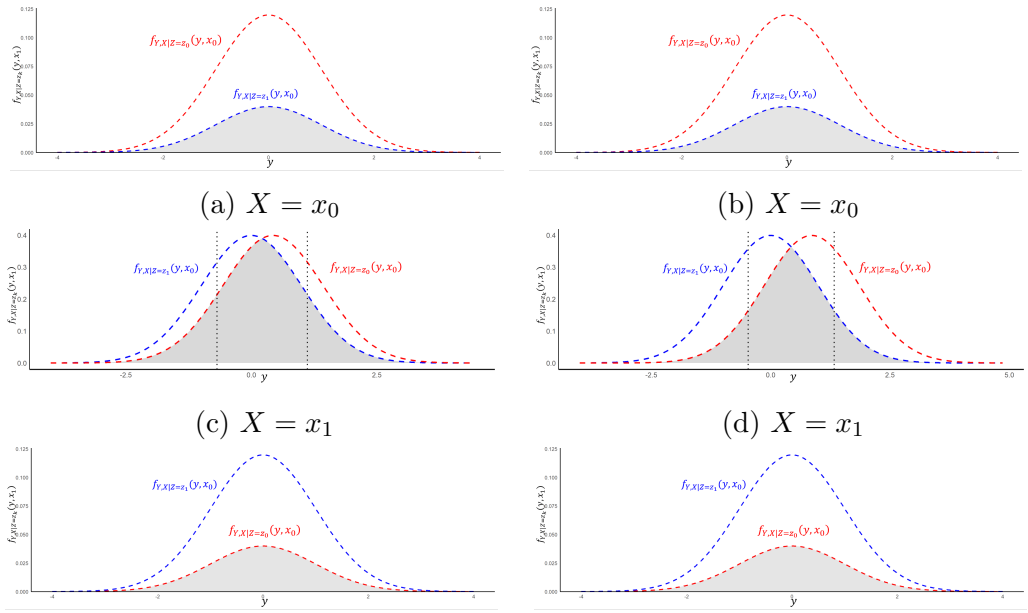
The vertical lines in the subfigures in the top (subfigures (a) and (b)) row depict the limits of the three rightmost discretization bins that would result if discretization was done using 5 bins constructed from unconditional quantiles of the observed density f_Y. The figures in the bottom row (subfigures (b) and (d)) only differ in that the vertical lines depict the three rightmost bins that would result if 25 discretization bins were used instead of 5.

Data generating processes (Finite grid integration failure): The second data generating process I consider simply lets $Y|Z,X$ follow a normal distribution so that $f_{Y,X|Z=z_k}(y,x_l)$ is proportional to the density of a normal random variable. In this DGP, the instrument is binary, and the treatment has three support points ($\mathcal{X}=\{x_0,x_1,x_2\}$). I let the mean of $Y|Z,X$ depend on both X and Z. Dependence on X represents treatment effects, and dependence of Z violations of the exclusion restriction.

When testing IV validity against this data generating process, I impose monotonicity. Moreover, I allow for the instrument to affect the outcome only when $X=x_1$, that is, a the intermediate treatment value $X=x_2$. These conditions imply three things: One, when $X=x_0$, the function $f_{Y,X|Z=z_0}$ strictly dominates the function $f_{Y,X|Z=z_1}$[17]. Similarly, when $X=x_2$, the function $f_{Y,X|Z=z_1}$ strictly dominates the function $f_{Y,X|Z=z_0}$. Finally, at $X=x_1$, neither of the two functions dominates the other, the two subdensities cross and their overlap (Ψ_{x_1}) potentially binds the prevalence of always-x_1-takers. If Ψ_{x_1} is sufficiently large, it does not bind and the model is not falsifibable. On the other hand, as it shrinks, it constraints the admissible share of always takers in the population and eventually implies that the data generating process is inconsistent with instrument independence, exclusion, and monotonicity. That is, if Ψ_{x_1} is small, the IV model is falsified.

For this data generating process, I parametrize violations of the IV model with the scalar τ_{di}. Depending on the value of τ_{di} the joint distribution of Y,X and Z might be jointly compatible or not with IV validity and mononicity. In particular, $\tau_{di}=1$ corresponds to the case in which the restriction $\mathbb{P}[\textit{Always takers}]\leq\Psi_{x_2}$ binds exactly, thus, $\tau_{di}\leq 1$ implies that IV validity, and monotonicity are not falsifiable, while $\tau_{di}>0$ implies falsifiability. The formal description of this data generating process is given in appendix 7. The following figure illustrates the functions $f_{Y,X|Z=z_0}$ at different values of X.

[17] Since x_1 is the minimal treatment and its take-up can only decrease when Z changes from $Z=z_0$ to $Z=z_1$.



(a) $X = x_0$ (b) $X = x_0$

(c) $X = x_1$ (d) $X = x_1$

(e) $X = x_2$ (f) $X = x_2$

Notes: *These figures depict the functions $f_{Y,X|Z}(y, x_l)$ over $\mathcal{Y} = \mathbb{R}$, for different fixed values of x_l. The first row (subfigures (a) and (b)) corresponds to x_0, the second (subfigures (c) and (d)) to x_1 and the third (subfigures (e) and (f)) to x_2. The data generating process satisfies monotonicity, that is, when Z changes from z_0 to z_1, units either realize the same treatment, or switch to one with a strictly higher index.*
In the first row, in both columns, treatment $X = x_0$ is more likely when $Z = z_0$, thus, the function $f_{Y,X|Z=z_0}$ dominates $f_{Y,X|Z=z_1}$. Equivalently, $X = x_2$ is more likely when $Z = z_1$ and thus, in the third row, $f_{Y,X|Z=z_1}$ dominates $f_{Y,X|Z=z_0}$. This is because x_0 and x_2 are the minimal and maximal treatment respectively and thus, the former is weakly disincentivized by the instrument, while the later is weakly incentivized. The second row corresponds to $X = x_1$. Since this is not a minimal or maximal treatment, both units that switch from x_0 to x_1, and from x_1 to x_2 might exist. Nonetheless, the population frequency of always-x_1-takers is bounded above by the overlap ψ_{x_2}. This constraint might be binding, and if ψ_{x_2} is sufficiently small, imply that the model is falsified. The figure in the second row, in the first column, corresponds to the case $\tau_{di} = 0.5$, while the one to the left to $\tau_{di} = 1.5$. In the first case, the restriction on always takers is not binding and thus the data generating process is compatible with independence, exogeneity and ordered monotonicity. In the second case, the restriction is binding, thus, IV validity is falsified.

Multiple outcomes and other simulations: The identification and inference theory I present in this paper is not restricted to one dimensional outcomes, it extends naturally to a muti-valued Y. In appendix 7 I show an additional set of simulations in which I implement the considering a two dimensional outcome. The results are qualitatively similar to the ones I show in this section.

Additionally, I asses the robustness of the method to the case in which some

treatments occur with low probability in the population and thus, their associated overlaps are effectively estimated from just a few observations. I find that the test retains correct size and some power against nulls that are inconsistent with IV validity.

Finally, to asses the quality of the inputs that the final step of the test receives (i.e. my implementation of the method by Cox and Shi (2023)), for all the data generating processes I consider in this section, in appendix 7 I also asses the bias, and confidence interval coverage of my overlap estimators. I find low bias, and coverages close to nominal rates.

Simulation results: I implement my test and the discretization-based test proposed by Kwon and Roth (2024) with a nominal size of 5%, and compare their perfomance in Montecarlo simulations. The simulations cover different combinations of tuning parameters and data generating processes primitives. The three main takeaways are that my test is robust to choices of the tuning parameters, that discretization-based tests can be very sensitive to the cardinality of the discretizing partition, and that, for the data generating processes considered, my overlaps-based test has better size control and more power.

In the simulations, I consider samples of size 500, 1000, and 2000. For every sample size and data generating process, I draw 500 samples, in every sample I implement multiple versions of both tests. For each version of the test, I compute its average rejection rate across simulated samples.

All results are summarized in table 2. Each panel corresponds to a different data generating process. In both panels, each column corresponds to a different test, and each row to a different combination of data generating process and sample size. I report results for 14 different tests grouped in two categories, one for my method using different combinations of tuning parameters, and one for the method proposed by Kwon and Roth (2024) varying the number of discretization bins.

The test of Kwon and Roth (2024) requires a method for inference on linear systems. They consider different alternatives such as Andrews et al. (2023), Fang et al. (2023), Cox and Shi (2023), or Cho and Russell (2024). Since the test I propose relies on the procedure by Cox and Shi (2023), I implement their method using the same procedure so that differences are not driven by the procedure used for inference on linear systems. Furthermore, I consider 5 discretization cardinalities, $N_{Disc} = 3$, $N_{Disc} = 5$, $N_{Disc} = 15$, $N_{Disc} = 25$, and $N_{Disc} = 50$. For my test, I consider the bandwidth sequences $h_n \propto n^{-1/3}, n^{-1/4}$ and $n^{-1/5}$, and splits $c_{1/3}, c_{1/2}$ and $c_{2/3}$.

The first 9 columns of table 2 report the rejection rates of my method under different tuning parameter values. The last 5 for the method by Kwon and Roth (2024) under different discretization regimes. The acronym NF indicates which rows correspond to data generating processes under which monotonicity, exclusion and exogeneity are not jointly falsified. The capital letter F indicates data generating processes under which these assumptions are jointly falsified.

The asymptotic theory from section 5 depends on the full sample size n. Nonetheless, the proposed overlap estimator, its variance, and conditional treatment probability estimatores (that is, all the parameters required to test IV validity) are functions of conditional probabilities, where the conditioning is on the different instrument values. Thus, in the tables, I also report the expected number of observations per instrument values, which I denote by $\mathbb{E}n_{z_k}$.

The first panel (A) of table 2 corresponds to the data generating process in which exclusion is only violated at $X = x_2$ (i.e. the highest treatment), and on the right tail of the distribution of the outcome. Recall that $\tau_{ud} = 0$ corresponds to distributions from which IV validity cannot be falsified. Consistent with this, both, my test and the one proposed by Kwon and Roth (2024) do not reject the null hypothesis more than 5% of the times.

However, when τ_{ud} is different from zero, the two tests differ. For a sample size of 500 (first four columns of panel A in table 2), the rejection rate of my method increases as τ_{ud} grows and violations of the assumptions become easier to detect, while discretization retains the exact same rejection rate of 5%.

Furthermore, as the sample size increases, the rejection rate of my test steadily increases. On the other hand, for the test of Kwon and Roth (2024), the relation between the rejection rate and the sample size depends on the number of bins in which the suport of the outcome is partitioned. For 3, 5 and 15 bins, the rejection rate does not increases with the sample size. Only for 25 and 50 bins the rejection rate increases.

When $n = 1000$ and $\tau_{ud} = 1$, my method rejectes between 14% and 24% of the times, while Kwon and Roth (2024) with 25 bins reject 13% of the times. If instead $\tau_{ud} = 1.5$ my test rejects between 15% to 30% of the times while discretization attains a maximum rate of 16%. That is, the best version of my test rejects almost twice as much as the best performing version of the test by Kwon and Roth (2024).

When the sample size is equal to 2000, a similar patern arises. In general, my

test using either $c_{1/2}$ or $c_{2/3}$ has a rejection rate that is between 3/2 and 2 times larger than the discretization-based test. The only exception is when $n = 2000$ and $\tau_{1.5}$. That is, for the largest simulated violation and with the largest simulated sample size. However, for that same sample size, and a less severe violation of the IV assumptions, my test with $c_{1/2}$ or $c_{2/3}$ attains rejection rates between 12% and 20% while discretization with 50 bins only rejects 6% of the times.

The second panel (B) of table 2 shows results for the data generating process in which falsifiability depends on how accurately the integral Ψ_{x_l} is estimated. As in panel A, for my tests and when the null of IV validity holds, the rejection rate is not higher than 5%. When the null is violated, the rejection rate is higher than 5%. Furthermore, the sample splitting regimes $c_{1/2}$ or $c_{2/3}$ ouperform the regime $c_{1/3}$.

On the other hand, the results in panel B of table 2 for the test proposed by Kwon and Roth (2024) show how discretization might lead to severe over-rejection. In particular, if 3 or 5 bins are used, their test rejects between 30 and 79% of the times under a data generating process under which IV validity is not falsifiable.

While this issue is alleviated with finer discretizations, the cost is that the test loses power against invalid nulls. Only for the test based on 15 bins rejection rates increase beyond the nominal size of 5% as violations of the IV assumptions become stronger. These rates, however, decline again when the number of bins increases to either 25 or 50. Furthermore, in all rows of panel B in which the model is falsifiable, my test attains higher rejection rates.

Throughout all simulations reported in panel A, three main patterns arise. The first pattern is that, my method has noticeably more power with $c_{1/2}$ or $c_{2/3}$, that is, when at least half the data is used to train the classifier, than under the sample splitting regime $c_{1/3}$. This is consistent with the discussion and results from the previous subsection: additional estimation data improves the classifier, while the empirical loss of the classifier is unbiased for its theoretical loss, and thus, on average, the gains from additional evaluation data are smaller.

The second, is that my test is robust to choices of tuning parameters, specially bandwidth choices. On the other hand, the simulations show that discretization-based tests are sensitive to the cardinality of the discretizting partition and conclusions might substantially differ depending on how which value of this tuning parameter is chosen.

Finally, the simulations show that, consistent with theory, the rejection rate of

my method increases both with the sample size, and when joint violations of the IV assumptions become stronger. On the other hand, discretization-based tests might not necesarilly improve with the sample size, and its power against stronger violations of the null depends on the choice of discretizing partition.

Table 2: Null hypothesis of IV validity rejection rates.

Panel A: Undetectable violations of the assumptions

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
		KDE overlaps									KR Discretization				
DGP		$h_n = n^{-1/3}$			$h_n = n^{-1/4}$			$h_n = n^{-1/5}$							
$n = 500$ $\mathbb{E}n_{z_k} = 250$		$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	b_3	b_5	b_{15}	b_{25}	b_{50}
$\tau_{ud} = 0$	NF	0.04	0.03	0.03	0.04	0.03	0.04	0.05	0.03	0.03	0.06	0.03	0.02	0.01	0.01
$\tau_{ud} = 0.5$	F	0.05	0.06	0.07	0.06	0.05	0.07	0.07	0.06	0.06	0.06	0.03	0.02	0.03	0.02
$\tau_{ud} = 1$	F	0.08	0.10	0.11	0.09	0.12	0.10	0.09	0.10	0.10	0.06	0.03	0.02	0.05	0.04
$\tau_{ud} = 1.5$	F	0.11	0.14	0.14	0.11	0.17	0.14	0.12	0.17	0.14	0.06	0.03	0.02	0.06	0.05
$n = 1000$ $\mathbb{E}n_{z_k} = 500$		$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	b_3	b_5	b_{15}	b_{25}	b_{50}
$\tau_{ud} = 0$	NF	0.05	0.05	0.03	0.05	0.05	0.03	0.05	0.05	0.04	0.07	0.04	0.04	0.01	0.00
$\tau_{ud} = 0.5$	F	0.09	0.10	0.08	0.08	0.10	0.08	0.09	0.11	0.09	0.07	0.04	0.03	0.06	0.02
$\tau_{ud} = 1$	F	0.14	0.20	0.17	0.14	0.21	0.16	0.14	0.24	0.18	0.07	0.04	0.03	0.13	0.07
$\tau_{ud} = 1.5$	F	0.15	0.24	0.23	0.17	0.28	0.22	0.19	0.30	0.24	0.07	0.04	0.03	0.13	0.16
$n = 2000$ $\mathbb{E}n_{z_k} = 1000$		$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	b_3	b_5	b_{15}	b_{25}	b_{50}
$\tau_{ud} = 0$	NF	0.04	0.05	0.04	0.03	0.05	0.04	0.03	0.03	0.05	0.06	0.05	0.03	0.02	0.01
$\tau_{ud} = 0.5$	F	0.09	0.17	0.12	0.08	0.20	0.14	0.09	0.18	0.15	0.06	0.05	0.03	0.18	0.06
$\tau_{ud} = 1$	F	0.19	0.38	0.28	0.19	0.40	0.30	0.23	0.43	0.31	0.06	0.05	0.03	0.27	0.51
$\tau_{ud} = 1.5$	F	0.24	0.48	0.36	0.26	0.51	0.39	0.28	0.52	0.39	0.06	0.05	0.03	0.27	0.83

Panel B: Discrete integration failure

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
		KDE overlaps									KR Discretization				
DGP		$h_n = n^{-1/3}$			$h_n = n^{-1/4}$			$h_n = n^{-1/5}$							
$n = 500$ $\mathbb{E}n_{z_k} = 250$		$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	b_3	b_5	b_{15}	b_{25}	b_{50}
$\tau_{di} = 0$	NF	0.02	0.01	0.02	0.01	0.01	0.01	0.03	0.01	0.02	0.53	0.31	0.01	0.00	0.00
$\tau_{di} = 0.5$	F	0.06	0.07	0.08	0.06	0.08	0.08	0.06	0.08	0.07	0.31	0.24	0.03	0.00	0.00
$\tau_{di} = 1$	F	0.14	0.20	0.15	0.13	0.21	0.18	0.13	0.21	0.17	0.22	0.29	0.10	0.01	0.00
$n = 1000$ $\mathbb{E}n_{z_k} = 500$		$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	b_3	b_5	b_{15}	b_{25}	b_{50}
$\tau_{di} = 0$	NF	0.02	0.02	0.02	0.03	0.02	0.02	0.02	0.03	0.03	0.63	0.38	0.00	0.00	0.00
$\tau_{di} = 0.5$	F	0.08	0.12	0.09	0.08	0.12	0.10	0.08	0.12	0.10	0.30	0.23	0.05	0.00	0.00
$\tau_{di} = 1$	F	0.21	0.39	0.31	0.22	0.39	0.33	0.21	0.42	0.32	0.32	0.51	0.29	0.05	0.00
$n = 2000$ $\mathbb{E}n_{z_k} = 1000$		$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	b_3	b_5	b_{15}	b_{25}	b_{50}
$\tau_{di} = 0$	NF	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.02	0.02	0.79	0.52	0.00	0.00	0.00
$\tau_{di} = 0.5$	F	0.12	0.20	0.17	0.12	0.22	0.18	0.12	0.22	0.18	0.29	0.28	0.12	0.01	0.00
$\tau_{di} = 1$	F	0.36	0.62	0.52	0.35	0.65	0.52	0.38	0.65	0.54	0.45	0.77	0.59	0.21	0.02

7 An application to the Oregon Health Experiment

In this section I illustrate the test in the context of the Oregon Health Experiment (OHE). An empirical setting in which IV methods within the framework of this paper have been used by Finkelstein et al. (2012) and Rose and Shem-Tov (2024) to study the effect of insurance on health and hospital utilization. Applying the test from this paper, I reject the extensive margin compliers only (EMCO) assumption invoked by Rose and Shem-Tov (2024). I find, however, that slight modifications of this assumption to allow for moderate intensive margin effects, are not rejected by the data. Leveraging these assumptions, I re-estimate the effect of insurance and obtain similar results to those of Finkelstein et al. (2012) and Rose and Shem-Tov (2024).

The OHE and plausible shape restrictions: Medicaid is a government program in the United States that provides health insurance for adults and children with limited income and resources, in the state of Oregon, in the united state, access to Medicaid is provided by the State government through the *Oregon Health Plan*. In January 2008, the government of the state of Oregon, in the United States, announced that it would extend the Oregon Health Plan, its goal was to enroll 10,000 new individuals into Medicaid. However, the state anticipated that demand would exceed this number, and thus, admittance into the program was randomized through a public lottery. Randomization, makes this empirical setting ideal to study the effects of health insurance among low income individuals.

Individuals selected by the lottery had the option to enroll into medicaid, but were not obliged to do so. Thus, the lottery is exogenous, but insurance is not. The endogeneity of insurance can be addressed using instrumental variables, but for this, independence is not sufficient. It must jointly hold along with exclussion and the shape restrictions that researchers might want to invoke.

By design, the lottery is binary. Furthermore, in the United States, the minimum frequency at which insurance is acquired and payed is monthly. Thus, the duration of insurance is, by definition, discrete. That is, the setting and variables in the OHE fall directly into the framework of this paper.

The first paper to study the OHE using IV methods was Finkelstein et al. (2012). In their analysis, they used the binary lottery results as an instrument for a binary treatment variable defined as an indicator that an individual had insurance for at least

one month between the randomization and the end of the study. Two of the outcomes they considered were indices that measured health and hospital utilization. Having a binary instrument and a binary treatment, they argued that, under an exclusion restriction plus a no-defiers/monotonicity assumption, the 2SLS estimand could be interpreted as LATE.

In a posterior paper, Rose and Shem-Tov (2024) argued that binarizing the treatment ignores changes in the duration of insurance. Absent the lottery, some households might have acquired insurance for a few months, but not for as many months as they would if given the option to enroll into Medicaid. Binarizing the treatment variable ignores these changes and its potential effects on health and hospital utilization. To recover a LATE interpretation for the 2SLS estimand, Rose and Shem-Tov (2024) proposed a different assumption: Extensive Margin Compliers Only (EMCO). This assumption imposes that winning the lottery can only affect insurance decisions on the extensive margin. That is, absent the experiment, inviduals would either acquire no insurance, or acquire it for the exact same duration for which they had Medicaid.

Rose and Shem-Tov (2024) argue that the EMCO assumption is plausible in the OHE. Their main argument is that, to be eligible for Medicaid, individuals are required to have been uninsured during at least 6 uninterrupted months. Thus, a person selected by the lottery who already had insurance cannot take advantage of Medicaid in order to retain insurance for a longer time. Nonetheless, shape restrictions in general, and the EMCO assumption in particular, do not only constraint realized treatments, but also potential treatments. For example, in a hypothetical scenario with no lottery, an individual might have acquired private insurance only for 3 months. If that same individual was selected by the lottery, thanks to Medicaid being affordable, that same individual might now be able to be insured for an entire year. The Medicaid requirements do not restrict hypothetical behaviors and thus, do not rule out this possibility[18].

An alternative, less restrictive assumption that might hold in the OHE, is ordered monotonicity. Angrist and Imbens (1995) show that, when treatments are ordered, under the assumption that the instrument can only induce units to adopt weakly

[18]Furthermore, prior to the lottery, the government of Oregon advertised that the Oregon Health Standard program was to be expanded so that all interested individuals could participate from the lottery. It is plausible that this advertising campaign increased awareness about insurance and its benefits which in turn might lead some individuals who where eligible for Medicaid, to acquire private insurance.

higher treatments, the 2SLS estimand can be interpreted as a weighted average of the individual treatment effects among the units whose treatment is changed by the instrument (i.e. among compilers). In the OHE, this is a plaussible assumption. Ordered monotonicity does not rule out the possibility that, absent the lottery, individuals who enroll into Medicaid might have still acquired insurance for at least some months. It simply requires that the assumption that they remain enrolled into Medicaid for at least as long as they would have retained private insurance. This is plaussible since Medicaid is just an additional, affordable, available alternative for lottery winners.

Figures 6 (a) and (b) show the histogram of the duration of insurance conditional on being selected ($Z = 1$), and not selected ($Z = 0$) by the lottery. Subfigure (a) shows all durations observed in the sample (from 0 to 19 months) and reveals two things, first, a large share of the population was not insured, even when they had the option to enroll into Medicaid, and second, it is clear that the share of individuals who never had insurance is much lower among lottery winners. This second finding confirms that a large portion of the lottery winners went from having no insurance, to having insurance for some positive number of months.

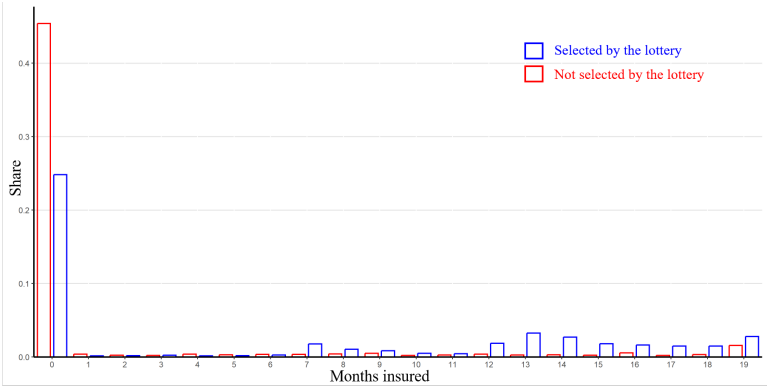
But, for EMCO to hold, it is not sufficient that the lottery had a large effect on the extensive margin. It also requires that there are no intensive margin effects. Subfigure 6 (b) shows the same histogram but now restricted to strictly positive durations of insurance. From this figure, it is clear that any durations between 1 and 7 months was slightly less prevalent among lottery winners. This is inconsistent with EMCO, if the intervention only affected the intensive margin, all positive insurance durations should be weakly more prevalent.

Note that while subfigure 6 (b) suggests that EMCO is not consistent with the data, it is unclear if these differences are statistically meaningful, or attributable to sampling noise. The need to distinguish these two possibilities further motivates testing the validity of shape restrictions on their own, and jointly with exclusion and IV independence.

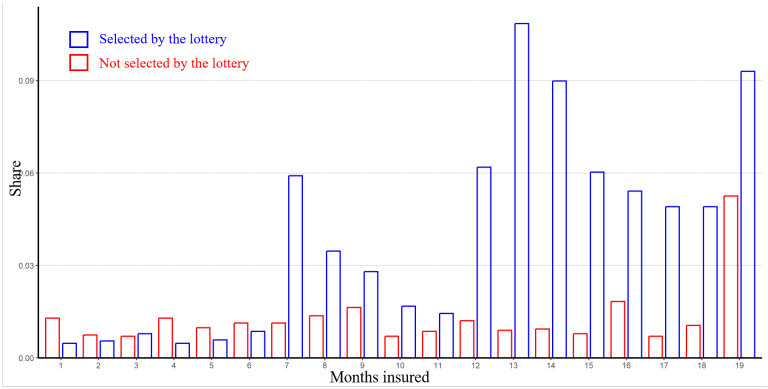
Figures 6 (a) and (b) suggest one additional, plausible restriction. While most of the effect of the lottery was on the extensive margin, there were also some individuals that, absent the experiment, would have acquired insurance but failed to retain it for more than $1 \leq t \leq 7$ months. On the other hand, it could be that individuals who had private insurance for more than t months are less financially constrained, and thus,

as EMCO imposes, would have held Medicaid (if available to them), for the exact same duration that they held private insurance. One advantage of this restriction is that, the valid binarization conditions from Rose and Shem-Tov (2024) hold, and thus, it restores a LATE interpretation for the the 2SLS estimand, but now as the effect among individuals who where affected on the extensive margin, or induced to retain insurance for more than t months.

Figure 6: Histograms of insurance duration in the OHE



(a) All durations.



(b) Strictly positive durations.

Notes: *These two figures show the histrograms of duration of insurance across the OHE sample. Duration of insurance is measured in months, and ranges from 0 to 19. The red bars represent the histogram for individuals who where not selected by the lotery, the blue bars represent the histrogram for individuals who where selected by the lottery. Sibfigure (a) shows the histogram including all observed durations. Subfigure (b) restricts the histrogams to strictly positive durations.*

Thus, I test multiple potentially valid shape restrictions: EMCO, Ordered Monotonicity, and the assumption described in the previous paragraph for multiple values

of t. The role of these assumptions is to facilitate identification of the LATE, to this end, the must hold jointly along with independence and the exclusion restriction. Hence, I do not only test the shape restrictions, but their joint validity along with independence and the exclusion.

By design, the lottery is randomized and thus, independent from potential treatments and outcomes. On the other hand, there are multiple reasons why exogeneity might not hold in the OHE and thus it might be tested. The firs reason is income effects. Since Medicaid is more affordable, enrolled individuals might have additional savings which in turn might have a positive income on their health if, for example, the additional resources are allow individuals to improve their diet. Furthermore, both Finkelstein et al. (2012) and Rose and Shem-Tov (2024) define the treatment variable as months of insurance. This definition underpins one assumption: that N months of private insurance have the same effect as N months of Medicaid. While plaussible, it is not obvious that this assumption hold. If it doesn't, then, the same unit might have different outcomes when $X = N$ and $Z = 0$ than when $X = N$ and $Z = 0$. This would be a violation of the exclusion restriction.

Test implementation: The outcomes of interest in Finkelstein et al. (2012) are multiple outcomes associated with health and hospital utilization. In the main analysis of Rose and Shem-Tov (2024), these outcomes are aggregated into two Z-score variables, one for health, and one for hospital utilization. In my implementations of the test, I consider these two variables as the outcome Y. I first apply the test to each of the two scores separately, and then, taking advantage of the fact that both the identification and inference results of this paper apply to multi-dimensional outcomes, I jointly apply the test to both indices.

I report results of the test for gaussian kernels with a bandwidth of $h_n \propto n^{-1/5}$, using half the sample to estiamte the classifier, and the other half to evaluate its loss and estimate conditional treatment probabilities. The reason I report the results for these values of the tuning parameters is that they achieved the best performance in the simulations from section 6. For alternative choices of the tuning parameters, I obtained qualitatively similar results which I omit in the interest of brevity.

Two features of the OHE make it different from the simulations from section 6. The fact that the outcome is two-dimensional, and the fact that the treatment takes 20 different values, and some of them occur with very low frequency in the sample. In appendix I asses the robustness of my method to data generating processes with these

features. I find that the test has adequate size control, and power against alternatives under which IV validity is falsified.

The expansion of the Oregon Health Standard occurred in 7 different waves. Across waves, the probability of wining the lottery was different. Hence, I apply the test separately to each wave. Additionally, I pool together waves 1, 2 and 3, as well as 4, 5 and 6. I choose this particular pooling because the hypothesis that the probability of wining the lottery is identical across all waves within each of the two the pooled waves is not rejected. Pooling allows to implement the test on a larger sample. The results of the test are consistent across waves and pooled waves. In other words, my findings are consistent across all waves of the OHE.

While the lottery was at the individual level, the option to enroll into Medicaid was given at the household level, thus, individuals in larger households have a higher probability of effectively wining the lottery. Moreover, some observations were assigned to an intensive-follow up survey program and thus, their outcomes are more like to be observed in the final sample. I restrict the testing sample to one person households who were not part of the intensive follow up survey. The resulting testing sample consist of 8,625 different individuals, and represents 58% of the entire sample. Since, within this subsample the lottery remains independent from potential treatments and outcomes, if the IV validity assumptions hold for the entire population, they must also hold for this subsample. Restricting to households who were not part of the intensive follow up survey simplifies implementation, but is not necessary to apply the test. Both the kernel density estimators used in the classifier, and the second sample empirical loss could be estimated using weights that reflect the different sample probabilities.

Results: I present the results of the test in table 3. I do not jointly reject ordered monotonicity along with the exclusion restriction and instrument independence. I reject however the EMCO assumptions, that is, I find that the lottery had an effect on the intensive margin too. Furthermore, I also fail to reject the hypotheses that, for any t between 2 and 7, the lottery allowed individuals that would have had insurance for less that t months absent the Medicaid insurance, to retain insurance for more than t months, but that the duration of insurance among individuals that would have been able to afford more than t months of private insurance was not affected.

The results also, how accounting for the fact that the outcome is multivalued, shrinks the set of admissible distributions of the data which in turn increases the

power of the test. This is clear in the case of the monotonicity assumption, where, separately testing IV validity for one outcome yields p-values as high as 1. While, jointly testing validity for the two outcomes substantially reduces the obtained p-values.

Table 3: Joint tests of IV validity.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Shapre R.	Mntn.	Ext.	$t=2$	$t=3$	$t=4$	$t=5$	$t=6$	$t=7$	$t=8$	$t=9$	$t=10$	$t=11$	$t=12$
Waves 1,2 and 3 (n=2615)													
Health	1	0.64	0.658	0.658	0.658	0.547	0.547	0.739	0.578	0.164	0	0	0
Hospital	1	0.734	0.851	0.898	0.849	0.74	0.377	0.64	0.012	0.002	0	0	0
Joint	0.188	0.563	0.455	0.455	0.455	0.437	0.698	0.671	0.041	0.006	0	0	0
Waves 4,5 and 6 (n=3920)													
Health	1	0.049	0.254	0.254	0.174	0.24	0.14	0.367	0.041	0.001	0	0	0
Hospital	1	0.044	0.235	0.172	0.11	0.147	0.081	0.078	0	0	0	0	0
Joint	0.729	0.007	0.012	0.023	0.041	0.18	0.15	0.237	0	0	0	0	0
Wave 1 (n=890)													
Health	1	0.252	0.252	0.217	0.316	0.979	1	1	0.048	0.004	0	0	0
Hospital	0.155	0.278	0.156	0.142	0.187	0.104	0.048	0.152	0.001	0	0	0	0
Joint	0.245	0.302	0.302	0.283	0.629	0.42	0.692	0.698	0.076	0.008	0	0	0
Wave 2 (n=876)													
Health	0.158	0.182	0.25	0.387	0.384	0.216	0.216	0.382	0.061	0.051	0	0	0
Hospital	0.158	0.205	0.282	0.159	0.266	0.328	0.421	0.185	0.01	0.014	0	0	0
Joint	0.082	0.054	0.067	0.017	0.049	0.138	0.168	0.123	0.009	0.018	0	0	0
Wave 3 (n=849)													
Health	0.711	0.683	0.683	0.583	0.622	0.529	0.433	0.377	0.12	0.017	0	0	0
Hospital	0.871	0.368	0.368	0.381	0.448	0.359	0.273	0.461	0.18	0.032	0	0	0
Joint	0.301	0.347	0.347	0.278	0.298	0.237	0.184	0.134	0.033	0.004	0	0	0
Wave 4 (n=1105)													
Health	0.212	0.207	0.277	0.277	0.254	0.362	0.903	0.903	0.91	0.833	0	0	0
Hospital	0.212	0.128	0.168	0.172	0.154	0.203	0.279	0.357	0.051	0.039	0	0	0
Joint	0.584	0.127	0.166	0.166	0.132	0.156	1	1	0.658	0.816	0	0	0
Wave 5 (n=1149)													
Health	1	0.133	0.209	0.285	0.389	0.317	0.465	0.548	0.017	0.003	0	0	0
Hospital	1	0.179	0.294	0.386	0.357	0.308	0.412	0.406	0.001	0	0	0	0
Joint	0.549	0.392	0.495	0.63	0.596	0.552	0.712	0.66	0.028	0.005	0	0	0
Wave 6 (n=1666)													
Health	0.139	0.108	0.07	0.054	0.03	0.026	0.026	0.026	0.035	0.034	0	0	0
Hospital	0.648	0.251	0.204	0.343	0.459	0.485	0.328	0.428	0.098	0.121	0	0	0
Joint	0.377	0.261	0.196	0.452	0.35	0.365	0.365	0.365	0.356	0.349	0	0	0

Notes: *This table shows the p-values for the test of joint validity of instrument independence, the exclusion restriction, and a different shape restrictions. Each column corresponds to a different shape restriction. The firs column corresponds to ordered monotonicity, the second to the EMCO assumption, columns 3 to 13 represent different versions of the assumption that winning the lottery promoted insurance duration strictly above t, for t ranging from 2 to 12. The table reports the reslts of applying the test wave by wave, as well as pooling waves 1,2 and 3 toghether, as well as waves 4,5 and 6. Within each wave or group of waves I report three versions of the test: first, applied just to the health index as the outcome, second, applied to just the hospital utilization index, and third, applied jointly to both the health and hospital utilization indices, treated as a two-dimensional outcome. Each row corresponds to a different wave-outcome(s) combination. The test was implemented using Gaussian kernels, a bandwidth $h_n \propto n^{-1/5}$ and Silverman's rule, and using half the data to estimate kernel densities and the non-parametric classifier.*

Finally, I report weighted LATE estimates under all the assumptions that were not rejected in at least one of the tests. For the EMCO assumption, and each assumption associated with a different threshold t, the estimands where obtained by recoding

the original, multi-valued treatment variable according to Rose and Shem-Tov (2024) and then running 2SLS on a constructed, binarized, treatment variable. For ordered monotonicity, the estimator was obtained using 2SLS on the original treatment variable. Note that in this monotone case (as Angrist and Imbens (1995) show), the 2SLS is a weighted average of local treatment effects among compliers, but, units whose potential treatment is increased the most (in terms of duration), receive larger weight in the final estimand. These results are shown in table 4 for both the full sample, and the sub-sample in which I implemented the test.

In all cases, I find that increasing insurance (either on the intensive margin, extensive margin, or both) has a positive and significant effect on health and hospital utilization. However, under ordered monotonicity the magnitude of the effect is about 10 times smaller than under the rest of constraints.

Finally, I find that for $t = 1, ..., 7$, the LATE estimnates are similar in magnitude to the estimate under EMCO, which corresponds to the original numbers reported both by Finkelstein et al. (2012) and Rose and Shem-Tov (2024). The reason is that, while the lottery had an effect on the intensiver margin, the share of intensive-margin compliers is small relative to the share of extensive-margin compliers. Consequently, the LATE estimator does not change substantially after accounting for the intensive-margin compliers.

If any of the restrictions associated with $t = 1, ..., 7$ hold, then, I find that the the effect of increasing insurance among all units that were either switched on the extensive margin, or allowed to retain insurance from less than t to at least $t + 1$ months was between 0.15 and 0.160 standard deviations for the health index, and between 0.24 and 0.26 standard deviations for hospital utilization.

Table 4: Weighted local average treatment effect estimates.

Panel A: Full sample										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Shape R	Mntn.	Ext.	$t=2$	$t=3$	$t=4$	$t=5$	$t=6$	$t=7$	$t=8$	$t=9$
Health										
WLATE	0.011***	0.150***	0.147***	0.147***	0.146***	0.145***	0.145***	0.144***	0.160***	0.170***
s.e.	(0.004)	(0.049)	(0.049)	(0.048)	(0.048)	(0.048)	(0.048)	(0.048)	(0.053)	(0.056)
Compliers share		0.292	0.296	0.297	0.298	0.300	0.301	0.301	0.273	0.257
Wave-Hh size FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hospital Utilization										
WLATE	0.007**	0.099**	0.098**	0.097**	0.097**	0.096**	0.096**	0.096**	0.106**	0.113**
s.e.	(0.003)	(0.045)	(0.045)	(0.045)	(0.044)	(0.044)	(0.044)	(0.044)	(0.049)	(0.052)
Compliers share		0.028	0.027	0.027	0.026	0.026	0.025	0.025	0.026	0.028
Observations	14,782	14,782	14,782	14,782	14,782	14,782	14,782	14,782	14,782	14,782
Wave-Hh size FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Panel B: One person households, standard follow up										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Shape R	Mntn.	Ext.	$t=2$	$t=3$	$t=4$	$t=5$	$t=6$	$t=7$	$t=8$	$t=9$
Health										
WLATE	0.013***	0.182***	0.180***	0.179***	0.180***	0.177***	0.176***	0.176***	0.192***	0.201***
s.e.	(0.004)	(0.050)	(0.049)	(0.049)	(0.049)	(0.049)	(0.048)	(0.048)	(0.053)	(0.055)
Compliers share		0.338	0.342	0.343	0.342	0.346	0.348	0.349	0.318	0.304
Wave-Hh size FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hospital Utilization										
WLATE	0.009***	0.122***	0.120***	0.120***	0.120***	0.119***	0.118***	0.117***	0.128***	0.134***
s.e.	(0.003)	(0.042)	(0.041)	(0.041)	(0.041)	(0.041)	(0.041)	(0.040)	(0.044)	(0.046)
Compliers share		0.027	0.026	0.025	0.025	0.024	0.023	0.022	0.024	0.025
Observations	8,625	8,625	8,625	8,625	8,625	8,625	8,625	8,625	8,625	8,625
Wave-Hh size FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: *This table shows LATE type estimates. That is, estimates of weighted averages of individual treatment effects among the complier groups that different shape restrictions allow. Column one is estimated by 2SLS of the outcomes of interest on duration of insurance, using the lottery as an instrument. Under ordered monotonicity, this can be interpreted as an estimate for a weighted average of the treatment effect among individuals who increased their duration of insurance as a result of the lottery (see Angrist and Imbens (1995)). Columns (2) to (10) report estimates for average treatment effect under the assumption that the lottery incentivized individuals to swich from an insurance level below a threshold t to an insurance strictly higher than t, but induced no switches within durations below t, nor within durations above t. The first column corresponds to $t = 1$, that is, the EMCO assumption. The estimates are obtained using 2SlS on a binarized constructed treatment variable as described in Rose and Shem-Tov (2024).*

8 Conclusion

This paper combines two novel results to obtain a sharp, general test of instrument validity. The first result is a novel characterization of the sharp observable implication of the IV model. This characterization reduces the set of infinite restrictions that the exclusion restriction combined with the distribution of the outcome impose on the model, to an equivalent, finite set of linear constrains on the selection model. That is, a set of additional inequalities on the simple, finite, discrete model that describes how the instrument affects treatment adoption.

The second novel result is a valid, joint inference procedure for the quantities involved in the finite characterization of the observable implications of the model. Some of these quantities are subdensity overlaps, that is, integrals of the pointwise minimum of conditional densities over some strict subset of their support. Naive plug in approaches to subdensity overlap estimation suffer two problems, one, whenever the two densities are equal, applying the minimum operator bias the estimators. Second, analyzing the behavior of the in-sample minimizers is technically challenging. I avoid this problems by noting that the target parameters can be recast as the expected loss of non-parametric classifiers. This insight allows me to obtain a valid, root-n consistent, asymptotically normal, asymptotically unbiased estimator.

These two novel results, combined with the inference method for linear systems proposed by Cox and Shi (2023) allows me to derive a sharp test of instrument validity that jointly accommodates continuous, potentially multi-valued outcomes, along with a general class of restrictions on how the instrument affects treatment adoption.

In an application to the OHE, I illustrate the test, and how it can be used to interpret and formulate plausible restrictions on how a lottery that randomized access to Medicaid affected insurance on the extensive and intensive margin. Leveraging these restrictions, I obtain estimates for the effect of increasing insurance duration on health and hospital utilization that are similar to those found by Finkelstein et al. (2012) and Rose and Shem-Tov (2024). I emphasize, however, that part of the effect came from changes in the duration of insurance on the intensive margin.

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A Proofs for section 3

Proof of proposition **??**.

First observe that for every measurable $B_Y \subset \mathcal{Y}$, every non-empty $\tilde{Z} \subset \mathcal{Z}$ and $x_l \in \mathcal{X}$, if $z_k \in \tilde{Z}$, then:

$$\mathbb{P}[Y \in B_Y, X = x_l | Z = z_k] \geq \mathbb{P}[Y(x_l) \in B_Y, X(\tilde{z}) = x_l \ \forall \tilde{z} \in \tilde{Z} | Z = z_k] \tag{46}$$

$$\implies \mathbb{P}[Y \in B_Y, X = x_l | Z = z_k] \geq \mathbb{P}[Y(x_l) \in B_Y, X(\tilde{z}) = x_l \ \forall \tilde{z} \in \tilde{Z}] \tag{47}$$

$$\implies \min_{z_k \in \tilde{Z}} \big\{ \mathbb{P}[Y \in B_Y, X = x_l | Z = z_k] \big\} \geq \mathbb{P}[Y(x_l) \in B_Y, X(\tilde{z}) = x_l \ \forall \tilde{z} \in \tilde{Z}] \tag{48}$$

$$\implies \min_{z_k \in \tilde{Z}} \Big\{ \int_{B_Y} f_{Y,X|Z=z_k}(y, x_l) d\mu_y \Big\} \geq \mathbb{P}[Y(x_l) \in B_Y, X(0) = x, X(\tilde{z}) = x_l \ \forall \tilde{z} \in \tilde{Z}] \tag{49}$$

The initial inequality is true because outcomes in B_Y might be realized by response types other than the sufficient-takers associated to $\tilde{Z}$. The first implication follows from exogeneity of potential outcomes, the second is true because the inequality must hold for every individual $z_k \in \tilde{Z}$. The last simply applies the definition of $f_{Y,X|Z=z_k}(y, x_l)$.

Note that equation 49 must hold for every measurable set $B_Y \in \mathcal{B}_Y$. The next step is to transform this collection of equations into a single restriction.

To this end, consider the set of all orderings over $\tilde{Z}$ denoted $\mathcal{O}(\tilde{Z}) = \big\{\kappa : \{0, 1, ..., |\tilde{Z}| - 1\} \longrightarrow \tilde{Z} : \exists \ \kappa^{-1}\big\}$. For compactness, for every $\pi \in \mathcal{O}(\tilde{Z})$ and every $k \in \{0, 1, ..., |\tilde{Z}| - 1\}$ I denote the $k+1$-th element of $\tilde{Z}$ according to κ as κ_k (instead of $\kappa(k)$).

Then, for every x_l and $\kappa \in \mathcal{O}(\tilde{Z})$, define the set $Y^{\pi}_{x_l}$ as follows:

$$Y^{\kappa}_{x_l} = \Big\{ y \in \mathcal{Y} : f_{Y,X|Z=\kappa_0}(y, x_l) \leq \min_{k=0,2,\ldots,|\tilde{Z}|-1-1} \{ f_{Y,X|Z=\kappa_k}(y, x_l) \} \Big\} \tag{50}$$

$$\bigcap \Big\{ y \in \mathcal{Y} : f_{Y,X|Z=\kappa_1}(y, x_l) \leq \min_{k=1,\ldots,|\tilde{Z}|-1-1} \{ f_{Y,X|Z=\kappa_k}(y, x_l) \} \Big\} \tag{51}$$

$$\ldots \bigcap \Big\{ y \in \mathcal{Y} : f_{Y,X|Z=\kappa_m}(y, x_l) \leq \min_{k=m,\ldots,|\tilde{Z}|-1} \{ f_{Y,X|Z=\kappa_k}(y, x_l) \} \Big\} \tag{52}$$

$$\ldots \bigcap \Big\{ y \in \mathcal{Y} : f_{Y,X|Z=\kappa_{K-2}}(y, x_l) \leq \min_{k=|\tilde{Z}|-1-2,|\tilde{Z}|-1-1} \{ f_{Y,X|Z=\kappa_k}(y, x_l) \} \Big\} \tag{53}$$

This is, the set in which the values that the functions $f_{Y,X|Z=z_k}(y, x_l)$ take are sorted by magnitude in exactly in the same order that κ sorts the instrument values. Note that since the functions $f_{Y,X|Z=z_k}(y, x_l)$ are point identified, the sets $Y^{\kappa}_{x_l}$ are identified too. Moreover, the sets $Y^{\kappa}_{x_l}$ are the intersection of sets defined by inequalities that depend on measurable functions, thus, they are measurable themselves.

Then:

$$\mathbb{P}[Y \in B_Y, X(\tilde{z}) = x_l \ \ \forall \tilde{z} \in \tilde{Z} | Z = z] = \sum_{\kappa \in \mathcal{O}(\tilde{Z})} \mathbb{P}[Y \in Y^{\kappa}_{x_l} \cap B_Y, X(\tilde{z}) = x_l \ \ \forall \tilde{z} \in \tilde{Z} | Z = z] \tag{54}$$

$$\leq \sum_{\kappa \in \mathcal{O}(\tilde{Z})} \min_{z_k \in \tilde{Z}} \Big\{ \int_{Y^{\kappa}_{x_l} \cap B_Y} f_{Y,X|Z=z_k}(y, x_l) d\mu_Y \Big\} \tag{55}$$

$$= \sum_{\kappa \in \mathcal{O}(\tilde{Z})} \int_{Y^{\kappa}_{x_l} \cap B_Y} f_{Y,X|Z=\kappa_0}(y, x_l)(y) d\mu_Y \tag{56}$$

$$= \sum_{\kappa \in \mathcal{O}(\tilde{Z})} \int_{Y^{\kappa}_{x_l} \cap B_Y} \psi_{x_l, \tilde{Z}}(y) d\mu_Y \tag{57}$$

The first line follows from the fact that the sets $Y^{\kappa}_{x_l}$ comprise a partition of $\mathcal{Y}$. The inequality in the second line follows from equation 49. The equality in the third line is crucial, it is true because the sets $Y^{\kappa}_{x_l}$ are constructed so that $f_{Y,X|Z=\kappa_0}(y, x_l)(y) \leq f_{Y,X|Z=z_k}(y, x_l)$ for every $y \in Y^{\kappa}_{x_l}$ and every $z_k \in \tilde{Z}$. Moreover, in $Y^{\kappa}_{x_l}$ it holds that $\psi_{x_l, \tilde{Z}}(y) f_{Y,X|Z=\kappa_0}(y, x_l)$ which proves the third line. Using again the fact that the sets

$Y^{\kappa}_{x_l}$ partition $\mathcal{Y}$ yields the last equality. □

Proof of theorem 3.1.

1⟹2:

The proof follows immediately from proposition **??**.

2⟹1:

Let $\boldsymbol{p}$ be a distribution over instrument response types such that:

$$A_Z \boldsymbol{p} \leq \boldsymbol{r}_Z \tag{58}$$

$$A_X \boldsymbol{p} = \boldsymbol{r}_X \tag{59}$$

$$A_Y \boldsymbol{p} \leq \boldsymbol{\Psi} \tag{60}$$

The goal is to define a distribution over response types that satisfies equation 6 and proposition 2.1. To this end, it is necessary to specify potential treatments and outcomes for every possible response types, including potential outcomes for types that never realize certain treatments and thus never realize the potential outcome associated to such treatment.

To simplify exposition, let $\tilde{p}_{x_l}$ denote the quantity $p\big(AT_{x_l}\big)$.

Let $t \in \Delta(\mathcal{T})$ be a response type. Define the set of treatments that response type $t = (t^Z, t^X)$ realizes as $R(t) = \{x_l \in \mathcal{X} : (t^Z)_{k'} = x_l \text{ for some } k' = 1, 2\}$. Note that by construction $|R(t)| \leq 2$

Let λ be an arbitrary fixed distribution over $\mathcal{Y}$. Then, define $g' : \Delta(\mathcal{T}) \longrightarrow \mathbb{R}$ as follows:

$$g'(t^Z, t^X) = \begin{cases} \Big[\frac{1}{\Psi_{x_l,\mathcal{Z}}}\psi_{x_l,\mathcal{Z}}\big((t^X)_l\big)\Big]\Big[\prod_{l':x_{l'}\notin R(t)}\lambda\big((t^X)_{l'}\big)\Big], & \text{if } R(t) = \{x_l\} \text{ and } \Psi_{x_l} > 0 \\ \Big[\prod_{l,k:x_l=(t^Z)_k}\frac{1}{P_{x_l|z_k}-\tilde{p}_{x_l}}\xi_{x_l}\big((t^X)_l\big)\Big]\Big[\prod_{l':x_{l'}\notin R(t)}\lambda\big((t^X)_{l'}\big)\Big], & \text{if } |R(t)| = 2 \text{ and } \Psi_{x_l} - \tilde{p}_{x_l} > 0 \\ 0, & \text{otherwise} \end{cases} \tag{61}$$

Where $\xi_{x_l}\big((t^X)_l\big) = f_{Y,X|Z=z_k}((t^X)_l, x_l) - \frac{\tilde{p}_{x_l}}{\Psi_{x_l,\mathcal{Z}}}\psi_{x_l,\mathcal{Z}}\big((t^X)_l\big)$ and $P_{x_l|z_k} = \mathbb{P}[X = x_l|Z = z_k]$.

Integrating the function g' allows to construct the desired distribution over response types. To this end, recall the following result from measure theory:

Lemma A.1. *Any measurable set of response types $B_{\mathcal{T}} \in \mathcal{B}_{\mathcal{T}}$ can be partitioned into slices as follows:*

$$B_{\mathcal{T}} = \bigcup_{t^Z \in \mathcal{T}_Z} \{t^Z\} \times B_{T_X}^{t^Z} \tag{62}$$

Where $B_{T_X}^{t^Z} = B_{Y_0}^{t^Z} \times B_{Y_1}^{t^Z} \times ... \times B_{Y_{L-1}}^{t^Z} \in \mathcal{Y}^L$ is potentially empty and measurable with respect to the product measure over the product of the L spaces that correspond to the potential outcomes at each different treatment value $x_l \in \mathcal{X}$.

Moreover, each individual set $= B_{Y_l}^{t^Z}$ for $l = 0, 1..., L-1$ is measurable with respect to μ_Y.

Then, using the lemma, define:

$$g\Big(B_{\mathcal{T}}\Big) = \sum_{t^Z \in \mathcal{T}_Z} \Big[p(t^Z) \int_{B_{T_X}^{t^Z}} g'(t^Z, t^X) d\mu_{T^X} \Big] \tag{63}$$

Where μ_{T^X} denotes the product measure $\mu_{T^X} = \mu_Y \otimes \mu_Y \otimes ... \otimes \mu_Y$ over the space of potential treatment-responses.

$$\int_{B_{T_X}^{t^Z}} g'(t^Z, t^X) d\mu_{T^X} = \int_{B_{Y_0}^{t^Z}} \int_{B_{Y_1}^{t^Z}} ... \int_{B_{Y_{K-1}}^{t^Z}} g'(t^Z, t^X) d\mu_Y \mu_Y ... \mu_Y \tag{64}$$

It remains to prove that $g\Big(B_{\mathcal{T}}\Big)$ is a valid distribution over reponse types, this is, it is non-negative, adds up to 1, satisfies countable additivity, and is consistent with the observed joint distribution of outcomes, treatments and instruments..

When $|R(t)| = 1$, the function $g'(t^Z, t^X)$ is the product of non-negative functions, therefore, it must be non-negative itself. When $|R(t)| = 2$ it is either the product of non-negative functions, or zero. Therefore, it must be non-negative. From here, it follows that $g\Big(B_{\mathcal{T}}\Big)$ is non-negative for every measurable $B_{\mathcal{T}} \in \mathcal{B}_{\mathcal{T}}$.

Moreover, note that:

$$\int_{\mathcal{Y}} \frac{p(t^Z)}{\Psi_{x_l,\mathcal{Z}}} \psi_{x_l,\mathcal{Z}}\big(y\big) d\mu_Y = p(t^Z) \tag{65}$$

$$\int_{\mathcal{Y}} \Big[\frac{p(t^Z)}{P_{x_l|z_k} - \tilde{p}_{x_l}} \big(f_{Y,X|Z=z_k}(y, x_l) - \frac{\tilde{p}_{x_l}}{\Psi_{x_l,\mathcal{Z}}} \psi_{x_l,\mathcal{Z}}\big((y)\big) \Big] = p(t^Z) \tag{66}$$

From here, which follows that:

$$g(\mathcal{T}) = \sum_{t^Z \in \mathcal{T}_Z} \Big[p(t^Z) \int_{\mathcal{T}_X} g'(t^Z, t^X) d\mu_{T^X} \Big] \tag{67}$$

$$= \sum_{t^Z \in \mathcal{T}_Z} \Big[p(t^Z) \Big] \tag{68}$$

$$= 1 \tag{69}$$

Countable additivity follows mechanically. Let $\{B^m_{\mathcal{T}}\}_{m \in M}$ be a countable collection of disjoint subsets of $\mathcal{T}$ indexed by $m \in M$. Applying lemma A.1, each sets $B^m_{\mathcal{T}}$ can be sliced as follows:

$$B^m_{\mathcal{T}} = \bigcup_{t^Z \in \mathcal{T}_Z} \{t^Z\} \times B^{t^Z,m}_{T_X} \tag{70}$$

Since $B^m \cap B^{m'} = \emptyset$ for all $m, m' \in M$ such that $m \neq m'$ then its must be that $B^{t^Z,m}_{T_X} \cap B^{t^Z,m'}_{T_X} = \emptyset$.

Then:

$$g\Big(\bigcup_{m\in M} B^m_{\mathcal{T}}\Big) = g\Big(\bigcup_{m\in M}\Big[\bigcup_{t^Z\in\mathcal{T}_Z}\Big\{\{t^Z\}\times B^{t^Z,m}_{T_X}\Big\}\Big]\Big) \tag{71}$$

$$= g\Big(\bigcup_{t^Z\in\mathcal{T}_Z}\Big[\bigcup_{m\in M}\Big\{\{t^Z\}\times B^{t^Z,m}_{T_X}\Big\}\Big]\Big) \tag{72}$$

$$= g\Big(\bigcup_{t^Z\in\mathcal{T}_Z}\Big[\{t^Z\}\times\Big\{\bigcup_{m\in M} B^{t^Z,m}_{T_X}\Big\}\Big]\Big) \tag{73}$$

$$= \sum_{t^Z\in\mathcal{T}_Z} p(t^Z) \int\limits_{\bigcup_{m\in M} B^{t^Z,m}_{T_X}} g'(t^Z,t^X)d\mu_{T_X} \tag{74}$$

$$= \sum_{t^Z\in\mathcal{T}_Z}\sum_{m\in M} p(t^Z) \int\limits_{B^{t^Z,m}_{T_X}} g'(t^Z,t^X)d\mu_{T_X} \tag{75}$$

$$= \sum_{m\in M}\sum_{t^Z\in\mathcal{T}_Z} p(t^Z) \int\limits_{B^{t^Z,m}_{T_X}} g'(t^Z,t^X)d\mu_{T_X} \tag{76}$$

$$= \sum_{m\in M} g(B^m_{\mathcal{T}}) \tag{77}$$

The first line is comes from application of lemma A.1 to every set $B^m_{\mathcal{T}}$, the second simply inverts the order of the unions, the third is a property of cartesian products, the fourth simply applies the definition of g and takes advantage of the fact that the set is already sliced, the fifth line uses countable additivity of the integral, the sixth changes the order of the sums, and the last one uses the definition of g but now applied to each set B^m_Y separately. This proves countable additivity.

It only remains to verify consistency with observed probabilities, let $B^{t^Z}_{T_X} = \mathcal{Y}\times ....\times B_Y\times ...\times\mathcal{Y}$ be the set obtained by taking the product of the outcome's support $L-1$ times and letting the $l'-th$ element of the product coincide with B_Y instead of $\mathcal{Y}$. Then:

$$g\Big(\big\{(t^Z,t^X)\in\mathcal{T}:(t^Z)_k=x_{l'},(t^X)_l\in B_Y\big\}\Big) \tag{78}$$

$$= \sum_{t^Z:(t^Z)_k=x_l}\Big[p(t^Z)\int\limits_{B^{t^Z}_{T_X}} g'(t^Z,t^X)d\mu_{T^X}\Big]$$

$$= \frac{\tilde{p}_{x_l}}{\Psi_{x_l,\mathcal{Z}}}\int\limits_{B_Y}\psi_{x_l,\mathcal{Z}}\big(y\big)d\mu_Y \tag{79}$$

$$+\frac{P_{x_l|z_k}-\tilde{p}_{x_l}}{P_{x_l|z_k}-\tilde{p}_{x_l}}\int\limits_{B_Y}\Big(f_{Y,X|Z=z_k}(y,x_l)-\frac{\tilde{p}_{x_l}}{\Psi_{x_l,\mathcal{Z}}}\psi_{x_l,\mathcal{Z}}(y)\Big)d\mu_Y$$

$$= \int\limits_{B_Y} f_{Y,X|Z=z_k}(y,x_l)d\mu_Y \tag{80}$$

The first equality is just the definition of g, the second uses the fact that at instrument value z_k there are only two instrument-response types that realize x_l (always-x_l takers, and $\{z_k\}$-IFF-x_l-takers) combined with the definition of g'. The last line simply re-arranges terms.

This completes the proof.

B Proofs for section 4

B.1 Proof for subsection 4.1

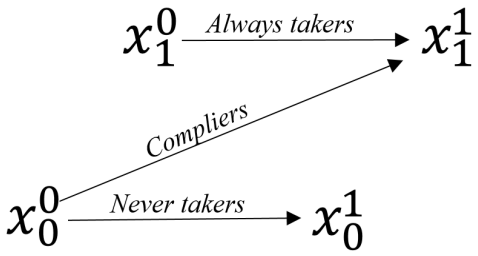
Before providing the proofs I formally define the transportation problem as well as the maximum flow and minimum cut problems.

First define the sets $\mathcal{X}_k$ for $k = 0, 1$ as copies of $\mathcal{X}$ but with elements $x_l^k \in \mathcal{X}_k$ labeled with the superscript k to make explicit that x_l^0 refers to treatment $x_l \in \mathcal{X}$ realized when $Z = 0$ and x_l^1 to treatment $x_l \in \mathcal{X}$ realized when $Z = 1$.

Next, define the directed graph with vertex set $\mathcal{X}_0 \cup \mathcal{X}_0$ and let the edges be $\mathcal{E}_R \subset \mathcal{X}_1 \times \mathcal{X}_2$, where the ordered pair $(x_l^0, x_{l'}^1) \in \mathcal{X}_0 \times \mathcal{X}_1$ belongs to $\mathcal{E}_R$ if and only if the instrument-response type $t^Z = (x_l, x_{l'})$ does not belong to R. In order words, an edge $(x_l^0, x_{l'}^1)$ belongs to the graph if the response type that realizes x_l at $Z = 0$ and $x_{l'}$ at $Z = 1$ is not ruled out by assumption 4.1.

For instance, in the case of the no defiers assumption, the set R would consist of the singleton $\{(x_1, x_0)\}$. Therefore, the edges $(x_0^0, x_0^1), (x_0^0, x_1^1)$ and (x_1^0, x_1^1) –corresponding to never takers, compliers, and always takers– would belong to $\mathcal{E}_R$, in contrast, the edge (x_1^0, x_0^1) would not. This particular case is illustrated in figure 7.

Figure 7: Graph representation of the no-defiers instrument-response type restriction.



Notes: *This figure depicts the graph representation of the no-defiers/monotonicity assumption from Imbens and Angrist (1994). The edges with a super-index $k = 0, 1$ represent treatment values realized at instrument values $Z = z_k$. The edges with sub-index $l = 0, 1$ represent treatments x_0 and x_1 respectively. Since there is no edge from x_1^0 to x_0^1, defiers are ruled out from the model.*

The transportation problem is as follows: every node $x_l^0 \in \mathcal{X}_0$ starts with a mass $\mathbb{P}[X_i = x_l | Z_i = z_0]$ of units. Each unit can travel from its origin node x_l^0 through the admissible edges $(x_l^0, x_{l'}^1) \in \mathcal{E}_R$ in order to reach some destination node $x_{l'}^1 \in \mathcal{X}_1$ subject to the constraint that each destination node $x_{l'}$ cannot receive more than $\mathbb{P}[X_i = x_{l'} | Z_i = z_1]$ units. If it is possible to empty all origin nodes x_l^0 so that all

destination nodes end at full capacity, then, the transportation problem is solved.

In what follows, I prove that if the transportation problem is solved, a distribution over response types exists that is consistent with both the observed joint distribution of (X, Y) and the imposed response type restriction.

Throught the definitions and proofs I alternate from edges of the graphs that define the transportation problem and response types. This is possible because there is a one-to-one correspondence between the response types t^Z that belong to $\mathcal{T}_Z \setminus R$ and edges $e \in \mathcal{E}_R$. This correspondence is given by $t^Z = (x_l, x_{l'}) \longrightarrow e = (x_l^0, x_{l'}^1)$. I denote the unique edge associated to $t^Z \in \mathcal{T}_Z$ as e_{t^z} and the unique instrument-response type associated to $e \in \mathcal{E}_R$ as t_e^Z.

Definition B.1 (**Circulation with demands problem associated to R**). *Here I introduce a version of the problem taylored to the instrument validity application. For a detailed discussion of discrete circulation and transportation problems see Kleinberg and Tardos (2006) chapter 7.5.*

Let $R \subset \mathcal{T}_Z$ be a collection of instrument-response types with associated graph $\mathcal{E}_R$. Define demands $d_{x_l^k}$ for $x_l^k \in \mathcal{X}_0 \cup \mathcal{X}_1$ as follows:

$$d_{(x,z)} = \begin{cases} -\mathbb{P}[X = x_l | Z = z_k] & \text{if } k = 0 \\ \mathbb{P}[X = x_l | Z = z_k] & \text{if } k = 1 \end{cases} \tag{81}$$

Furthermore, every edge $e = (x_l^0, x_{l'}^1) \in \mathcal{E}_R$ has a maximum capacity $c_e \in [0, 1]$ defined as $c_e = c_e((x_l^0, x_{l'}^1))$.

A candidate solution to the circulation with capacities problem is a function $\tilde{p} : E_R \longrightarrow \mathbb{R}_+$ such that, for each directed edge $e = (x_l^0, x_{l'}^1) \in \mathcal{E}_R$, the number $\tilde{p}(e)$, represents the flow from node x_l^0 to node $x_{l'}^1$. The flow $\tilde{p}$ solves the problem if:

1. *Capacity constraints are satisfied:*

$$0 \leq \tilde{p}(e) \leq c_e \text{ for all } e \in E_R \tag{82}$$

2. *Demand is satisfied at every node (note that supply is represented as negative*

demand):

$$d_{x_l^0} = -\sum_{e \in E^{out}(x_l^0)} \tilde{p}(e) \qquad \text{for all } l = 1, 2, ..., L \tag{83}$$

$$d_{x_{l'}^1} = \sum_{e \in E^{in}(x_{l'}^1)} \tilde{p}(e) \qquad \text{for all } l = 1, 2, ..., L \tag{84}$$

Where $E^{out}(x_l^0) = \{e \in \mathcal{E}_R : e = (x_l^0, x_{l'}^1) \text{ for some } x_{l'}^1 \in \mathcal{X}_1\}$ *and* $E^{in}(x_{l'}^1) = \{e \in \mathcal{E}_R : e = (x_l^0, x_{l'}^1) \text{ for some } x_l^1 \in \mathcal{X}_0\}$ *represent the sets of edges that depart from* x_l^0 *and end in* $x_{l'}^1$ *respectively.*

Definition B.2 (**Maximum flow problem associated to R**). *Let* $R \subset \mathcal{T}_Z$ *be a collection of instrument-response types with associated graph* $\mathcal{E}_R$*. Let* $(\mathcal{X}_0 \cup \mathcal{X}_1, \mathcal{E}_R)$ *be the graph from definition B.1. From theorem 7.50 in Kleinberg and Tardos (2006) that any circulation with demands problem is equivalent to a maximum flow problem. The maximum flow problem associated to* R *is obtained by applying this principle to the associated circulation with demands problem.*

The directed graph for the maximum flow problem is defined as follows:

Two vertices are added: a source s^* *and a sink* t^* *to obtain the set of vertices* $\mathcal{X}_0 \cup \mathcal{X}_1 \cup \{s^*, t^*\}$*.*

A directed edge points from the source to every node x_l^1 *and from every node* $x_{l'}^1$ *to the sink:*

$$\mathcal{E}_R^{MaxFlow} = \mathcal{E}_R \bigcup \Big\{(x_l^0, t^*) : x \in \mathcal{X}\Big\} \bigcup \Big\{(s^*, x_{l'}^1\ ) : x \in \mathcal{X}\Big\} \tag{85}$$

The capacity of all edges of the form $e = (x_l^0, t^*)$ *is set to* $c_e = \mathbb{P}[X = x_l | Z = z_0]$*, the capacity of all edges of the form* $e = (s^*, x_{l'}^1\ )$ *is set to* $c_e = \mathbb{P}[X = x_{l'} | Z = z_1]$*. All remaining edges have the same capacities as in definition B.1.*

A flow is a function $\tilde{p} : \mathcal{E}_R^{MaxFlow} \longrightarrow \mathbb{R}_+$ *such that every intermediate node (i.e. all nodes except* s^* *and* t^* *has exactly the same inflow and outflow). Formally:*

1. *For every* $x_l^0 \in \mathcal{X}_0$*:*

$$\sum_{e' \in E^{out}(x_l^0)} \tilde{p}(e') = \tilde{p}\big((s^*, x_l^0)\big) \tag{86}$$

2. *For every* $x_{l'}^1 \in \mathcal{X}_1$*:*

$$\sum_{e' \in E^{in}(x_{l'}^1)} \tilde{p}(e') = \tilde{p}\big((x_{l'}^1, t^*)\big) \tag{87}$$

Let $\tilde{P}$ *denote the set of all flows. The problem is:*

$$\underset{\tilde{p} \in \tilde{P}}{max.} \sum_{x_l^0 \in \mathcal{X}_0} \tilde{p}\big((s^*, x_l^0)\big) \tag{88}$$

$$\text{subject to: } \tilde{p}(e) \leq c_e. \tag{89}$$

Definition B.3 ($s^* - t^*$ **cut of** $\big(\mathcal{X}_0 \cup \mathcal{X}_1 \cup \{s^*, t^*\}, \mathcal{E}_R^{MaxFlow}\big)$)**.** *Let* $\big(\mathcal{X}_0 \cup \mathcal{X}_1 \cup \{s^*, t^*\}, \mathcal{E}_R^{MaxFlow}\big)$ *be the graph from definition B.2.*

1. *A cut* $C = (S_C, T_C)$ *is a partition (with elements* S *and* T*) of the set* $\mathcal{X}_0 \cup \mathcal{X}_1 \cup \{s^*, t^*\}$ *such that* $s^* \in S_C$ *and* $t^* \in T_C$*. I denote the set of all cuts of* $\big(\mathcal{X}_0 \cup \mathcal{X}_1 \cup \{s^*, t^*\}, \mathcal{E}_R^{MaxFlow}\big)$ *as* $\mathcal{C}(R, \mathcal{X})$*.*

2. *The cut set of* $C = (S_C, T_C)$ *denoted* Ξ_C*, is defined as the set of edges that, if removed, would disconect* S_C *from* T_C*, formally:*

$$\Xi_C = \{e = (e_1, e_2) \in \mathcal{E}_R^{MaxFlow} | e_1 \in S_C \text{ and } e_2 \in T_C\} \tag{90}$$

3. *The capacity of a cut* C*, denoted* $Cap(C)$*, is defined as the sum of the capacities of the edges in* Ξ_C*:*

$$Cap(C) = \sum_{e \in \Xi_C} c_e \tag{91}$$

4. *The minimum cut associated to* R *is* $\underset{C \in \mathcal{C}(R, \mathcal{X})}{max} Cap(C)$

Proposition B.1. *The following statements are equivalent:*

1. *A distribution over instrument response types exists that satisfies assumptions (2.1), (4.1) and* $p(x_l, x_{l'}) \leq c_e = c_e((x_l^0, x_{l'}^1))$ *exists.*

2. *A solution to the circulation with demands problem from definition B.1 exists.*

3. *The maximum flow attainable in the problem from definition B.2 is equal to 1.*

4. *The minimum cut attainable in definition B.3 is equal to 1.*

Proof of proposition B.1. I prove the equivalence between 1 and 2. The equivalence of 2, 3 and 4 follows from well known results in operations research (See Kleinberg and Tardos (2006) theorems 7.9 and 7.50).

for reference.

$2 \implies 1$:

Let $\tilde{p}$ be a flow that satisfies the circulation with demands problem associated to R.

For any instrument-response type $t^Z = (x_l, x_{l'}) \in \mathcal{T}_Z$ define the distribution over instrument-response types p as follows:

$$p(t^Z) = p\big((x_l, x_{l'})\big) := \begin{cases} \tilde{p}(e_{t^Z}) & \text{if } e_{t^Z} \in \mathcal{E}_R \\ 0 & \text{otherwise} \end{cases} \tag{92}$$

Consider any node x_l^0:

$$-\mathbb{P}[X = x|Z = 0] = d_{x_l^0} \tag{93}$$

$$= -\sum_{e \in E^{out}(x_l^0)} \tilde{p}(e) \tag{94}$$

$$= -\sum_{x_{l'} \in \mathcal{X}} p((x_l, x_{l'})) \tag{95}$$

The first two equalities follow from the definition of the transportation problem, the last from $p(t^Z) = \tilde{p}(e_{t^Z})$.

An analogous argument shows that for $z = 1$:

$$\mathbb{P}[X = x_l|Z = 1z] = \sum_{x' \in \mathcal{X}} p((x_{l'}, x_l)) \tag{96}$$

Which shows that p is consistent with observed conditional treatment probabilities.

Capacity constraints are trivially satisfied since $\tilde{p}(e_{t^Z}) \leq c_e = c_e((x_l^0, x_{l'}^1))$.

Finally, from the definition of $\mathcal{E}_R$ it follows that $\tilde{p}$ only assigns non-negative flows to edges in $e \in \mathcal{E}_R$, therefore, p only assigns non-negative probabilities to response types $t^Z \in R$. This shows consistency with assumption 4.1.

$\underline{1 \implies 2:}$

Let p be a distribution over response types that satisfies assumptions (2.1) and (4.1). For every edge $e = (x_l^0, x_{l'}^1)$ define the flow $\tilde{p}$ as follows:

$$\tilde{p}(e) = \tilde{p}\big((x_l^0, x_{l'}^1)\big) = p(x_l, x_{l'}) \tag{97}$$

Note that e is always well defined since every response type is assigned non-negative probability.

Now, consider an arbitrary node $x_l^0 \in \mathcal{X}_0$:

$$\begin{aligned} d_{x_l^0} &= -\mathbb{P}[X = x_l | Z = z_0] && (98)\\ &= -\sum_{x_{l'} \in \mathcal{X}} p((x_l, x_{l'})) && (99)\\ &= -\sum_{e \in E^{out}(x_l^0)} \tilde{p}(e) && (100) \end{aligned}$$

An analogous argument shows that for $x_{l'}^1$:

$$d_{(x,z)} = \sum_{e \in E^{in}(x_{l'}^1)} \tilde{p}(e) \tag{101}$$

Thus, demand is satisfied at every node. Again, capacity constraints are trivially satisfied since $\tilde{p}((x_l^0, x_{l'}^1)) \leq c_e = c_e((x_l^0, x_{l'}^1))$. This completes the proof.

□

Proof of theorem 4.1.

Part 1: First note that the restriction $c_{tz} = 1$ simply states that instrument-response probabilities are not bounded from above other than by the requirement that they are less than 1. This is, response types in R can occur with arbitrary frequency in the population.

$\Longrightarrow$

$$\mathbb{P}[X \in S|Z = z_0] = p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \mathcal{X}\}) \tag{102}$$

$$= \sum_{x_{l'} \in \mathcal{X}} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \tag{103}$$

$$= \sum_{x_{l'} \in \{x_{l'} \in \mathcal{X} : (s, x_{l'}) \in R \text{ and } s \in S\}} (\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \tag{104}$$

$$= \sum_{x_{l'} \in L_{\geq^R}(S)^C} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \tag{105}$$

$$\leq \sum_{x_{l'} \in L_{\geq^R}(S)^C} \mathbb{P}[X = x_{l'}|Z = 1] \tag{106}$$

$$= \mathbb{P}[X \in L_{\geq^R}(S)^C|Z = z_1] \tag{107}$$

The first equality comes from the consistency of p with observed probabilities, the second from decomposing the probability in terms of all the different response types that realize x_l when $Z = 0$, the third equality comes from the fact that p assigns zero probability to response types outside of R, and the fourth noting that if $(x_l, x_{l'}) \in R$ then $\not x_l \geq^R x_{l'}$ which implies $x_{l'} \notin L_{\geq^R}(S)$. The inequality comes from noting that p is consistent with observed probabilities and $\{x_{l'}\}$ could also be realized when $Z = 1$ by types whose potential outcome when $Z = 0$ lies outside of S. The last equality comes from adding disjoint probabilities across the elements of $L_{\geq^R}(S)^C$.

$\Longleftarrow$

I prove the counterpositive, this is, if no p that satisfies the hypotheses exists, then, a violation of equation 19 must exist. To this end I use the minimum cut characterization.

First, note that by setting $C = \{e \in \mathcal{E}_R : e = (s^*, x_l^0) \textit{ for some } x_l \in \mathcal{X}_0\}$ a cut of capacity one can always be attained. Therefore, edges of the form $(x_l^0, x_{l'}^1)$ will never be part of a minimum cut since their capacity is 1 and removing a single edge will not induce a cut. Thus, minimum cut sets will only contain edges of the form (s^*, x_l^0) or $(x_{l'}^1, t^*)$.

Suppose that C is a minimum cut such that $Cap(C) \leq 1$. Let S be the set of all $x_l \in \mathcal{X}$ such that the edge (s^*, x_l^0) belongs to Ξ_C. Then, by construction, any node of the form $(s^*, x_{l'}^0)$ for $x_{l'} \in S^C = \mathcal{X} \setminus S'$ does not belong to the cut.

Define $S' := \mathcal{X} \setminus S$ and note that by construction for any $x_{l'} \in S'$ and any $x_{l''} \in \mathcal{X}$ such that $(x_{l'}, x_{l''}) \in R$, the edge $(x^1_{l''}, t^*)$ belongs to C. Otherwise, the path $(s^*, x_{l'}), (x_{l'}, x_{l''}), (x_{l''}, t^*)$ would connect t^* and s^*.

But the set of elements $x_{l''} \in \mathcal{X}$ such that $(x_{l'}, x_{l''}) \in R$ for some $x_{l'} \in S'$ is actually the set $L_{\geq^R}(S')^C$.

Therefore:

$$\sum_{c_e \in \Xi_C} c_e < 1 \tag{108}$$

$$\implies \sum_{e \in \mathcal{E}_R \cap \{s^*\} \times S} c_e + \sum_{e \in \mathcal{E}_R \cap L_{\geq^R}(S') \times \{t^*\}} c_e < 1 \tag{109}$$

$$\implies \sum_{e \in \mathcal{E}_R \cap L_{\geq^R}(S') \times \{t^*\}} c_e < 1 - \sum_{e \in \mathcal{E}_R \cap \{s^*\} \times S\}} c_e \tag{110}$$

$$\implies \sum_{e \in \mathcal{E}_R \cap L_{\geq^R}(S') \times \{t^*\}} c_e < \sum_{e \in \mathcal{E}_R \cap \{s^*\} \times S'} c_e \tag{111}$$

$$\implies \mathbb{P}[X \in L_{\geq^R}(S') | Z = z_1] < \mathbb{P}[X \in S' | Z = z_0] \tag{112}$$

The first implication follows from the previous discussion. The second from re-arranging terms. The third from $S' = S^C$ and the fourth from the definition of the capacities. The last equation is a violation of equation 19.

Part 2: $\underline{\implies:}$ Suppose that a distribution over instrument-response types $p \in \Delta(\mathcal{T}_Z)$ exists that satisfies assumptions 1 and 3.

For $\emptyset \neq S \subset \mathcal{X}$, let (Λ_S, Λ'_S) be a partition of $L_{\geq^R}(S)^C$ such that the following three conditions are satisfied:

1. $\emptyset \neq \Lambda'_S \subset S$.
2. For every $x_{l'} \in \Lambda'_S$ the set $L_{\geq^R}(x_{l'})^C \setminus \{x_{l'}\}$ is contained in Λ_S.
3. $L_{\geq^R}(S \setminus \Lambda'_S)^C \subset \Lambda_S$.

Take equation 105:

$$\mathbb{P}[X \in S|Z = z_0] = \sum_{x_{l'} \in L_{\geq R}(S)^C} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \tag{113}$$

$$= \sum_{x_{l'} \in \Lambda} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \tag{114}$$

$$+ \sum_{x_{l'} \in \Lambda'} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\})$$

$$= \sum_{x_{l'} \in \Lambda} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \tag{115}$$

$$+ \sum_{x_{l'} \in \Lambda'} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in (S \setminus \Lambda'_S) \times \{x_{l'}\}\})$$

$$+ \sum_{x_{l'} \in \Lambda'} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in \Lambda'_S \times \{x_{l'}\}\})$$

The second term from the last equation is equal to zero by condition 3. Condition 2 implies that the only possible value that x_l can take in the last term is $x_{l'}$. Therefore, the equation becomes:

$$\mathbb{P}[X \in S|Z = z_0] = \sum_{x_{l'} \in \Lambda} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \tag{116}$$

$$+ \sum_{x_{l'} \in \Lambda'} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_l) \text{ such that } x_l \in \Lambda'\})$$

$$\leq \mathbb{P}[X \in \Lambda|Z = z_1] + \sum_{x_l \in \Lambda'} p((x_l, x_l)) \tag{117}$$

$$\leq \mathbb{P}[X \in \Lambda|Z = z_1] + \sum_{x_l \in \Lambda'} \Psi_{x_l} \tag{118}$$

The first inequality follows from consistency of p with observed probabilities and the fact that response types that realize treatments outside of S when $Z = 0$ might also realize treatments in Λ when $Z = 1$ along with the fact that $L_{\geq R}(x_l)^C \setminus \{x_l\} \subset \Lambda$ for $x_l \in \Lambda'$. The second inequality follows from theorem 3.1.

$\Longleftarrow$

As for part 1, I prove the contrapositive statement. That is, I suppose that a cut with capacity less than 1 exists and show that it implies that inequality (20) from theorem (4.1) is not satisfied for some $S \subset \mathcal{X}$.

Now the capacities of edges of the form (x_l^0, x_l^1) are potentially binding. For cuts that only contain edges connected to s^* or t^* the argument from part 1 still applies. Edges of the form $(x_l^0, x_{l'}^1)$ for $x_l \neq x_{l'}$ still have capacity 1 and therefore will not belong to a minimum cut that is strictly less than one.

Suppose that C is a minimum cut with at least one element of the form (x_l^0, x_l^1).

Let S' be the set of elements in $x_l \in \mathcal{X}$ such that $(s^*, x_l^0) \in C$. Define $S = \mathcal{X} \setminus S'$. Note that for any $x_l \in S$ and any $x_{l'} \in L_{\geq^R}(x_l)^C$ one of the following sets of conditions must hold:

1. The following holds:
   - $(x_{l'}^1, t^*) \in C$
2. The following three conditions hold
   - $x_l = x_{l'} \in S \cap L_{\geq^R}(S)^C$
   - $(x_{l'}^0, x_{l'}^1) \in C$.
   - $(x_{l''}^1, t^*) \in C$ for all $x_{l''} \in L_{\geq^R}(x_{l'})^C \setminus \{x_{l'}\}$

The first condition prevents paths of the form $(s^*, x_l^0), (x_l^0, x_{l'}^1), (x_{l'}^1, t^*)$ as in part 1.

If condition 1 is not satisfied and $(x_{l'}^1, t^*) \notin C$, then, the path $(s^*, x_l^0), (x_l^0, x_{l'}^1), (x_{l'}^1, t^*)$ must be cut at the middle at edge $(x_l^0, x_{l'}^1)$. Because edges associated to $x_l \neq x_{l'}$ cannot belong to the cut, it must be that $x_l = x_{l'} \in S' \cap L_{\geq^R}(S')^C$. Furthermore, for every $x_{l''} \in L_{\geq^R}(x_l)^C \setminus \{x_l\}$, the path $(s^*, x_l^0), (x_l^0, x_{l''}^1), (x_{l''}^1, t^*)$ must be blocked at some point. By construction, this can only happen at $(x_{l''}^1, t^*)$, thus, $(x_{l''}^1, t^*) \in C$.

In other words, a path $(s^*, x_l^0), (x_l^0, x_{l'}^1), (x_{l'}^1, t^*)$ might be blocked by condition 1, if not, then, conditions 2 must hold.

Let Λ'_S be the set $\Lambda'_S = \{x_l \in \mathcal{X} : (x_l^0, x_l^1) \in C\} \subset L_{\geq^R}(S)^C \cap S$. Set $\Lambda_S = L_{\geq^R}(S)^C \setminus \Lambda'$. By construction the sets Λ'_S and Λ_S partition $L_{\geq^R}(S)^C$.

Note that $\{x_{l'} \in \mathcal{X} : (x_{l'}, t^*) \in C\}$ must be a subset of $L_{\geq^R}(S)^C$, otherwise, $x_{l'}^1$ would not be reachable from $\{x_l^0 : x_l \in S\}$ and eliminating $(x_{l'}, t^*)$ from C would attain a cut with strictly lower capacity. Thus, $\{x_{l'} \in \mathcal{X} : (x_{l'}, t^*) \in C\} \subset \Lambda_S$.

Similarly, note that if $x_{l'} \in \Lambda_S$ it must be that $x_l \in \{x_{l'} \in \mathcal{X} : (x_{l'}, t^*) \in C\}$, otherwise, a path through some $x_l \in S$ such that $L_{\geq^R}(x_l)^C \cap \Lambda_S \neq \emptyset$ to $x_{l'}$ would

connect s^* to t^*. Hence:

$$\{x_{l'} \in \mathcal{X} : (x_{l'}, t^*) \in C\} = \Lambda \tag{119}$$

Finally, suppose that for some $x_l \in S \setminus \Lambda'_S$ exists some $x_{l'} \in L_{\geq^R}(x_l)^C \cap \Lambda'_S \neq \emptyset$. By definition it must be that $x_{l'} \neq x_l$. Then, the path $(s^*, x_l), (x_l, x_{l'}), (x_{l'}, t^*)$ would connect s^* to t^*. Hence, it must be that:

$$L_{\geq^R}(S \setminus \Lambda'_S)^C \subset \Lambda_S \tag{120}$$

Then:

- $\emptyset \neq \Lambda' \subset S$ (By conditions 2).
- For all $x_{l'} \in \Lambda'$, $L_{\geq^R}(x'_l)^C \setminus \{x_{l'}\} \subset \Lambda$ (By conditions 2 and equation (119)).
- $L_{\geq^R}(S \setminus \Lambda'_S)^C \subset \Lambda_S$ by expression (120).

Moreover, it must be that:

$$\sum_{c_e \in \Xi_C} c_e < 1 \tag{121}$$

$$\implies \sum_{c \in C \cap \{s^*\} \times S'} c_e + \sum_{\{c \in C : c = (x_l^0, x_l^1) \ s.t. \ x_l \in \Lambda'\}} c_e + \sum_{\{c \in C : c = (x_l^1, t^*) \ s.t \ x_l \in \Lambda\}} c_e < 1 \tag{122}$$

$$\implies \sum_{\{c \in C : c = (x_l^0, x_l^1) \ s.t. \ x_l \in \Lambda'\}} c_e + \sum_{\{c \in C : c = (x_l^1, t^*) \ s.t \ x_l \in \Lambda\}} c_e < 1 - \sum_{c \in C \cap \{s^*\} \times S'} c_e \tag{123}$$

$$\implies \sum_{\{c \in C : c = (x_l^0, x_l^1) \ s.t. \ x_l \in \Lambda'\}} c_e + \mathbb{P}[X \in \Lambda | Z = 1] < \mathbb{P}[X \in S | Z = z_0] \tag{124}$$

Where the first implication follows from the previous discussion, the second from rearranging terms and the last from the definition of the capacities.

Setting $c_e((x_l^0, x_{l'}^1)) = \Psi_{x_l}$ completes the proof.

□

B.2 Proofs for subsection 4.2

To prove theorem 4.2 I begin by formally defining latent preference types. To this end, let $\mathcal{L}_s(\mathcal{X})$ denote the set of all strict preference relations over $\mathcal{X}$ and let $\mathcal{N}(x_l) =$

$\{\succ \in \mathcal{L}_s(\mathcal{X}) : x_l \succ x_{l'} \text{ for all } x_{l'} \in \mathcal{X}\}$ denote the subset of strict preference relations that has x_l as a maximizer.

Definition B.4 (Latent preference types)**.** *A latent preference type $t^{\succ} \in \mathcal{T}_{\succ}$ is a function that maps instrument realizations to strict preference profiles: $t^{\succ} : \mathcal{Z} \longrightarrow \mathcal{L}_s(\mathcal{X})$.*

Note that $t^{\succ}(z)$ is a preference relation. Therefore, the equation $x_l t^{\succ}(z) x_{l'}$ reads as "latent preference types $t^{\succ}$ strictly prefer x_l over $x_{l'}$ when $Z = z$". To improve readibility, I denote $t^{\succ}(z)$ as $\succ^{t(z)}$ so that the previous equation becomes $x_l \succ^{t(z)} x_{l'}$.

Definition B.5 (Distribution over latent preference types)**.** *A distribution over latent preference types is a function $h : \mathcal{H} \longrightarrow [0,1]$ that satisfies:*

$$h(t^{\succ}) \geq 0 \text{ for all } t^{\succ} \in \mathcal{T}_{\succ} \tag{125}$$

$$\sum_{t^{\succ} \in \mathcal{T}_{\succ}} h(t^{\succ}) = 1 \tag{126}$$

Remark B.1. *Let $h : \mathcal{T}_{\succ} \longrightarrow [0,1]$ be a distribution over latent preference types. The distribution over response types implied by h is the function $p_h : \Omega \longrightarrow [0,1]$ defined as:*

$$p_h(x_l, x_{l'})) = \sum_{t^{\succ} \in N(x_l, x_{l'})} h(t^{\succ}) \tag{127}$$

Where $N(x_l, x_{l'}) = \{t^{\succ} \in \mathcal{T}_{\succ} | t^{\succ}(0) \in \mathcal{N}(x_l) \text{ and } t^{\succ}(1) \in \mathcal{N}(x_{l'})\}$.

Definition B.6. *A distribution over latent preference types h is $\geq^*$-Submonotone if $h(t^{\succ}) = 0$ for every $t^{\succ}$ that violates point 3 of assumption 4.2. This is, types $t^{\succ}$ such that $x_l \geq^* x_{l'}$ and $x_l \succ^{t(z_0)} x_{l'}$ but $x_{l'} \succ^{t(z_1)} x_l$ are not allowed.*

Assumption B.1 (**No S always takers**)**.** *Let $S \subset \mathcal{X}$ and $R \subset \mathcal{T}_Z$ a collection of instrument-response types . R has no S always takers if $(x_l, x_l) \notin R$ for all $x_l \in S$.*

Lemma B.1. *Let h be a distribution over latent preference types. Let R be a collection of instrument-response types. Let $\geq^*$ be a binary relation (not necesarilly transitive or complete) over $\mathcal{X}$. Let $Ref_{\geq^*} = \{x_l \in \mathcal{X} | x_l \geq^* x_l\}$ denote the elements that belong to the reflexive part of $\geq^*$. Let $\geq^*_{Ref}$ and $\geq^*_{Irr}$ denote the reflexive and irreflexive part of $\geq^*$ respectively. Then:*

1. *h satisfies* $\geq^*_{Irr}$*-Submonotonicity if and only if h satisfies* $\geq^*$*-Submonotonicity.*

2. *If* $\geq^*_{Irr} \setminus \geq^R \neq \emptyset$*, then, a distribution over latent preference types h exists such that h satisfies* $\geq^R$*-Submonotonicity and* $p_h(t^Z) = 0$ *for all* $t^Z \in R$ *but h is not* $\geq^*-$*Submonotone.*

3. *If* $\geq^R \setminus \geq^*_{Irr} \neq \emptyset$*, then, a distribution over latent latent preference types h exists that is* $\geq^*$*-Submonotone but* $p_h(t^Z) > 0$ *for some* $t^Z \in R$*.*

4. *If* $\geq^R_{Ref} \neq \emptyset$ *a distribution over latent response types exists that is* $\geq^R$*-Submonotone but* $p_h(t^Z) > 0$ *for some* $t^Z \in R$*.*

5. *If* $\geq^R$*-Submonotonicity holds and no* $R_{\geq^M}$ *always takers holds, then,* $p(t^Z) = 0$ *for all* $t^Z \notin R$*.*

6. *If a distribution over response types p that satisfies equation* **??** *and* $p(t^Z) = 0$ *for all* $t^Z \in R$ *exists, then, a distribution h over latent preference types exists such that h is* $\geq^R$*-Submonotone and* $p = p_h$*. This last equality implies that h is consistent with the data and that if p satisfies no S always takers, so does* p_h*.*

Proof of lemma B.1.

Part 1:

Suppose $x_l \geq^* x_{l'}$ and $x_l \succ_1^{(0)} x_{l'}$. If $x_l = x_{l'}$, then $x_l \succ_1^{(1)} x_{l'}$ holds trivially, otherwise, $\geq^*_{Irr} -QM$ implies that $x_l \succ_1^{(1)} x_{l'}$. The converse is trivial since $\geq^*_{Irr} \subset \geq^*$.

Part 2:

Take the ordering induced by the indexing of the elements of $\mathcal{X}$. Let $\succ^{x_l,x_{l'},*}$ be the preference relation that ranks x_l as the best alternative, $x_{l'}$ as the second best, and all remaining alternatives according to their index: $x_l \succ^{x_l,x_{l'},*} x_{l'} \succ^{x_l,x_{l'},*} x_1 \succ^{x_l,x_{l'},*} ... \succ^{x_l,x_{l'},*} x_{L-1}$.

Suppose that $x_l \neq x_{l'}$ and $(x_l, x_{l'}) \in \geq^*_{Irr} \setminus \geq^R$. Let $t^\succ$ be the unique latent preference type that satisfies $t^\succ(0) = \succ^{x_l,x_{l'},*}$ and $t^\succ(1) = \succ^{x_{l'},x_l,*}$. Define h as:

$$h(t^\succ) = 1 \tag{128}$$

$$h(t^{\succ'}) = 0 \qquad \text{for all } t^{\succ'} \neq t^\succ \tag{129}$$

Note that $(x_l, x_{l'}) \notin \geq^R$ implies that h is $\geq^R$ $-$Submonotone, but h is not $\geq^*$ $-$Submonotone (since preference reversals between x_l and $x_{l'}$ occur with positive

probability). Moreover, note that p_h assigns zero probability to response types outside of R.

Part 3:

If $Ref_{\geq^R}$ is non-empty, a reasoning analogous to part 1 proves the claim. Otherwise, pick some $x_l \in Ref_{\geq^*}$ and define h as follows:

$$h((\succ^{x_l,*}, \succ^{x_l,*})) = 1 \tag{130}$$

$$h(t^Z) = 0 \qquad \text{for all } t^Z \neq (\succ^{x_l,*}, \succ^{x_l,*}) \tag{131}$$

h is $\geq^*$-QM but p_h assigns positive probability to the response type $(x_l, x_l) \notin R$.

Part 4

This part follows immediately from part 3 by setting $\geq^*_{Irr} = \emptyset$.

Part 5:

Let R' be the set of response types allowed by $\geq^R$-QM and no stayers at $Ref_{\geq^R}$. The goal is to show that $R \subset R'$.

Suppose that $(x_l, x_{l'}) \notin R$. This implies that $x_l \geq^R x_{l'}$.

If $x_l \neq x_{l'}$, consider a decision maker i such that $X(0) = x_{l'}$, then, rationality implies that $x_{l'} = X(0) \succ^{(0)} x_l$. Therefore, by $\geq^R$-Submonotonicity, it must be that $x_{l'} \succ^{(1)} x_l$. This implies that $x_{l'}$ is not chosen by decision maker i when $Z = 1$ a.s.. Therefore $(x_l, x_{l'}) \notin R'$.

If $x_{l'} = x_l$, then, no stayers at $Ref_{\geq^R}$ implies that $(x_l, x_{l'}) \notin R'$.

Therefore, $t^Z \notin R \Longrightarrow t^Z \notin R'$ which is equivalent to $t^Z \in \mathbb{R}' \Longrightarrow t^Z \in R$.

Part 6:

Let $t^{\succ}_{x_l,x_{l'},*}$ be the latent preference type such that $t^{\succ}_{x_l,x_{l'},*}(0) = \succ^{x_l,x_l{}',*}$ and $t^{\succ}_{x_l,x_{l'},*}(1) = \succ^{x_{l'},x_l,*}$. Define h as follows:

$$h(t^{\succ}) = \begin{cases} p(x_l, x_{l'}) & \text{if } t^{\succ} = t^{\succ}_{x_l,x_{l'},*} \\ 0 & \text{otherwise} \end{cases} \tag{132}$$

Note that h is $\geq^R$-QM and that for all $(x_l, x_l') \in R_Z$:

$$p_h((x_l, x_l')) = \sum_{t^{\succ} \in N(x_l, x_l')} h(t^{\succ}) = h(t^{\succ}_{x_l, x_{l'}, *}) = p(x_l, x_{l'}) \tag{133}$$

□

Now I comment Lemma 1. The first part shows that only the irreflexive part of binary relations is relevant for submonotonicity. Part 2 shows that when a binary relation $\geq^*$ is stronger than the one induced by R, restricting response types to R does not guarantee $\geq^*$-submonotonicity. Parts 3 and 4 show the converse, that imposing weaker than than $\geq^R$ dos not guarantee that response types outside of R are ruled out. Part 5 shows that R can be guaranteed by imposing $\geq^R$-submonotonicity and ruling out a specific set of always takers. 6 shows that $\geq^R$-submonotonicity is only falsified if the response type restriction associated to R is falsified. Thus, in the binary instrument case $\geq^R$-submonotonicity is just an interpretation for response type restrictions but yields no additional identifying power.

Theorem 4.2 follows immediately from lemma 6 by noting there is a unique correspondence between R and $\geq^R$ and therefore their cardinalities.

C Proofs for section

C.1 Estimation algorithm 1

1. Partition the sample in two splits: S_1 and S_2 of size n_1 and n_2 respectively.
2. Using the first sample, train a non-parametric classifier $\eta^*_{n_1}(y, x)$ as follows:
   (a) Define:

$$\phi_{z_k,n}(y, x_l) := P^{-1}_{k,n_1}\Big[\frac{1}{n_1 h^d_n}\sum_{i=1}^{n_1}\mathbf{1}\{X_i = x_l, Z_i = z_k\}K_n\big(y - Y_i\big)\Big] \tag{134}$$

   Where $n_{k,1} = |\{i \in S_1 : Z_i = z_k\}|$ and $P_{k,n_1} = n_{k,1}/n_1$.

   (b) Then, define:

$$\Delta_{n_1}(y, x_l) := \phi_{z_1,n}(y, x_l) - \phi_{z_0,n}(y, x_l) \tag{135}$$

   (c) And:

$$\eta^*_{n_1}(y, x_l) := \mathbf{1}\{\Delta_{n_1}(y, x_l) \geq 0\}. \tag{136}$$

3. In the second sample, estimate the probabilities $\mathbb{P}[Z = z_k]$ for $k = 0, 1$. Denote the estimators as P_{k,n_2}.
4. In the second sample, evaluate the weighted empirical loss:

$$L_{n_2}(\eta^*_{n_1}, x_l) := \frac{1}{n_2}\sum_{i=1}^{n_2}\Big(\frac{\mathbf{1}\{\eta^*_{n_1}(Y_i, X_i) = 1, Z_i = z_0, X_i = x_l\}}{P_{0,n_2}} + \frac{\mathbf{1}\{\eta^*_{n_1}(Y_i, X_i) = 0, Z_i = z_1, X_i = x_l\}}{P_{1,n_2}}\Big) \tag{137}$$

C.2 Proofs for subsection 5.1

Decompose:

$$\sqrt{n_2}\big(L_{n_2}(\eta^*_{n_1}, x_l) - \Psi_{x_l}\big) = \underbrace{\sqrt{n_2}\big(L_{n_2}(\eta^*_{n_1}, x_l) - L(\eta^*_{n_1}, x_l)\big)}_{:=A_{1,n}(x_l)} + \underbrace{\sqrt{n_2}\big(L(\eta^*_{n_1}, x_l) - L(\eta^*, x_l)\big)}_{:=A_{2,n}(x_l)} \tag{138}$$

And:

$$n_2 Var\Big(L_{n_2}(\eta^*_{n_1}, x_l) - \Psi_{x_l}\Big) = \underbrace{n_2 Var\Big(L_{n_2}(\eta^*_{n_1}, x_l) - L(\eta^*_{n_1}, x_l)\Big)}_{:=V_1} + \underbrace{n_2 Var\Big(L(\eta^*_{n_1}, x_l) - L(\eta^*, x_l)\Big)}_{:=V_2} \tag{139}$$

$$+ \underbrace{2n_2 Cov\Big(L_{n_2}(\eta^*_{n_1}, x_l) - L(\eta^*_{n_1}, x_l), L(\eta^*_{n_1}, x_l) - L(\eta^*, x_l)\Big)}_{:=V_3} \tag{140}$$

The first term corresponds to how well the empirical loss evaluated in the sub-sample S_2 approximates the expected loss of the classifier $\eta^*_{n_1}$, the second term corresponds to the loss difference between the estimated classifier $\eta^*_{n_1}$ and the optimal unfeasible classifier η^*.

Propositions C.1 shows that, under assumptions 5.1 and 5.2, the classifier trained in sub-sample S_1 is sufficiently good so that its expected missclassification loss (and its variance) are small relative to the expected loss of the optimal classifier. Thus, the asymptotic behavior of the estimator is governed by the term $A_{1,n}(x_l)$.

Proposition C.1.

1. $A_{2,n}(x_l) = o_p(1)$.
2. $Cov(A_{1,n}(x_l), A_{2,n}(x_l)) = 0$.
3. $Var(A_{2,n}(x_l)) = o(1)$.

Proof of proposition 5.1.

$$L(\eta^*, x_l) = \frac{\mathbb{P}[\eta^*(Y,X)=0, Z=z_1, X=x_l]}{\mathbb{P}[Z=z_1]} + \frac{\mathbb{P}[\eta^*(Y,X)=1, Z=z_0, X=x_l]}{\mathbb{P}[Z=z_0]} \tag{141}$$

$$= \mathbb{P}[X=x_l]\Big(\frac{\mathbb{P}[\eta^*(Y,X)=0, Z=z_1|X=x_l]}{\mathbb{P}[Z=z_1]} + \frac{\mathbb{P}[\eta^*(Y,X)=1, Z=z_0|X=x_l]}{\mathbb{P}[Z=z_0]}\Big) \tag{142}$$

$$= \mathbb{P}[X=x_l]\int_{\mathcal{Y}} \Big(\frac{\mathbb{P}[\eta^*(y,x_l)=0, Z=z_1|Y=y, X=x_l]}{\mathbb{P}[Z=z_1]} \tag{143}$$

$$+ \frac{\mathbb{P}[\eta^*(y,x_l)=1, Z=z_0|Y=y, X=x_l]}{\mathbb{P}[Z=z_0]}\Big) f_{Y|X=x_l}(y)dy$$

$$= \mathbb{P}[X=x_l]\int_{\mathcal{Y}} \Big(\mathbf{1}\{\eta^*(y,x_l)=0\}\frac{\mathbb{P}[Z=z_1|Y=y, X=x_l]}{\mathbb{P}[Z=z_1]} \tag{144}$$

$$+ \mathbf{1}\{\eta^*(y,x_l)=1\}\frac{\mathbb{P}[Z=z_0|Y=y, X=x_l]}{\mathbb{P}[Z=z_0]}\Big) f_{Y|X=x_l}(y)dy \tag{145}$$

$$= \mathbb{P}[X=x_l]\int_{\mathcal{Y}} \Big(\mathbf{1}\{\eta^*(y,x_l)=0\}\frac{f(y,x_l,z_1)}{\mathbb{P}[Z=z_1]f(y,x_l)} \tag{146}$$

$$+ \mathbf{1}\{\eta^*(y,x_l)=1\}\frac{f(y,x_l,z_0)}{\mathbb{P}[Z=z_0]f(y,x_l)}\Big) f(y,x_l)dy \tag{147}$$

$$= \int_{\mathcal{Y}} \Big(\mathbf{1}\{\eta^*(y,x_l)=0\} f_{Y,X|Z=z_1}(y,x_l) + \mathbf{1}\{\eta^*(y,x_l)=1\} f_{Y,X|Z=z_0}(y,x_l)\Big)dy \tag{148}$$

$$= \Psi_{x_l} \tag{149}$$

□

Proof of proposition C.1.

Proof of part 1: Consider the difference between the loss of the estimated and target classifier:

$$A_{2,n}(x_l) = L(\eta^*_{n_1}, x_l) - L(\eta^*, x_l) \tag{150}$$

$$= \int_{\mathcal{Y}} \Big(\mathbf{1}\{\eta^*_{n_1}(y, x_l) = 0\} f_{Y,X|Z=z_1}(y, x_l) + \mathbf{1}\{\eta^*_{n_1}(y, x_l) = 1\} f_{Y,X|Z=z_0}(y, x_l) \Big)$$
$$- \Big(\mathbf{1}\{\eta^*(y, x_l) = 0\} f_{Y,X|Z=z_1}(y, x_l) + \mathbf{1}\{\eta^*(y, x_l) = 1\} f_{Y,X|Z=z_0}(y, x_l) \Big) dy \tag{151}$$

$$= \int_{\mathcal{Y}} \mathbf{1}\{\eta^*_{n_1}(y, x_l) \neq \eta^*(y, x_l)\} \big| \Delta(y, x_l) \big| dy \tag{152}$$

Where $\Delta(y, x_l) := f_{Y,X|Z=z_1}(y, x_l) - f_{Y,X|Z=z_0}(y, x_l)$.

Partition the support of the outcome as follows:

$$\mathcal{Y} = \mathcal{S}_{x_l,n_1} \cup \mathcal{M}_{x_l,n_1} \cup \mathcal{C}_{x_l} \tag{153}$$

Where:

$$\mathcal{S}_{x_l,n_1} = \Big\{ y \in \mathcal{Y} : |\Delta(y, x_l)| > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}} \Big\} \tag{154}$$

$$\mathcal{M}_{x_l,n_1} = \Big\{ y \in \mathcal{Y} : 0 < |\Delta(y, x_l)| \leq C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}} \Big\} \tag{155}$$

$$\mathcal{C}_{x_l} = \Big\{ y \in \mathcal{Y} : |\Delta(y, x_l)| = 0 \Big\} \tag{156}$$

Then, equation 152 becomes:

$$A_{2,n}(x_l) = \int_{\mathcal{S}_{x_l,n_1}} \Big(\mathbf{1}\{\eta^*(y, x_l) \neq \eta^*_{n_1}(y, x_l)\} |\Delta(y, x_l)| \Big) dy \tag{157}$$
$$+ \int_{\mathcal{M}_{x_l,n_1}} \Big(\mathbf{1}\{\eta^*(y, x_l) \neq \eta^*_{n_1}(y, x_l)\} |\Delta(y, x_l)| \Big) dy$$
$$+ \int_{\mathcal{C}_{x_l}} \Big(\mathbf{1}\{\eta^*(y, x_l) \neq \eta^*_{n_1}(y, x_l)\} |\Delta(y, x_l)| \Big) dy$$

Now I show that $A_{2,n}(x_l) = o_p(n^{-\frac{1}{2}})$. For the first term I use the classic results on rates of uniform convergence for kernel density estimators from Audibert and

Tsybakov and Talagrand to prove that, when the sub-densities are well separated, misslasification occurs with low probability. For the second term I invoke the margin condition from assumption 5.1.3 combined with the fact that miss-classification loss near the boundary of the contact set is small. The third term is exactly zero since $|\Delta(y, x_l)| = 0$ for every $y \in \mathcal{C}_{x_l}$.

Consider the term associated with $\mathcal{S}_{x_l,n_1}$:

$$\int_{\mathcal{S}_{x_l,n_1}} \Big(\mathbf{1}\{\eta^*(y, x_l) \neq \eta^*_{n_1}(y, x_l)\}|\Delta(y, x_l)|\Big)dy \tag{158}$$

$$\leq \int_{\mathcal{S}_{x_l,n_1}} \mathbf{1}\{\eta^*(y, x_l) \neq \eta^*_{n_1}(y, x_l)\}\Big(f_{Y,X|Z=z_1}(y, x_l) + f_{Y,X|Z=z_0}(y, x_l)\Big)dy$$

$$\leq B^{-1} \int_{\mathcal{S}_{x_l,n_1}} \mathbf{1}\{\eta^*(y, x_l) \neq \eta^*_{n_1}(y, x_l)\}f_Y(y)dy \tag{159}$$

$$= B^{-1}\mathbb{P}[\eta^*(Y, x_l) \neq \eta^*_{n_1}(Y, x_l), Y \in \mathcal{S}_{x_l,n_1}|S_1] \tag{160}$$

Where the first line follows from the triangle inequality, the second from the definition of $f_{Y,X|Z=z_k}$ combined with assumption 5.1.1, and the third is a standard property of indicator expectations.

Note that for any $y \in \mathcal{S}_{x_l,n_1}$, $|\Delta(y, x_l)| > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$. Hence, if $\|\Delta_{n_1}(\cdot, x_l) - \Delta(\cdot, x_l)\|_\infty \leq r_{n_1}$, then $\Delta_{n_1}(y, x_l)$ has the same sign as $\Delta(y, x_l)$ for every $y \in \mathcal{S}_{x_l,n_1}$. Therefore:

$$\mathbb{P}\big[\eta^*(Y, x_l) \neq \eta^*_{n_1}(Y, x_l), Y \in \mathcal{S}_{x_l,n_1} \mid S_1\big] \leq \mathbf{1}\Big\{\|\Delta_{n_1}(\cdot, x_l) - \Delta(\cdot, x_l)\|_\infty > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big\}. \tag{161}$$

By Lemma C.3, the right-hand side is $o_p(n^{-1})$. Hence

$$\mathbb{P}\big[\eta^*(Y, x_l) \neq \eta^*_{n_1}(Y, x_l), Y \in \mathcal{S}_{x_l,n_1}|S_1\big] = o_p(n^{-1}). \tag{162}$$

Combining 160 and 162 yields:

$$\int_{\mathcal{S}_{x_l,n_1}} \Big(\mathbf{1}\{\eta^*(y, x_l) \neq \eta^*_{n_1}(y, x_l)\}|\Delta(y, x_l)|\Big)dy = o_p(n^{-1}) \tag{163}$$

Now consider the term associated with $\mathcal{M}_{x_l,n_1}$. Assumption **??** implies that the conditions from the first part of lemma C.4 is satisfied for $r_n = \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$ and $\mathcal{M}_{x_l}(r_n) = \mathcal{M}_{x_l,n_1}$. Therefore:

$$\int_{\mathcal{M}_{x_l,n_1}} \Big(\mathbf{1}\{\eta^*(y,x_l) \neq \eta^*_{n_1}(y,x_l)\}|\Delta(y,x_l)|\Big)dy = o_p(n^{-1/2}) \tag{164}$$

Thus, $A_{2,n}(x_l) = o_p(n^{-1/2})$. This completes the proof of part 1.

Proof of part 2:

Lemma C.5 implies that:

$$Cov\Big(L_{n_2}(\eta^*_{n_1},x_l), L(\eta^*_{n_1},x_l)\Big) = \mathbb{E}\Big[L_{n_2}(\eta^*_{n_1},x_l)L(\eta^*_{n_1},x_l)\Big] - \mathbb{E}\Big[L_{n_2}(\eta^*_{n_1},x_l)\Big]\mathbb{E}\Big[L(\eta^*_{n_1},x_l)\Big] \tag{165}$$

$$= Var\Big(L(\eta^*_{n_1},x_l)\Big) \tag{166}$$

Then:

$$Cov\Big(L_{n_2}(\eta^*_{n_1},x_l) - L(\eta^*_{n_1},x_l), L(\eta^*_{n_1},x_l) - L(\eta^*,x_l)\Big) \tag{167}$$

$$= Cov\Big(L_{n_2}(\eta^*_{n_1},x_l), L(\eta^*_{n_1},x_l)\Big) - Var\Big(L_{n_2}(\eta^*_{n_1},x_l)\Big)$$

$$= 0 \tag{168}$$

Proof of part 3:

The argument replicates the idea from part 1, but now applied to the variance. In the contact set, any classification decision yields the exact same loss, therefore, even if the classifier is random inside the contact set, the loss associated to such randomness is deterministic. On a shrinking region near the boundary of the contact set, classification mistakes occur with non-negligible probability. Nonetheless, the difference in loss between the optimal classifier and a random classifier is small and so is the variance of such loss, furthermore, the margin condition implies that such region shrinks at a sufficiently fast rate. Finally, far from the closure of the contact set, missclassification events are unlikely.

First note that:

$$
\begin{aligned}
Var(A_{2,n}(x_l)) =& Var\Big(\int_{\mathcal{Y}} \mathbf{1}\{\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l)\} \big|\Delta(y,x_l)\big| dy \Big) && (169)\\
=& Cov\Big(\int_{\mathcal{Y}} \mathbf{1}\{\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l)\} \big|\Delta(y,x_l)\big| dy, && (170)\\
& \qquad \int_{\mathcal{Y}} \mathbf{1}\{\eta^*_{n_1}(y',x_l) \neq \eta^*(y',x_l)\} \big|\Delta(y',x_l)\big| dy' \Big) \\
=& \int_{\mathcal{Y}} \int_{\mathcal{Y}} Cov\big(\mathbf{1}\{\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l)\} \big|\Delta(y,x_l)\big|, && (171)\\
& \qquad \mathbf{1}\{\eta^*_{n_1}(y',x_l) \neq \eta^*(y',x_l)\} \big|\Delta(y',x_l)\big|\big) dy dy' \\
=& \int_{\mathcal{Y}} \int_{\mathcal{Y}} \Big(\mathbb{E}\big[\mathbf{1}\{\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l)\}\mathbf{1}\{\eta^*_{n_1}(y',x_l) \neq \eta^*(y',x_l)\}\big] && (172)\\
& \qquad - \mathbb{E}\big[\mathbf{1}\{\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l)\}\big]\mathbb{E}\big[\mathbf{1}\{\eta^*_{n_1}(y',x_l) \neq \eta^*(y',x_l)\}\big] \Big) \\
& \qquad \times \big|\Delta(y,x_l)\big|\big|\Delta(y',x_l)\big| dy dy' \\
=& \int_{\mathcal{Y}} \int_{\mathcal{Y}} \Big(\mathbb{P}\big[\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l), \eta^*_{n_1}(y',x_l) \neq \eta^*(y',x_l)\big] && (173)\\
& \qquad - \mathbb{P}\big[\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l)\big]\mathbb{P}\big[\eta^*_{n_1}(y',x_l) \neq \eta^*(y',x_l)\big] \Big) && (174)\\
& \qquad \times \big|\Delta(y,x_l)\big|\big|\Delta(y',x_l)\big| dy dy' \\
=& \int_{\mathcal{Y}} \int_{\mathcal{Y}} \boldsymbol{C_p}(y,x_l) dy dy' && (175)\\
\leq& \int_{\mathcal{Y}} \int_{\mathcal{Y}} |\boldsymbol{C_p}(y,x_l)| dy dy' && (176)
\end{aligned}
$$

Where:

$$
\begin{aligned}
C_P(y,y',x_l) :=& \mathbb{P}\big[\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l), \eta^*_{n_1}(y',x_l) \neq \eta^*(y',x_l)\big] && (177)\\
& - \mathbb{P}\big[\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l)\big]\mathbb{P}\big[\eta^*_{n_1}(y',x_l) \neq \eta^*(y',x_l)\big]\big|\Delta(y,x_l)\big|\big|\Delta(y',x_l)\big|
\end{aligned}
$$

Now decompose 176 as follows:

$$Var(A_{2,n}(x_l)) \leq \int\limits_{\mathcal{S}_{x_l,n_1}} \int\limits_{\mathcal{Y}_{x_l,n_1}} |C_P(y,y',x_l)|dydy' + \int\limits_{\mathcal{M}_{x_l,n_1}} \int\limits_{\mathcal{M}_{x_l,n_1}} |C_P(y,y',x_l)|dydy' \tag{178}$$
$$+ \int\limits_{\mathcal{M}_{x_l,n_1}} \int\limits_{\mathcal{C}_{x_l}} |C_P(y,y',x_l)|dydy' + \int\limits_{\mathcal{C}_{x_l}} \int\limits_{\mathcal{C}_{x_l}} |C_P(y,y',x_l)|dydy'$$

Where the sets $\mathcal{S}_{x_l,n_1}$, $\mathcal{M}_{x_l,n_1}$ and $\mathcal{C}_{x_l,n_1}$ are as defined in part 1 of the proof.

All the terms that involve integrals over $\mathcal{C}_{x_l,n_1}$ are exactly zero, thus:

$$Var(A_{2,n}(x_l)) \leq \int\limits_{\mathcal{S}_{x_l,n_1}} \int\limits_{\mathcal{Y}_{x_l,n_1}} |C_P(y,y',x_l)|dydy' + \int\limits_{\mathcal{M}_{x_l,n_1}} \int\limits_{\mathcal{M}_{x_l,n_1}} |C_P(y,y',x_l)|dydy' \tag{179}$$

Let $M(y,x_l)$ denote the event $\{\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l)\}$. Observe that the term $\mathbb{P}\big[\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l), \eta^*_{n_1}(y',x_l) \neq \eta^*(y',x_l)\big] - \mathbb{P}\big[\eta^*_{n_1}(y,x_l) \neq \eta^*(y,x_l)\big]\,\mathbb{P}\big[\eta^*_{n_1}(y',x_l) \neq \eta^*(y',x_l)\big]$ is the covariance of the random variables $\upsilon_{y,x_l} := M(y,x_l)$ and $\upsilon_{y',x_l} := M(y',x_l)$ that indicate missclassification events at fixed values of Y and X with respect to the distribution of sub-sample S_1.

Thus, Cauchy-Schwarz's inequality implies:

$$|Cov(\upsilon_{y,x_l}, \upsilon_{y',x_l})| \leq Var(\upsilon_{y,x_l})Var(\upsilon_{y',x_l}) \tag{180}$$
$$= \big(\mathbb{P}[M(y,x_l)] - \mathbb{P}[M(y,x_l)]^2\big)\big(\mathbb{P}[M(y',x_l)] - \mathbb{P}[M(y',x_l)]^2\big) \tag{181}$$
$$\leq \mathbb{P}[M(y,x_l)]\mathbb{P}[M(y',x_l)] \tag{182}$$

Therefore:

$$\int_{\mathcal{S}_{x_l,n_1}}\int_{\mathcal{Y}_{x_l,n_1}} |C_P(y,y',x_l)|dydy' \leq \int_{\mathcal{S}_{x_l,n_1}}\int_{\mathcal{Y}_{x_l,n_1}} \mathbb{P}[M(y,x_l)]\mathbb{P}[M(y',x_l)]\big|\Delta(y,x_l)\big|\big|\Delta(y',x_l)\big|dydy' \tag{183}$$

$$\leq B\int_{\mathcal{S}_{x_l,n_1}} \mathbb{P}[M(y,x_l)]dy \tag{184}$$

$$\leq B\mathbb{P}\Big[\|\Delta_{n_1}(\cdot,x_l)-\Delta(\cdot,x_l)\|_\infty > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big] \tag{185}$$

$$= o_p(n^{-1}) \tag{186}$$

Where the first inequality follows from 182, the second from the fact that $\mathbb{P}[M(y',x_l)]\big|\Delta(y,x_l)\big|\big|\Delta(y',x_l)$ is uniformly bounded, and the last from the fact that for any $y \in \mathcal{S}_{x_l,n_1}$, $|\Delta(y,x_l)| > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$. Hence, if $\|\Delta_{n_1}(\cdot,x_l)-\Delta(\cdot,x_l)\|_\infty \leq r_{n_1}$, then $\Delta_{n_1}(y,x_l)$ has the same sign as $\Delta(y,x_l)$ for every $y \in \mathcal{S}_{x_l,n_1}$. The final identity follows from lemma C.3.

Now consider the term associated with $\mathcal{M}_{x_l,n_1}$:

$$\int_{\mathcal{M}_{x_l,n_1}}\int_{\mathcal{M}_{x_l,n_1}} |C_P(y,y',x_l)|dydy' \leq \int_{\mathcal{M}_{x_l,n_1}}\int_{\mathcal{M}_{x_l,n_1}} \mathbb{P}[M(y,x_l)]\mathbb{P}[M(y,x_l)]\big|\Delta(y,x_l)\big|\big|\Delta(y',x_l)\big|dydy' \tag{187}$$

But, part 2 of lemma C.4 implies that the right hand side is $o(n^{-1})$. This completes the proof.

□

Proof of proposition 5.1.

Proposition C.1 implies that:

$$\sqrt{n_2}\big(\boldsymbol{\theta}_n - \boldsymbol{\mu}_\theta\big) = \sqrt{n_2}\big(\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\mu}_\theta\big) + o(1) \tag{188}$$

Where $\tilde{\boldsymbol{\theta}}_n = (\boldsymbol{P}^T_{\boldsymbol{\mathcal{X}_0,n}}, \boldsymbol{P}^T_{\boldsymbol{\mathcal{X}_1,n}}, n_2^{-1/2}\boldsymbol{A}^T_{1,n})$ and $\boldsymbol{A}^T_{1,n} = (n_2^{-1/2}A_{1,n}(x_0),...,A_{1,n}(x_{L'-1}))$. Furthermore, lemma C.5 and proposition C.1 imply that $\mathbb{E}[\tilde{\boldsymbol{\theta}}_n] - \mu_\theta = o(n^{-1})$.

Define the random variable $\xi(Y,X,Z;z_k,x_l) = \mathbf{1}\{\eta^*_{n_1}(Y,X,Z;z_k,x_l) \neq k, X =$

$x_l, Z = z_k\}$ and note that:

$$A_{1,n}(x_l) = \frac{\sqrt{n_2}}{n_2} \sum_{i=1}^{n_2} \left(\frac{n_2}{n_{2,0}} \xi(Y_i, X_i, Z_i; z_0, x_l) + \frac{n_2}{n_{2,1}} \xi(Y_i, X_i, Z_i; z_1, x_l) - L(\eta^*_{n_1}, x_l) \right) \tag{189}$$

Let the vector $\boldsymbol{P}_{n_2}$ and the function $\boldsymbol{g}$ be as defined in lemmas C.6 and C.7 respectively. Equation 189 implies that $\tilde{\boldsymbol{\theta}}_n$ is a subvector of $\boldsymbol{g}(\boldsymbol{P}_{n_2})$. Formally, if J denotes the set of indices associated with all the entries of $\boldsymbol{g}(\boldsymbol{P}_{n_2})$ that comprise $\tilde{\boldsymbol{\theta}}_n$, and e_j is the $j-th$ canonical vector in $\mathbb{R}^{3L}$. Then, the selector matrix $S_{\boldsymbol{\theta}}$ is the matrix that stacks the vectors e_j for every $j \in J$ so that $\tilde{\boldsymbol{\theta}}_n = S_{\boldsymbol{\theta}} \boldsymbol{g}(\boldsymbol{P}_{n_2})$.

By construction, every component of $(\boldsymbol{P}^T_{\boldsymbol{\chi_0,n}}, \boldsymbol{P}^T_{\boldsymbol{\chi_1,n}})$ is linearly independent from the rest asymptotically, and all the components of $\boldsymbol{A}_{1,n}$ are linearly independent from each other with probability approaching one. Moreover, assumption **??** g uarantees that every component of $A_{1,n}$ of $\boldsymbol{A}_{1,n}$ is linearly independent from $P_{x_l|z_0,n_2}$ and $P_{x_l|z_0,n_2}$ with probability approaching one. Thus, all the elements of $\tilde{\boldsymbol{\theta}}_n$ are asymptotically linearly independent with probability approaching one.

Hence, with probability approaching one, the matrix $S_{\boldsymbol{\theta}} \boldsymbol{V}_{S_{1,n}} S^T_{\boldsymbol{\theta}}$ is symmetric positive-definite and, for every non-zero $\boldsymbol{c} \in \mathbb{R}^m$, a fixed constant $\epsilon_{\boldsymbol{c}}$ that depends on $\boldsymbol{c}$ exists such that the probability $\mathbb{P}[\boldsymbol{c}^T S_{\boldsymbol{\theta}} \boldsymbol{V}_{S_{1,n}} S^T_{\boldsymbol{\theta}} \boldsymbol{c} > \epsilon_{\boldsymbol{c}}]$ tends to one.

Then, lemma C.7 implies that:

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P}\left[\frac{\boldsymbol{c}^T \sqrt{n_2} \left(\boldsymbol{\theta}_n - \boldsymbol{\mu}_\theta \right)}{\sqrt{\boldsymbol{c}^T S_{\boldsymbol{\theta}} \boldsymbol{V}_{S_{1,n}} S^T_{\boldsymbol{\theta}} \boldsymbol{c}}} \leq t \middle| S_1 \right] - \Phi(t) \right| \xrightarrow{p} 0, \ \forall \boldsymbol{c} \in \mathbb{R}^m, \boldsymbol{c} \neq \boldsymbol{0}, \tag{190}$$

and the Cramér-Wold device yields the desired result.

□

Lemma C.1 (KDE bias)**.** *Let* $\tilde{\Delta}_{n_1}(Y, x_l) = \frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} H_n(Y_i, X_i, Z_i; x_l, y)$*, then:*

$$\sup_{y \in \mathcal{Y}} \left|\left| \mathbb{E}[\tilde{\Delta}_{n_1}(Y, x_l)] - \Delta(y, x_l) \right|\right| = O(h_n^2) \tag{191}$$

Lemma C.2. *For any* $r_n > 0$ *such that* $\frac{n_1 r_n^2}{\log(n_1)} \longrightarrow \infty$ *and* $r_n \longrightarrow 0$*:*

$$\mathbb{P}\left(\left|\left| R_n(y, x_l) \right|\right|_\infty \geq r_n \right) = o(n_1^{-1}) \tag{192}$$

Lemma C.3. *Let K be a kernel density function satisfying assumption 5.2.1, h_n a bandwidth sequence satisfying assumption 5.2.2 and suppose that $f_{Y|X=x_l,Z=z_k}$ satisfies assumption 5.1.2. Then, a constant $C_0 > 0$ exists such that:*

$$\mathbb{P}\Big[\Big|\Big|\Delta_{n_1}(y,x_l) - \Delta(y,x_l)\ \Big|\Big|_\infty > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big] = o(n_1^{-1}) \tag{193}$$

Lemma C.4 (Margin missclassification loss)**.**
Let r_n be a sequence such that $r_n \longrightarrow 0$ and $r_n^{1+\alpha}\sqrt{n_1} \longrightarrow 0$, and let $\mathcal{M}_{x_l}(r_n)$ be the set:

$$\mathcal{M}_{x_l}(r_n) = \{y \in \mathcal{Y} : 0 < |\Delta(y,x_l)| \leq r_n\} \tag{194}$$

Then:

$$\int\limits_{\mathcal{M}_{x_l}(r_n)} \mathbf{1}\{M(y,x_l)\}|\Delta(y,x_l)|dy = o(n^{-1/2}) \tag{195}$$

And:

$$\int\limits_{\mathcal{M}_{x_l}(r_n)}\int\limits_{\mathcal{M}_{x_l}(r_n)} \mathbb{P}[M(y,x_l)]\mathbb{P}[M(y,x_l)]\big|\Delta(y,x_l)\big|\big|\Delta(y',x_l)\big|dydy' = o(n^{-1}) \tag{196}$$

Where $M(y,x_l)$ denotes the event $\eta^*(y,x_l) \neq \eta^*_{n_1}(y,x_l)$

Lemma C.5 (The expected empirical loss of $\eta^*_{n_1}$ is its theoretical loss)**.**

$$\mathbb{E}\Big[L_{n_2}(\eta^*_{n_1},x_l)|S_1\Big] = L(\eta^*_{n_1},x_l) \tag{197}$$

$$\tag{198}$$

Lemma C.6.

For $i \in S_2$, let $\boldsymbol{\pi}_{i,n}$ denote the random vector:

$$\begin{aligned}\boldsymbol{\pi}_{i,n} =&(\mathbf{1}\{Z_i = z_0\}, \mathbf{1}\{Z_i = z_1\},\\ &\mathbf{1}\{Z_i = z_0, X_i = x_0\}), ..., \mathbf{1}\{Z_i = z_0, X_i = x_{L-1},\\ &\mathbf{1}\{Z_i = z_1, X_i = x_1\}), ..., \mathbf{1}\{Z_i = z_1, X_i = x_{L-1}\},\\ &\xi(Y_i, X_i, Z_i; z_0, x_0), ..., \xi(Y_i, X_i, Z_i; z_0, x_{L-1})\\ &\xi(Y_i, X_i, Z_i; z_1, x_0), ..., \xi(Y_i, X_i, Z_i; z_1, x_{L-1}))\end{aligned} \tag{199}$$

With mean $\boldsymbol{\mu}_{S_1,n} := \mathbb{E}[\boldsymbol{\pi}_{i,n}|S_1]$ and variance $\boldsymbol{\Sigma}_{S_1,n} := Var(\boldsymbol{\pi}_{i,n}|S_1)$

Let $\boldsymbol{P}_{n_2}$ denote the sample average:

$$\boldsymbol{P}_{n_2} := \frac{1}{n_2}\sum_{i=1}^{n_2} \boldsymbol{\pi}_{i,n} \tag{200}$$

For any measurable vector-valued function $\boldsymbol{b}_{S_1,n} = \boldsymbol{b}_{S_1,n}(S_1, n)$ mapping realizations of S_1 to R^{2+3L} for which constants M_b and c_b exist such that $\mathbb{P}[||\boldsymbol{b}_{S_1,n}||_1 \leq M_b] \longrightarrow 1$ and $\mathbb{P}[\boldsymbol{b}_{S_1,n}^T \boldsymbol{\Sigma}_{S_1,n} \boldsymbol{b}_{S_1,n} > c_b] \longrightarrow 1$:

$$\sup_{t\in\mathbb{R}} \left| \mathbb{P}\left[\sqrt{n_2} \frac{\boldsymbol{b}_{S_1,n}^T \left(\boldsymbol{P}_{n_2} - \boldsymbol{\mu}_{S_1,n}\right)}{\sqrt{\boldsymbol{b}_{S_1,n}^T \boldsymbol{\Sigma}_{S_1,n} \boldsymbol{b}_{S_1,n}}} \leq t \middle| S_1 \right] - \Phi(t) \right| \xrightarrow{p} 0 \tag{201}$$

Where $\Phi(\cdot)$ is the cumulative distribution of a standard normal random variable.

Lemma C.7. *Consider the sample averages that comprise the vector $\boldsymbol{P}_{n_2}$ defined in lemma C.6:*

$$P_{z_k,n_2} = \frac{1}{n_2}\sum_{i=1}^{n_2} \mathbf{1}\{Z_i = z_k\} \tag{202}$$

$$P_{z_k,x_l,n_2} = \frac{1}{n_2}\sum_{i=1}^{n_2} \mathbf{1}\{X_i = x_l, Z_i = z_k\} \tag{203}$$

$$P_{\xi,z_k,x_l,n_2} = \frac{1}{n_2}\sum_{i=1}^{n_2} \xi(Y_i, X_i, Z_i; z_0, x_l) \tag{204}$$

Let $\boldsymbol{P}_{n_2}$, $\boldsymbol{\mu}_{S_1,n}$ and $\boldsymbol{\Sigma}_{S_1,n}$ be defined as in Lemma C.6. Define the map $\boldsymbol{g} : \mathbb{R}^{2+3L} \rightarrow$

$\mathbb{R}^{3L-2}$ *as:*

$$\begin{aligned}\boldsymbol{g}(\boldsymbol{P}_{n_2}) = \Big(&r_{0,0}(\boldsymbol{P}_{n_2}), \ldots, r_{0,L-1}(\boldsymbol{P}_{n_2}),\\ &r_{1,0}(\boldsymbol{P}_{n_2}), \ldots, r_{1,L-1}(\boldsymbol{P}_{n_2}),\\ &g_0(\boldsymbol{P}_{n_2}), \ldots, g_{L-1}(\boldsymbol{P}_{n_2}),\Big)^T,\end{aligned}$$

where, for each $l = 0, \ldots, L-1$,

$$g_l(\boldsymbol{P}_{n_2}) := \frac{P_{\xi,z_0,x_l,n_2}}{P_{z_0,n_2}} + \frac{P_{\xi,z_1,x_l,n_2}}{1 - P_{z_0,n_2}}, \tag{205}$$

and, for $k \in \{0,1\}$ *and* $l = 0, \ldots, L-1$,

$$r_{k,l}(\boldsymbol{P}_{n_2}) := \begin{cases} \frac{P_{z_0,x_l,n_2}}{P_{z_0,n_2}} & \text{if } k = 0 \\ \frac{P_{z_1,x_l,n_2}}{1-P_{z_0,n_2}} & \text{if } k = 1 \end{cases}. \tag{206}$$

Let $\boldsymbol{J}_{S_1,n}$ *denote the Jacobian of* $\boldsymbol{g}$ *evaluated at* $\boldsymbol{\mu}_{S_1,n}$*, and define* $\boldsymbol{V}_{S_1,n} := \boldsymbol{J}_{S_1,n}\boldsymbol{\Sigma}_{S_1,n}\boldsymbol{J}_{S_1,n}^T$.

Suppose that $P[Z = z_k] > c_z > 0$ *for* $k = 0, 1$*. Then, for any fixed* $\boldsymbol{a} \in \mathbb{R}^{3L}$ *such that there exists* $c_a > 0$ *that satisfies* $\mathbb{P}\left[\boldsymbol{a}^T\boldsymbol{V}_{S_1,n}\boldsymbol{a} > c_a\right] \longrightarrow 1$*:*

$$\sup_{t\in\mathbb{R}} \left| \mathbb{P}\left[\frac{\boldsymbol{a}^T\sqrt{n_2}\left(\boldsymbol{g}(\boldsymbol{P}_{n_2}) - \boldsymbol{g}(\boldsymbol{\mu}_{S_1,n})\right)}{\sqrt{\boldsymbol{a}^T\boldsymbol{V}_{S_1,n}\boldsymbol{a}}} \leq t \right] - \Phi(t) \right| \xrightarrow{p} 0. \tag{207}$$

Lemma C.8.

For every $x_l \in \mathcal{X}$ and $z_k \in \mathcal{Z}$:

$$n_2 Var\big(\Psi_{x_l,n} - \Psi_{x_l} \big| S_1\big) = \sum_{k=0,1} \frac{P_{\xi,x_l|z_k,n_2}(1 - P_{\xi,x_l|z_k,n_2})}{\mathbb{P}[Z = z_k]} + o(1) \tag{208}$$

$$n_2 Cov\big(\Psi_{x_l,n} - \Psi_{x_l}, \Psi_{x_l,n} - \Psi_{x_l'} \big| S_1\big) = -\sum_{k=0,1} \frac{P_{\xi,x_l|z_k,n_2} P_{\xi,x_{l'}|z_k,n_2}}{\mathbb{P}[Z = z_k]} + o(1) \tag{209}$$

$$n_2 Cov\big(\Psi_{x_l,n} - \Psi_{x_l}, P_{x_{l'}|z_{k'},n_2} - P_{x_{l'}|z_{k'}} \big| S_1\big) = \frac{P_{\xi,x_l|z_k,n_2}(\mathbf{1}\{l = l'\} - P_{x_l|z_k,n_2})}{\mathbb{P}[Z = z_k]} + o(1) \tag{210}$$

$$n_2 Cov\big(P_{x_l|z_k,n_2} - P_{x_l|z_k}, P_{x_{l'}|z_{k'},n_2} - P_{x_{l'}|z_{k'}} \big| S_1\big) = \mathbf{1}\{k = k'\} \frac{P_{x_l|z_k}(\mathbf{1}\{l = l'\} - P_{x_l|z_{k'}})}{\mathbb{P}[Z = z_k]} + o(1) \tag{211}$$

Where $P_{\xi,x_l|z_k,n_2} := \mathbb{P}[\eta^*_{n_1}(X,Y) \neq z_k, X = x_l | Z = z_k, S_1]$.

Proof of lemma C.1.
Note that:

$$\begin{aligned}
\mathbb{E}[\tilde{\Delta}_{n_1}(y,x_l)] - \Delta(y,x_l) =& \mathbb{P}[X = x_l|Z = z_1]\mathbb{E}\big[\frac{K_n(y-Y)}{h_n^d}|X = x_l, Z = z_1\big] \qquad (212)\\
&- \mathbb{P}[X = x_l|Z = z_0]\mathbb{E}\big[\frac{K_n(y-Y)}{h_n^d}|X = x_l, Z = z_0\big] - \Delta(y,x_l)\\
=& \mathbb{P}[X = x_l|Z = z_1]\big(\mathbb{E}\big[\frac{K_n(y-Y)}{h_n^d}|X = x_l, Z = z_1\big] - f_{Y|X=x_l,Z=z_1}(y)\big) \qquad (213)\\
&- \mathbb{P}[X = x_l|Z = z_0]\big(\mathbb{E}\big[\frac{K_n(y-Y)}{h_n^d}|X = x_l, Z = z_0\big] - f_{Y|X=x_l,Z=z_0}(y)\big)
\end{aligned}$$

Where the first equality follows from the law of total probability and the definition of H_n, and the second equality from re-writing $\Delta(y,x_l)$ in terms of conditional distributions.

Then, an exact Taylor expansion of $f_{Y|X=x_l,Z=z_k}$, assumptions 5.2.1, 5.2.2, 5.1.2, and standard properties of kernel density estimators imply that if $\sup_{||u||=1} ||(\nabla^2 f_{Y|X=x_l,Z=z_k})u||$

is bounded, then:

$$\mathbb{E}\big[\frac{K_n(y-Y)}{h_n^d}|X=x_l,Z=z_k\big]-f_{Y|X=x_l,Z=z_k}(y)=O(h_n^2) \tag{214}$$

uniformly over y, x_l and z_k.

□

Proof of lemma C.2. First note that:

$$\mathbb{P}\Big[\Big|\Big|R_n(y,x_l)\Big|\Big|_\infty\geq r_n\Big]\leq\mathbb{P}\Big[\Big|\Big|R_n^1(y,x_l)\Big|\Big|_\infty+\Big|\Big|R_n^0(y,x_l)\Big|\Big|_\infty\geq r_n\Big] \tag{215}$$

$$\leq\mathbb{P}\Big[\Big|\Big|R_n^1(y,x_l)\Big|\Big|_\infty\geq\frac{r_n}{2}\Big]+\mathbb{P}\Big[\Big|\Big|R_n^0(y,x_l)\Big|\Big|_\infty\geq\frac{r_n}{2}\Big] \tag{216}$$

To bound the previous probabilities, define the class of functions:

$$\mathcal{K}_n=\Big\{\kappa_n(y,x,z;t,x_l,z_k)=\mathbf{1}\{X=x_l,Z=z_k\}K_n\big(t-y\big):x_l\in\mathcal{X},z_k\in\mathcal{Z},t\in\mathcal{Y}\Big\} \tag{217}$$

And the associated sub-density estimators:

$$f_{z_k,n}(y,x_l):=\Big[\frac{1}{n_1h_n^d}\sum_{i=1}^{n_1}\mathbf{1}\{X_i=x_l,Z_i=z_k\}K_n\big(y-Y_i\big)\Big] \tag{218}$$

Note that if $K(\cdot)$ is bounded, for any constant $B'>0$:

$$\mathbb{P}\Big[\Big|\Big|f_{z_k,n}(y,x_l)-\mathbb{E}f_{z_k,n}(y,x_l)\Big|\Big|_\infty\geq B'\Big]$$

$$=\mathbb{P}\Big[\Big|\Big|\frac{1}{n_1h_n^d}\sum_{i=1}^{n_1}\big(\kappa_n(Y_i,X_i,Z_i;y,x_l,z_k)-\mathbb{E}[\kappa_n(Y,X,Z;y,x_l,z_k)]\big)\Big|\Big|_\infty\geq B'\Big] \tag{219}$$

$$=\mathbb{P}\Big[\Big|\Big|\sum_{i=1}^{n_1}\big(\kappa_n(Y_i,X_i,Z_i;y,x_l,z_k)-\mathbb{E}[\kappa_n(Y,X,Z;y,x_l,z_k)]\big)\Big|\Big|_\infty\geq B'n_1h_n^d\Big] \tag{220}$$

Let $\sigma_{\kappa_n}^2=\sup_{\kappa\in\mathcal{K}_n}Var\big(\kappa(Y,X,Z)\big)=O(h_n^d)$ and $U=\sup_{\kappa\in\mathcal{K}_n}||\kappa||_\infty$. Note that for any $(t,x_l,z_k)\in\mathcal{Y}\times\mathcal{X}\times\mathcal{Z}$ the function $\kappa_n(\cdot,\cdot,\cdot;t,x_l,z_k)$ satisfies the conditions from assumption 5.2.1. Moreover, the map $(y,x,z)\mapsto\kappa_n(y,x,z;t,x_l,z_k)$ is jointly

measurable and, for a sufficiently large n, the condition $0 < \sigma^2_{\kappa_n} < U/2$ is satisfied. Thus, corollary 2.2. from Gine and Gillou 2002 implies that for any sufficiently large B' and n, positive constants C and L depending only on the VC class of the family $\mathcal{K}_n$ exist, such that for every λ such that for $B'h_n^d \leq \frac{\lambda}{U}\sigma^2_{\kappa_n}$:

$$\mathbb{P}\Big[\big|\big|f_{z_k,n}(y,x_l) - \mathbb{E}f_{z_k,n}(y,x_l)\big|\big|_\infty \geq B'\Big] \leq L\exp\Big(\frac{-\log(1+\lambda/(4L))}{\lambda L}\frac{n_1(B'h_n^d)^2}{\sigma^2_{\kappa_n}}\Big) = o(n_1^{-1}) \tag{221}$$

Moreover, since $K(\cdot)$ and f are bounded, the sequence $\big|\big|\mathbb{E}f_{z_k,n}(y,x_l)\big|\big|_\infty$ is also bounded. Thus, for some $B \geq B'$:

$$\mathbb{P}\Big[\big|\big|f_{z_k,n}(y,x_l)\big|\big|_\infty \geq B\Big] = \mathbb{P}\Big[\big|\big|f_{z_k,n}(y,x_l) - \mathbb{E}f_{z_k,n}(y,x_l) + \mathbb{E}f_{z_k,n}(y,x_l)\big|\big|_\infty \geq B\Big] \tag{222}$$

$$\leq \mathbb{P}\Big[\big|\big|f_{z_k,n}(y,x_l) - \mathbb{E}f_{z_k,n}(y,x_l)\big|\big|_\infty + \big|\big|\mathbb{E}f_{z_k,n}(y,x_l)\big|\big|_\infty \geq B\Big] \tag{223}$$

$$\leq \mathbb{P}\Big[\big|\big|f_{z_k,n}(y,x_l) - \mathbb{E}f_{z_k,n}(y,x_l)\big|\big|_\infty \geq B'\Big] \tag{224}$$

Now let P_k denote $\mathbb{P}[Z = z_k]$ and note that for any $B > 0$:

$$\mathbb{P}\Big[\Big|\frac{1}{P_{k,n_1}} - \frac{1}{P_k}\Big|B \geq \frac{r_n}{2}\Big] = \mathbb{P}\Big[\Big|\frac{P_k - P_{k,n_1}}{P_k P_{k,n_1}}\Big|B \geq \frac{r_n}{2}\Big] \tag{225}$$

$$= \mathbb{P}\Big[\Big|\frac{P_k - P_{k,n_1}}{P_k P_{k,n_1}}\Big|B \geq \frac{r_n}{2}, P_{k,n_1} \geq \frac{P_k}{2}\Big] + \mathbb{P}\Big[\Big|\frac{P_k - P_{k,n_1}}{P_k P_{k,n_1}}\Big|B \geq \frac{r_n}{2}, P_{k,n_1} < \frac{P_k}{2}\Big] \tag{226}$$

$$\leq \mathbb{P}\Big[|P_k - P_{k,n_1}| \geq \frac{P_k^2 r_n}{2B}\Big] + \mathbb{P}\Big[|P_{k,n_1} - P_k| \geq \frac{P_k}{2}\Big] \tag{227}$$

Where the second equality follows from the law of total probability and the inequality from the fact that $P_{k,n_1} < \frac{P_k}{2} \Rightarrow |P_{k,n_1} - P_k| \geq \frac{P_k}{2}$.

Thus, for a sufficiently large n_1, Hoeffding's inequality implies:

$$\mathbb{P}\Big[\Big|\frac{1}{P_{k,n_1}} - \frac{1}{P_k}\Big|B \geq \frac{r_n}{2}\Big] \leq 2\mathbb{P}\Big[|P_k - P_{k,n_1}| \geq \frac{P_k^2 r_n}{2B}\Big] \tag{228}$$

$$\leq 4\exp\big(-\frac{P_k^4}{2B^2}n_1 r_n^2\big) = o(n_1^{-1}). \tag{229}$$

Then, constants B, B' and λ must exist such that $B'h_n^d \leq \frac{\lambda}{U}\sigma^2_{\kappa_n}$, $B \geq B'$ and:

$$\mathbb{P}\Big[\Big|\Big|R_n^k(y,x_l)\Big|\Big|_\infty \geq \frac{r_n}{2}\Big] = \mathbb{P}\Big[\Big|\Big|\Big(\frac{1}{P_{k,n_1}} - \frac{1}{P_k}\Big) f_{z_k,n}(y,x_l)\Big|\Big|_\infty \geq \frac{r_n}{2}\Big] \tag{230}$$

$$= \mathbb{P}\Big[\Big|\frac{1}{P_{k,n_1}} - \frac{1}{P_k}\Big| ||f_{z_k,n}(y,x_l)||_\infty \geq \frac{r_n}{2}\Big] \tag{231}$$

$$\leq \mathbb{P}\Big[\Big|\frac{1}{P_{k,n_1}} - \frac{1}{P_k}\Big| B \geq \frac{r_n}{2}\Big] + \mathbb{P}\Big[||f_{z_k,n}(y,x_l)||_\infty \geq B\Big] \tag{232}$$

$$\leq 4\exp\big(-\frac{P_k^4}{2B^2} n_1 r_n^2\big) + L\exp\Big(\frac{-\log(1+\lambda/(4L))}{\lambda L}\frac{n_1 (B'h_n^d)^2}{\sigma^2_{\kappa_n}}\Big) \tag{233}$$

$$= o(n_1^{-1}) \tag{234}$$

Where the last inequality follows from 224 and 229. □

Proof of lemma C.3.

This lemma shows that the asymptotic behavior of the difference $\Delta_{n_1}(y,x_l)$ is determined by the kernel density estimator component, and that the difference $|\Delta_{n_1}(y,x_l) - \Delta_n(y,x_l)|$ exceeds the rate $\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$ with probability $o(n^{-1})$.

The result uses lemmas C.1 and C.2 which prove that the bias from kernel density estimation and the randomness from estimating conditional treatment and instrument probabilities are negligible at the rate $\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$.

For a given value of $x_l \in \mathcal{X}$, define:

$$H_n(y',x,z;x_l,y) = \Big[\frac{\mathbf{1}\{x = x_l, z = z_1\}}{\mathbb{P}[Z = z_1]} - \frac{\mathbf{1}\{x = x_l, z = z_0\}}{\mathbb{P}[Z = z_0]}\Big] K_n(y - y') \tag{235}$$

Then, $\Delta_{n_1}(y,x_l)$ can be decomposed into a main component depending on $H_n(y',x,z;x_l,y)$ plus a bias term and a residual that depends on the difference $\Big(\frac{1}{P_{1,n_1}} - \frac{1}{\mathbb{P}[Z=z_1]}\Big)$:

$$\Delta_{n_1}(y,x_l)-\Delta(y,x_l) = P_{1,n_1}^{-1}\Big[\frac{1}{n_1 h_n^d}\sum_{i=1}^{n_1}\mathbf{1}\{X_i=x_l,Z_i=z_1\}K_n\big(y-Y_i\big)\Big] \tag{236}$$

$$- P_{0,n_1}^{-1}\Big[\frac{1}{n_1 h_n^d}\sum_{i=1}^{n_1}\mathbf{1}\{X_i=x_l,Z_i=z_0\}K_n\big(y-Y_i\big)\Big]-\Delta(y,x_l)$$

$$= \frac{1}{n_1 h_n^d}\sum_{i=1}^{n_1} H_n(Y_i,X_i,Z_i;x_l,y)-\Delta(y,x_l) \tag{237}$$

$$+ \underbrace{\Big(\frac{1}{P_{1,n_1}}-\frac{1}{\mathbb{P}[Z=z_1]}\Big)\Big[\frac{1}{n_1 h_n^d}\sum_{i=1}^{n_1}\mathbf{1}\{X_i=x_l,Z_i=z_1\}K_n\big(y-Y_i\big)\Big]}_{:=R_n^1(y,x_l)}$$

$$- \underbrace{\Big(\frac{1}{P_{0,n_1}}-\frac{1}{\mathbb{P}[Z=z_0]}\Big)\Big[\frac{1}{n_1 h_n^d}\sum_{i=1}^{n_1}\mathbf{1}\{X_i=x_l,Z_i=z_0\}K_n\big(y-Y_i\big)\Big]}_{:=R_n^0(y,x_l)}$$

$$= \underbrace{\frac{1}{n_1 h_n^d}\sum_{i=1}^{n_1} H_n(Y_i,X_i,Z_i;x_l,y)}_{:=\tilde{\Delta}_{n_1}(y,x_l)} -\Delta(y,x_l)+R_n(y,x_l) \tag{238}$$

$$= \Big(\tilde{\Delta}_{n_1}(y,x_l)-\mathbb{E}[\tilde{\Delta}_{n_1}(Y,x_l)]\Big)+\underbrace{\Big(\mathbb{E}[\tilde{\Delta}_{n_1}(Y,x_l)]-\Delta(y,x_l)\Big)}_{Bias}+R_n(y,x_l) \tag{239}$$

Where $R_n(y,x_l)=R_n^0(y,x_l)-R_n^1(y,x_l)$.

Define the class of functions:

$$\mathcal{H}_n=\Big\{H_n(y,x,z;x,tl):x_l\in\mathcal{X},t\in\mathcal{Y}\Big\} \tag{240}$$

Let $\sigma_{H_n}^2=\sup_{H\in\mathcal{H}_n} Var\big(H(Y,X,Z)\big)=O(h_n^d)$ and $U=\sup_{H\in\mathcal{H}_n}||H||_\infty$. Note that for any $(t,x_l)\in\mathcal{Y}\times\mathcal{X}$ the function $H_n(\cdot,\cdot,\cdot;t,x_l)$ satisfies the conditions from assumption 5.2.1. Moreover, the map $(y,x,z)\mapsto H_n(y,x,z;x_l,t)$ is jointly measurable and, for a sufficiently large n_1, the condition $0<\sigma_{H_n}^2<U/2$ is satisfied. Thus, part 2 of corollary 2.2. from Gine and Gillou 2002 implies that positive constants C and L

depending only on the VC class of the family $\mathcal{H}_n$ exist, such that for every $C' > C$:

$$\mathbb{P}\Big[\Big|\Big|\sum_{i=1}^{n_1}\big(H_n(Y_i,X_i,Z_i;x_l,y)-\mathbb{E}[H_n(Y_i,X_i,Z_i;x_l,y)]\big)\Big|\Big|_\infty > C'\sqrt{n_1\sigma_{H_n}^2}\sqrt{\log\big(U\sigma_{H_n}^{-1}\big)}\Big] \tag{241}$$

$$\leq L\exp\Big\{-\frac{C'\log(1+C'/(4L))}{L}\log(U\sigma_{H_n}^{-1})\Big\} \tag{242}$$

$$= L\exp\Big\{-K(C')\log(U\sigma_{H_n}^{-1})\Big\} \tag{243}$$

$$= L\big[U\sigma_{H_n}^{-1}\big]^{-K(C')} \tag{244}$$

$$= M\big[\sigma_{H_n}^{K(C')}\big] \tag{245}$$

Where M is positive and depends on L and C', and $K'(C')$ denotes the term $\frac{C'\log(1+C'/(4L))}{L}$.

Moreover, by standard properties of kernel density estimators:

$$\mathbb{E}\big[H_n(Y,X,Z;x_l,y)^2\big] = O(h_n^d) \tag{246}$$

Thus, $\sigma_{H_n} = O(h_n^{d/2})$ and $\sigma_{H_n}^{K(C')} = O(n_1^{-\gamma dK(C')/2})$.

Observe that the function $K(C')$ is strictly increasing in C', and thus, for a sufficiently large C', the upper bound from 245 is $o(n_1^{-1})$.

For any fixed C' such that $\sigma_{H_n}^{K(C')} = o(n_1^{-1})$, elementary algebra shows that a positive constant $C_0 \geq C'$ and a fixed integer N exist such that $\frac{C_0}{2}\sqrt{n_1 h_n^d \log(h_n^{-1})} \geq C'\sqrt{n_1\sigma_{H_n}^2\log(U\sigma_{H_n}^{-1})}$ for any $n_1 \geq N$.

Fix such C_0 and consider the probability $\mathbb{P}\Big[\Big|\Big|\Delta_{n_1}(y,x_l)-\Delta(y,x_l)\Big|\Big|_\infty > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big]$. Then, lemma C.1 applied to the bias term in equation 239 yields:

$$\mathbb{P}\Big[\Big|\Big|\Delta_{n_1}(y,x_l)-\Delta(y,x_l)\Big|\Big|_\infty > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big]$$

$$= \mathbb{P}\Big[\Big|\Big|\Big(\tilde{\Delta}_{n_1}(y,x_l)-\mathbb{E}[\tilde{\Delta}_{n_1}(y,x_l)]\Big)+O(h_n^2)+R_n(y,x_l)\Big|\Big|_\infty > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big] \tag{247}$$

Lemma C.2 applied with $r_n = \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$ implies that:

$$\mathbb{P}\Big[\Big|\Big|\Delta_{n_1}(y,x_l) - \Delta(y,x_l)\Big|\Big|_\infty > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big] \tag{248}$$

$$= \mathbb{P}\Big[\Big|\Big|\Big(\tilde{\Delta}_{n_1}(y,x_l) - \mathbb{E}[\tilde{\Delta}_{n_1}(y,x_l)]\Big) + O(h_n^2) + o\Big(\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big)\Big|\Big|_\infty > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big] + o(n_1^{-1})$$

$$\leq \mathbb{P}\Big[\Big|\Big|\Big(\tilde{\Delta}_{n_1}(y,x_l) - \mathbb{E}[\tilde{\Delta}_{n_1}(y,x_l)]\Big)\Big|\Big|_\infty + \Big|\Big|o\Big(\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big)\Big|\Big|_\infty > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big] + o(n_1^{-1}) \tag{249}$$

Where the last line follows from assumption 5.2.2 (which implies $h_n^2 = o\Big(\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big)$) and the triangle inequality.

Then, for some integer N', if $n_1 \geq \max\{N, N'\}$, the bound from 245 yields:

$$\mathbb{P}\Big[\Big|\Big|\Delta_{n_1}(y,x_l) - \Delta(y,x_l)\Big|\Big|_\infty > C_0\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\Big] \tag{250}$$

$$\leq \mathbb{P}\Big[\Big|\Big|\sum_{i=1}^{n_1}\Big(H_n(Y_i,X_i,Z_i;x_l,y) - \mathbb{E}[H_n(Y_i,X_i,Z_i;x_l,y)]\Big)\Big|\Big|_\infty > \frac{C_0}{2}\sqrt{n_1 h_n^d \log(h_n^{-1})}\Big] + o(n_1^{-1})$$

$$= o(n_1^{-1}) \tag{251}$$

□

Proof of lemma C.4.

For a sufficiently large n:

$$\int_{\mathcal{M}_{x_l}(r_n)} \Big(\mathbf{1}\{\eta^*(y,x_l) \neq \eta^*_{n_1}(y,x_l)\}|\Delta(y,x_l)|\Big)dy \leq \int_{\mathcal{M}_{x_l}(r_n)} \Big(\mathbf{1}\{\eta^*(y,x_l) \neq \eta^*_{n_1}(y,x_l)\}r_n\Big)dy \tag{252}$$

$$= \big(\mu\{\mathcal{M}_{x_l}(r_n)\}\big)r_n \tag{253}$$

$$\leq B r_n^{1+\alpha} = O(r_n^{1+\alpha}) \tag{254}$$

Where the firs inequality follows from the definition of the set $\mathcal{M}_{x_l}(r_n)$, and the second inequality from direct application of assumption 5.1.3.

For part 2, observe that:

$$\int_{\mathcal{M}_{x_l}(r_n)} \int_{\mathcal{M}_{x_l}(r_n)} \mathbb{P}[M(y,x_l)]\mathbb{P}[M(y,x_l)]\big|\Delta(y,x_l)\big|\big|\Delta(y',x_l)\big|dydy' \leq \big(\mu\{\mathcal{M}_{x_l}(r_n)\}\big)^2 r_n^2 dydy' \tag{255}$$

$$= O(r_n^{2\alpha+2}) \tag{256}$$

□

Proof of lemma C.5.

$$\mathbb{E}\Big[L_{n_2}(\eta^*_{n_1},x_l)|S_1\Big] \tag{257}$$

$$= \mathbb{E}\Big[\frac{1}{n_2}\sum_{i=1}^{n_2}\Big(\frac{\mathbf{1}\{\eta^*_{n_1}(Y_i,X_i)=0,Z_i=z_1,X_i=x_l\}}{P_{1,n_2}} + \frac{\mathbf{1}\{\eta^*_{n_1}(Y_i,X_i)=1,Z_i=z_0,X_i=x_l\}}{P_{0,n_2}}\Big)\Big|S_1\Big] \tag{258}$$

$$= \mathbb{E}\Big[\mathbb{E}\Big\{\frac{1}{n_2}\sum_{i=1}^{n_2}\Big(\frac{\mathbf{1}\{\eta^*_{n_1}(Y_i,X_i)=0,Z_i=z_1,X_i=x_l\}}{P_{1,n_2}} + \frac{\mathbf{1}\{\eta^*_{n_1}(Y_i,X_i)=1,Z_i=z_0,X_i=x_l\}}{P_{0,n_2}}\Big)\Big|\{Z_i\}_{i=1}^{n_2}\Big\}\Big|S_1\Big] \tag{259}$$

$$= \mathbb{E}\Big[\mathbb{E}\Big\{\frac{1}{n_{2,z_1}}\sum_{i\in S_2,Z_i=z_1}^{n_2}\mathbf{1}\{\eta^*_{n_1}(Y_i,X_i)=0,Z_i=z_1,X_i=x_l\} + \frac{1}{n_{2,z_0}}\sum_{i\in S_2,Z_i=z_0}^{n_2}\mathbf{1}\{\eta^*_{n_1}(Y_i,X_i)=1,Z_i=z_0,X_i=x_l\}\Big|\{Z_i\}_{i=1}^{n_2},S_1\Big\}\Big|S_1\Big] \tag{260}$$

$$= \mathbb{E}\Big[\mathbb{E}\Big\{\mathbb{P}[\eta^*_{n_1}(Y,X)=1,X=x_l|Z=z_0,S_1] + \mathbb{P}[\eta^*_{n_1}(Y,X)=0,X=x_l|Z=z_1,S_1]\Big|\{Z\}_{i=1}^{n_2},S_1\Big\}\Big|S_1\Big] \tag{261}$$

$$= \mathbb{P}[\eta^*_{n_1}(Y,X)=1,X=x_l|Z=z_0,S_1] + \mathbb{P}[\eta^*_{n_1}(Y,X)=0,X_i=x_l|Z=z_1,S_1] \tag{262}$$

$$= \frac{\mathbb{P}[\eta^*_{n_1}(Y,X)=1,Z=z_0,X=x_l|S_1]}{\mathbb{P}[Z=z_0]} + \frac{\mathbb{P}[\eta^*_{n_1}(Y,X)=0,Z=z_1,X=x_l|S_1]}{\mathbb{P}[Z=z_1]} \tag{263}$$

$$= L(\eta^*_{n_1},x_l) \tag{264}$$

□

Proof of lemma C.6.

Note that all the coordinates of $\boldsymbol{P}_i$ are indicator variables, thus, its support is bounded, and all its conditional moments exist.

Given the measurable function $\boldsymbol{b}_{S_1,n}$, define:

$$U_{i,n} = \boldsymbol{b}_{S_1,n}^T\big(\boldsymbol{\pi}_{i,n} - \boldsymbol{\mu}_{S_1,n}\big) \tag{265}$$

By definition, $\mathbb{E}[U_{i,n}|S_1] = 0$ and $\sigma_{b,S_1,n} := Var(U_i|S_1) = \boldsymbol{b}_{S_1,n}^T\boldsymbol{\Sigma}_{S_1,n}\boldsymbol{b}_{S_1,n}$.

Now consider the standardized sum:

$$\sqrt{n_2}\frac{\boldsymbol{b}_{S_1,n}^T\big(\boldsymbol{\pi}_{n_2} - \boldsymbol{\mu}_{S_1,n}\big)}{\sqrt{\boldsymbol{b}_{S_1,n}^T\boldsymbol{\Sigma}_{S_1,n}\boldsymbol{b}_{S_1,n}}} = \frac{1}{\sqrt{n_2}}\sum_{i=1}^{n_2}\frac{U_{i,n}}{\sigma_{b,S_1,n}} \tag{266}$$

Let $s_n^2 = \sum_{i=1}^{n_2}\sigma_{b,S_1,n}^2 = n_2\sigma_{b,S_1,n}^2$. Then, on the events $\sigma_{b,S_1,n}^2 > c_b$ and $||\boldsymbol{b}_{S_1,n}||_1 \leq M_b$, for any fixed $\epsilon > 0$ and a sufficiently large n:

$$\frac{1}{s_n^2}\sum_{i=1}^{n_2}\mathbb{E}\big[U_{i,n}^2\mathbf{1}\{|U_i| \geq \epsilon s_n\}|S_1\big] \leq \frac{4||\boldsymbol{b}_{S_1,n}||_1^2}{s_n^2}\sum_{i=1}^{n_2}\mathbb{E}\big[\mathbf{1}\{|U_i| \geq \epsilon s_n\}|S_1\big] \tag{267}$$

$$= \frac{4M_b^2}{s_n^2}\sum_{i=1}^{n_2}\mathbb{P}[|U_i| \geq \epsilon s_n|S_1] \tag{268}$$

$$= 0 \tag{269}$$

Where the inequality follows from the fact that $|U_2| \leq 2||\boldsymbol{b}_{S_1,n}||_1 \leq 2M_b$ which in turn implies that $\mathbb{P}[|U_i| \geq \epsilon s_n|S_1] = 0$.

Thus, the Lindeberg condition holds which implies that, conditional on S_1, $\sigma_{b,S_1,n}^2 > c_b$ and $||\boldsymbol{b}_{S_1,n}||_1 \leq M_b$, the CDF of the sum $\frac{1}{\sqrt{n_2}}\sum_{i=1}^{n_2}\frac{U_{i,n}}{\sigma_{b,S_1,n}}$ converges uniformly to $\Phi(\cdot)$.

Hence:

$$\sup_{t\in\mathbb{R}}\left|\mathbb{P}\left[\sqrt{n_2}\frac{\boldsymbol{b}_{S_1,n}^T\big(\boldsymbol{P}_{n_2} - \boldsymbol{\mu}_{S_1,n}\big)}{\sqrt{\boldsymbol{b}_{S_1,n}^T\boldsymbol{\Sigma}_{S_1,n}\boldsymbol{b}_{S_1,n}}} \leq t\middle|S_1\right] - \Phi(t)\right| \xrightarrow{p} 0 \tag{270}$$

□

Proof of Lemma C.7. The proof follows from the Delta method and Lemma C.6.

First note that, by construction:

$$g_l(\boldsymbol{\pi}_{n_2}) = \frac{\pi_{\xi,z_0,x_l,n_2}}{\pi_{z_0,n_2}} + \frac{\pi_{\xi,z_1,x_l,n_2}}{\pi_{z_1,n_2}} = L_{n_2}(\eta^*_{n_1}, x_l), \tag{271}$$

and

$$g_l(\boldsymbol{\mu}_{S_1,n}) = \frac{\mu_{\xi,z_0,x_l,S_1,n}}{P[Z = z_0]} + \frac{\mu_{\xi,z_1,x_l,S_1,n}}{P[Z = z_1]} = L(\eta^*_{n_1}, x_l). \tag{272}$$

Similarly:

$$r_{k,l}(\boldsymbol{\pi}_{n_2}) = \frac{\pi_{z_k,x_l,n_2}}{\pi_{z_k,n_2}} \tag{273}$$

is the sample analogue of $P[X = x_l \mid Z = z_k]$, and:

$$r_{k,l}(\boldsymbol{\mu}_{S_1,n}) = \frac{P[X = x_l, Z = z_k]}{P[Z = z_k]} = P[X = x_l \mid Z = z_k]. \tag{274}$$

Since $P[Z = z_k] > c_z > 0$ for $k = 0, 1$, the map $\boldsymbol{g}$ is continuously differentiable in a neighborhood of $\boldsymbol{\mu}_{S_1,n}$ with probability approaching one. Moreover, by the law of large numbers conditional on S_1,

$$\boldsymbol{\Pi}_{n_2} - \boldsymbol{\mu}_{S_1,n} = o_p(1). \tag{275}$$

Therefore, a first-order Taylor expansion yields

$$\sqrt{n_2}\,(\boldsymbol{g}(\boldsymbol{\Pi}_{n_2}) - \boldsymbol{g}(\boldsymbol{\mu}_{S_1,n})) = \boldsymbol{J}_{S_1,n}\sqrt{n_2}\,(\boldsymbol{\Pi}_{n_2} - \boldsymbol{\mu}_{S_1,n}) + \boldsymbol{R}_n, \tag{276}$$

where, for every fixed $\boldsymbol{a} \in \mathbb{R}^{3L}$, $\boldsymbol{a}^T\boldsymbol{R}_n = o_p(1)$.

Fix $\boldsymbol{a} \in \mathbb{R}^{3L}$ and define

$$\boldsymbol{b}_{S_1,n} := \boldsymbol{J}^T_{S_1,n}\boldsymbol{a}. \tag{277}$$

Because $\boldsymbol{J}_{S_1,n}$ is a function of $\boldsymbol{\mu}_{S_1,n}$, and $\boldsymbol{\mu}_{S_1,n}$ is measurable with respect to S_1, the vector $\boldsymbol{b}_{S_1,n}$ is S_1-measurable. Furthermore, since all coordinates of $\boldsymbol{\mu}_{S_1,n}$ are probabilities and $P[Z = z_k]$ is bounded away from zero, there exists a finite constant

M_b such that

$$\mathbb{P}\left[\|\boldsymbol{b}_{S_1,n}\|_1 \leq M_b\right] \longrightarrow 1. \tag{278}$$

Also,

$$\boldsymbol{b}_{S_1,n}^T \boldsymbol{\Sigma}_{S_1,n} \boldsymbol{b}_{S_1,n} = \boldsymbol{a}^T \boldsymbol{J}_{S_1,n} \boldsymbol{\Sigma}_{S_1,n} \boldsymbol{J}_{S_1,n}^T \boldsymbol{a} = \boldsymbol{a}^T \boldsymbol{V}_{S_1,n} \boldsymbol{a}. \tag{279}$$

By assumption,

$$\mathbb{P}\left[\boldsymbol{b}_{S_1,n}^T \boldsymbol{\Sigma}_{S_1,n} \boldsymbol{b}_{S_1,n} > c_a\right] \longrightarrow 1. \tag{280}$$

Thus, Lemma C.6 applies to the random direction $\boldsymbol{b}_{S_1,n} = \boldsymbol{J}_{S_1,n}^T \boldsymbol{a}$ and implies

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P}\left[\frac{\boldsymbol{a}^T \boldsymbol{J}_{S_1,n} \sqrt{n_2} (\boldsymbol{\Pi}_{n_2} - \boldsymbol{\mu}_{S_1,n})}{\sqrt{\boldsymbol{a}^T \boldsymbol{V}_{S_1,n} \boldsymbol{a}}} \leq t \middle| S_1 \right] - \Phi(t) \right| \xrightarrow{p} 0. \tag{281}$$

Finally, since:

$$\boldsymbol{a}^T \sqrt{n_2} \left(\boldsymbol{g}(\boldsymbol{\Pi}_{n_2}) - \boldsymbol{g}(\boldsymbol{\mu}_{S_1,n}) \right) = \boldsymbol{a}^T \boldsymbol{J}_{S_1,n} \sqrt{n_2} (\boldsymbol{\Pi}_{n_2} - \boldsymbol{\mu}_{S_1,n}) + o_p(1), \tag{282}$$

and since: $\boldsymbol{a}^T \boldsymbol{V}_{S_1,n} \boldsymbol{a}$ is bounded away from zero with probability approaching one, Slutsky's theorem gives:

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P}\left[\frac{\boldsymbol{a}^T \sqrt{n_2} \left(\boldsymbol{g}(\boldsymbol{\Pi}_{n_2}) - \boldsymbol{g}(\boldsymbol{\mu}_{S_1,n}) \right)}{\sqrt{\boldsymbol{a}^T \boldsymbol{V}_{S_1,n} \boldsymbol{a}}} \leq t \middle| S_1 \right] - \Phi(t) \right| \xrightarrow{p} 0. \tag{283}$$

Now set $T_n := \frac{\boldsymbol{a}^T \sqrt{n_2} \left(\boldsymbol{g}(\boldsymbol{\Pi}_{n_2}) - \boldsymbol{g}(\boldsymbol{\mu}_{S_1,n}) \right)}{\sqrt{\boldsymbol{a}^T \boldsymbol{V}_{S_1,n} \boldsymbol{a}}}$ and note that:

$$\begin{aligned} \sup_t \left| \mathbb{P}[T_n = t] - \boldsymbol{\Phi}(t) \right| &= \sup_t \left| \mathbb{E}[\mathbb{P}[T_n \leq t | S_1]] - \boldsymbol{\Phi}(t) \right| && (284) \\ &\leq \mathbb{E}\left[\sup_t \left| \mathbb{P}[T_n \leq t | S_1] - \boldsymbol{\Phi}(t) \right| \right] && (285) \end{aligned}$$

Thus, 283 becomes:

$$\sup_{t\in\mathbb{R}}\left|\mathbb{P}\left[\frac{\boldsymbol{a}^T\sqrt{n_2}\left(\boldsymbol{g}(\boldsymbol{\Pi}_{n_2})-\boldsymbol{g}(\boldsymbol{\mu}_{S_1,n})\right)}{\sqrt{\boldsymbol{a}^T\boldsymbol{V}_{S_1,n}\boldsymbol{a}}}\leq t\right]-\Phi(t)\right|\xrightarrow{p}0. \tag{286}$$

This proves the claim. □

Proof of lemma C.8.

First consider $Var(A_{1,n}(x_l))$:

$$\begin{aligned}Var(A_{1,n}(x_l)) &= Var(\mathbb{E}[A_{1,n}(x_l)|S_1]) + \mathbb{E}(Var[A_{1,n}(x_l)|S_1]) &(287)\\ &= \mathbb{E}(Var[A_{1,n}(x_l)|S_1]) &(288)\end{aligned}$$

Where the first line follows from the law of total variance and the second from lemma 5.1.

Note that:

$$\begin{aligned}E[L_{n_2}(\eta^*_{n_1},x_l)|S_1] =&\mathbb{E}\Big[\sum_{i=1}^{n_2}\Big(\sum_{k=0,1}\frac{1}{n_{2,k}}\xi(Y_i,X_i,Z_i;z_k,x_l)\Big)\Big|S_1\Big] &(289)\\ =&\mathbb{E}\Big\{\mathbb{E}\Big[\sum_{i=1}^{n_2}\Big(\sum_{k=0,1}\frac{1}{n_{2,k}}\xi(Y_i,X_i,Z_i;z_k,x_l)\Big)\Big|\{Z_i\}_{i\in S_2},S_1\Big]\Big|S_1\Big\} &(290)\\ =&\mathbb{E}\Big\{\sum_{k=0,1}\mathbb{E}[\xi(Y_i,X_i,Z_i;z_k,x_l)|Z_i=z_k,S_1]\Big|S_1\Big\} &(291)\\ =&\mathbb{P}\big[\eta^*_{n_1}(Y,X,Z;z_0,x_l)\neq k,X=x_k|Z=z_0,S_1\big] &(292)\\ &+\mathbb{P}\big[\eta^*_{n_1}(Y,X,Z;z_1,x_l)\neq k,X=x_k|Z=z_1,S_1\big]\end{aligned}$$

Where the second line comes from conditioning on realized instrument values, the third from the fact that observations are i.i.d. conditional on instrument values, and the last from the definition of $\xi(\cdot)$.

And:

$$\mathbb{E}\Big[L_{n_2}(\eta^*_{n_1}, x_l)^2|S_1\Big] =\mathbb{E}\Big[\sum_{i=1}^{n_2}\sum_{j=1}^{n_2}\Big(\sum_{k=0,1}\frac{1}{n_{2,k}}\xi(Y_i, X_i, Z_i; z_k, x_l)\Big)\Big(\sum_{k=0,1}\frac{1}{n_{2,k}}\xi(Y_j, X_j, Z_j; z_k, x_l)\Big)\Big|S_1\Big] \tag{293}$$

$$=\mathbb{E}\Big[\sum_{i=1}^{n_2}\Big(\sum_{k=0,1}\frac{1}{n^2_{2,k}}\xi(Y_i, X_i, Z_i; z_k, x_l) \tag{294}$$

$$+\sum_{j\neq i, j\in S_2}\sum_{k'=0,1}\frac{1}{n_{2,k}n_{2,k'}}\xi(Y_i, X_i, Z_i; z_k, x_l)\xi(Y_j, X_j, Z_j; z_k, x_l)\Big)\Big|S_1\Big]$$

Now expand the second term and condition on instrument realizations:

$$\mathbb{E}\Big[\sum_{i=1}^{n_2}\sum_{k=0,1}\sum_{j\neq i, j\in S_2}\sum_{k'=0,1}\frac{1}{n_{2,k}n_{2,k'}}\xi(Y_i, X_i, Z_i; z_k, x_l)\xi(Y_j, X_j, Z_j; z_k, x_l)\Big)\Big|S_1\Big] \tag{295}$$

$$=\mathbb{E}\Big\{\mathbb{E}\Big[\sum_{i=1}^{n_2}\sum_{k=0,1}\sum_{j\neq i, j\in S_2}\sum_{k'=0,1}\frac{1}{n_{2,k}n_{2,k'}}\xi(Y_i, X_i, Z_i; z_k, x_l) \tag{296}$$

$$\times\,\xi(Y_j, X_j, Z_j; z_k, x_l)\Big)\Big|\{Z_i\}_{i\in S_2}, S_1\Big]\Big|S_1\Big\}$$

$$=\mathbb{E}\Big[\frac{n_{2,0}(n_{2,0}-1)}{n^2_{2,0}}\mathbb{E}\big[\xi(Y, X, Z; z_0, x_l)|Z=z_0, S_1\big]^2+\frac{n_{2,1}(n_{2,1}-1)}{n^2_{2,1}}\mathbb{E}\big[\xi(Y, X, Z; z_1, x_l)|Z=z_1, S_1\big]^2 \tag{297}$$

$$+2\big(\mathbb{E}\big[\xi(Y, X, Z; z_0, x_l)|Z=z_0, S_1\big]\mathbb{E}\big[\xi(Y, X, Z; z_1, x_l)|Z=z_1, S_1\big]\big)\Big|S_1\Big]$$

Combining equations 293, 295 and 298 yields:

$$Var(A_{1,n}(x_l)|S_1) =\mathbb{E}\Big[\frac{n_2}{n_{2,k}}\sum_{k=0,1}\mathbb{P}[\eta^*_{n_1}(X, Y)\neq z_k, X=x_l|Z=z_k, S_1] \tag{298}$$

$$\times\,\big(1-\mathbb{P}[\eta^*_{n_1}(X, Y)\neq z_k, X=x_l|Z=z_k, S_1]\big)\Big|S_1\Big]$$

Moreover, the terms $\mathbb{P}[\eta^*_{n_1}(X, Y)\neq z_k, X=x_l|Z=z_k]\times\big(1-\mathbb{P}[\eta^*_{n_1}(X, Y)\neq z_k, X=x_l|Z=z_k]\big)4$ for $k=0,1$ are bounded and the difference $\mathbb{E}\big[\frac{n_2}{n_{2,k}}\big]-\frac{1}{\mathbb{P}[Z=z_k]}$ is $o(1)$ and

independent from S_1, thus:

$$Var(A_{1,n}(x_l)|S_1) = \mathbb{E}\Big[\sum_{k=0,1} \frac{\mathbb{P}[\eta^*_{n_1}(X,Y) \neq z_k, X = x_l | Z = z_k, S_1]}{\mathbb{P}[Z = z_k]} \tag{299}$$

$$\times \big(1 - \mathbb{P}[\eta^*_{n_1}(X,Y) \neq z_k, X = x_l | Z = z_k, S_1]\big)\Big| S_1\Big] + o(1)$$

$$= \sum_{k=0,1} \frac{\mathbb{P}[\eta^*_{n_1}(X,Y) \neq z_k, X = x_l | Z = z_k, S_1]}{\mathbb{P}[Z = z_k]} \tag{300}$$

$$\times \big(1 - \mathbb{P}[\eta^*_{n_1}(X,Y) \neq z_k, X = x_l | Z = z_k, S_1]\big) + o(1) \tag{301}$$

$$= \frac{P_{\xi,x_l|z_0,n_2}(1 - P_{\xi,x_l|z_0,n_2})}{\mathbb{P}[Z = z_0]} + \frac{P_{\xi,x_l|z_1,n_2}(1 - P_{\xi,x_l|z_1,n_2})}{\mathbb{P}[Z = z_1]} + o(1) \tag{302}$$

Where $P_{\xi,x_l|z_k,n_2} := \mathbb{P}[\eta^*_{n_1}(X,Y) \neq z_k, X = x_l | Z = z_k, S_1]$ is the missclassification probability for the pre-trained classifier associated with $X = x_l$, conditional on $Z = z_k$

To derive $Cov(A_{1,n}(x_l), A_{1,n}(x_{l'}))$ consider the variance of the sum $A_{1,n}(x_l) + A_{1,n}(x_{l'})$, define $\zeta(Y, X, Z; z_l, x_l, x'_l) := \xi(Y, X, Z; z_l, x_l) + \xi(Y, X, Z; z_l, x'_l)$ and note that:

$$\mathbb{E}\big[n_2^{-1/2}\big(L_{n_2}(\eta^*_{n_1}, x_l) + L_{n_2}(\eta^*_{n_1}, x_{l'})\big)|S_1\big] = \mathbb{E}\Big[\sum_{i=1}^{n_2}\Big(\sum_{k=0,1}\frac{1}{n_{2,k}}\zeta(Y_i, X_i, Z_i; z_k, x_l)\Big)|S_1\Big] \tag{303}$$

Then, a derivation analogous to the variance of $A_{1,n}(x_l)$ yields:

$$Var(A_{1,n}(x_l) + A_{1,n}(x_{l'})|S_1) = \frac{(P_{\xi,x_l|z_0,n_2} + P_{\xi,x_{l'}|z_0,n_2})(1 - P_{\xi,x_l|z_0,n_2} - P_{\xi,x_{l'}|z_0,n_2})}{\mathbb{P}[Z = z_0]} \tag{304}$$

$$+ \frac{(P_{\xi,x_l|z_1,n_2} + P_{\xi,x_{l'}|z_1,n_2})(1 - P_{\xi,x_l|z_1,n_2} - P_{\xi,x_{l'}|z_1,n_2})}{\mathbb{P}[Z = z_1]} + o(1)$$

Thus:

$$Cov(A_{1,n}(x_l), A_{1,n}(x_{l'})|S_1) = -\frac{P_{\xi,x_l|z_0,n_2}P_{\xi,x_{l'}|z_0,n_2}}{\mathbb{P}[Z = z_0]} - \frac{P_{\xi,x_l|z_1,n_2}P_{\xi,x_{l'}|z_1,n_2}}{\mathbb{P}[Z = z_1]} + o(1) \tag{305}$$

Now consider the conditional treatment probability estimators $P_{x_l|z_k,n_2} := \frac{P_{z_k,n_2}}{P_{z_k,x_l,n_2}}$. Note that:

$$\mathbb{E}\Big[L_{n_2}(\eta^*_{n_1}, x_l)P_{x_{l'}|z_k,n_2}\Big] \tag{306}$$

$$=\mathbb{E}\Big[\sum_{i=1}^{n_2}\sum_{j=1}^{n_2}\Big(\sum_{k'=0,1}\frac{1}{n_{2,k}}\xi(Y_i, X_i, Z_i; z_{k'}, x_l)\Big)\Big(\frac{1}{n_{2,k}}\mathbf{1}\{X_j = x_{l'}, Z_j = z_k\}\Big)\Big|S_1\Big]$$

$$=\mathbb{E}\Big[\sum_{i=1}^{n_2}\Big(\frac{\mathbf{1}\{l=l'\}}{n_{2,k}^2}\xi(Y_i, X_i, Z_i; z_k, x_l) \tag{307}$$

$$+\sum_{j\neq i, j\in S_2}\sum_{k'=0,1}\frac{1}{n_{2,k}n_{2,k'}}\xi(Y_i, X_i, Z_i; z_{k'}, x_l)\mathbf{1}\{X_j = x_{l'}, Z_j = z_k\}\Big)\Big|S_1\Big]$$

Again, expand the second term and condition on instrument realizations to obtain:

$$\mathbb{E}\Big[\sum_{i=1}^{n_2}\sum_{j\neq i, j\in S_2}\sum_{k'=0,1}\frac{1}{n_{2,k}n_{2,k'}}\xi(Y_i, X_i, Z_i; z_{k'}, x_l)\mathbf{1}\{X_j = x_{l'}, Z_j = z_k\}\Big|S_1\Big]\Big] \tag{308}$$

$$= \mathbb{E}\Big[\sum_{k'=0,1}\frac{(n_{2,k} - \mathbf{1}\{k=k'\})n_{2,k'}}{n_{2,k}n_{2,k'}}P_{\xi,x_l|z_k,n_2}P_{x'_l|z_0,n_2}\Big|S_1\Big]$$

Therefore:

$$Cov\Big(A_{1,n}(x_l), \sqrt{n_2}\big(P_{x_{l'}|z_{k'},n_2} - P_{x_{l'}|z_{k'}}\big)\Big) = \frac{P_{\xi,x_l|z_k,n_2}(\mathbf{1}\{l=l'\} - P_{x_l|z_k,n_2})}{\mathbb{P}[Z=z_k]} + o(1) \tag{309}$$

Finally, the delta method yields:

$$Cov\Big(\sqrt{n_2}\big(P_{x_l|z_k,n_2} - P_{x_l|z_k}\big), \sqrt{n_2}\big(P_{x_{l'}|z_{k'},n_2} - P_{x_{l'}|z_{k'}}\big)\Big) \tag{310}$$

$$= \mathbf{1}\{k=k'\}\frac{\mathbf{1}\{l=l'\}P_{x_l|z_k} - P_{x_l|z_k}P_{x_l|z_{k'}}}{\mathbb{P}[Z=z_k]} + o(1)$$

Equations 302, 305, 309 and 310 characterize the entire variance-covariance matrix of the vector $\mathbf{\Pi}_{n_2}$ conditional on S_1.

□

C.3 Estimation algorithm 2

1. Choose a measurable set $D \subset \mathbb{R}^d$.
2. Partition the sample in two $\tilde{K}$ folds $\kappa = 1, ..., \tilde{K}$ of size $\tilde{n} = \left\lfloor n/\tilde{K} \right\rfloor$.
3. For every $\kappa = 1, ..., \tilde{K}$:
   (a) In J_κ^C, train a non-parametric classifier $\eta^d_{\kappa,n}(y, x)$ as follows:
      i. Define:

$$\phi_{z_k,\kappa,n}(y, x_l) := P^{-1}_{k,\kappa,n}\Big[\frac{1}{(\tilde{K}-1)\tilde{n}h_n^d}\sum_{i=J_\kappa^C}\mathbf{1}\{X_i = x_l, Z_i = z_k\}K_n\big(y - Y_i\big)\Big] \tag{311}$$

      Where $n_{\kappa,1} = |\{i \in J_\kappa^C : Z_i = z_k\}|$ and $P_{k,n_\kappa} = n_{k,\kappa}/\tilde{n}$.
      ii. Then, define:

$$\Delta_{\kappa,n}(y, x_l) := \phi_{z_1,\kappa,n}(y, x_l) - \phi_{z_0,\kappa,n}(y, x_l) \tag{312}$$

      iii. And:

$$\eta^d_{\kappa,n}(y, x_l) := \begin{cases} \mathbf{1}\{\Delta_{\kappa,n}(y, x_l) \geq 0\} & \text{if } |\Delta_{\kappa,n}(y, x_l)| \geq d_n \\ \mathbf{1}\{y \in D\} & \text{otherwise} \end{cases} \tag{313}$$

   (b) In J_κ , estimate the probabilities $\mathbb{P}[Z = z_k]$ for $k = 0, 1$. Denote the estimators as $P_{k,\kappa,n}$.
   (c) In J_κ, evaluate the weighted empirical loss:

$$L_\kappa(\eta^d_{\kappa,n}, x_l) := \frac{1}{\tilde{n}}\sum_{i\in J_\kappa}\Big(\frac{\mathbf{1}\{\eta^d_{\kappa,n}(Y_i, X_i) = 1, Z_i = z_0, X_i = x_l\}}{P_{0,\kappa,n}} + \frac{\mathbf{1}\{\eta^d_{\kappa,n}(Y_i, X_i) = 0, Z_i = z_1, X_i = x_l\}}{P_{1,\kappa,n}}\Big) \tag{314}$$

4. Calculate:

$$\Psi^d_{x_l} = \frac{1}{\tilde{K}}\sum_{\kappa=1}^{\tilde{K}} L_\kappa(\eta^d_{\kappa,n}, x_l) \tag{315}$$

C.4 Proofs for subsection ??

Proposition C.2.

Decompose:

$$\sqrt{\tilde{n}}\big(L_\kappa(\eta^d_{\kappa,n}, x_l) - \Psi_{x_l}\big) = \underbrace{\sqrt{\tilde{n}}\big(L_\kappa(\eta^d_{\kappa,n}, x_l) - L(\eta^d_{\kappa,n}, x_l)\big)}_{:=A^d_{1,n}(x_l)} + \underbrace{\sqrt{\tilde{n}}\big(L(\eta^d_{\kappa,n}, x_l) - L(\eta^d, x_l)\big)}_{:=A^d_{2,n}(x_l)} \tag{316}$$

1. $A^d_{2,n} = o_p(1)$.
2. $Cov(A^d_{1,n}, A^d_{2,n}) = 0$.
3. $Var(A^d_{2,n}) = o(1)$.

Proof of proposition C.2.

The proof follows by the same arguments from C.1 but now considering a region around the contact set that shrinks at a strictly slower rate than $r_n = \sqrt{\frac{\log(n_1)}{n_1 h_n}}$. In this region, the classifier is intentionally defined so that it incurs in classification mistakes. The only additional requirement is a stronger margin condition necessary to bound the potential additional loss from worsening the classifier.

As in the proof of C.1: for any possible realization of the classifier $\eta^d_{n_1}$ i) the classification loss from the contact set $\mathcal{C}_{x_l}$ is invariant, and ii) the margin condition implies that the classification loss near the boundary of the contact set is $\{y \in \mathcal{Y} : |\Delta(y, x_l)| \leq d_n\}$ is of order $o(n^{-1})$. Thus, asymptotic properties of the classifier $\eta^d_{n_1}$ might only differ from those of $\eta^*_{n_1}$ if its missclassification loss far from the boundary of the contact set is non-negligible.

For any $y \in \mathcal{Y}$ such that $|\Delta(y, x_l)| \geq d_n$, missclassification might occur if $|\Delta_(y, x_l) - \Delta(y, x_l)| \geq d_n \geq r_n$ or if $|\Delta_n(y, x_l)| < d_n$. If $|\Delta_n(y, x_l)| < d_n$, there are two cases: i) when $|\Delta(y, x_l)| - r_n \geq d_n$, and ii) when $d_n \geq |\Delta(y, x_l)| - r_n$.

All the cases listed but the last imply $|\Delta(y, x_l) - \Delta_n(y, x_l)| \geq r_n$, and thus, are covered by the same argument used in the proof of proposition C.1 which uses lemma C.3 to bound the probability that $||\Delta(y, x_l) - \Delta_n(y, x_l)||_\infty \geq r_n$.

In the last case:

$$d_n \geq |\Delta(y, x_l)| - r_n \tag{317}$$

$$\implies d_n + r_n \geq |\Delta(y, x_l)| \tag{318}$$

$$\implies 2d_n \geq |\Delta(y, x_l)| \tag{319}$$

This implies that additional classification mistakes can only occur in the set $\{y \in \mathcal{Y} : |\Delta(y, x_l)| \leq 2d_n\}$. Thus, the arguments from the proof of proposition C.1 can be replicated replacing the rate $\sqrt{\frac{\log(n_1)}{n_1 h_n}}$ with the rate $d'_n = 2d_n$.

Partition the support of the outcome as follows:

$$\mathcal{Y} = \mathcal{S}_{x_l, d'_n} \cup \mathcal{M}_{x_l, d'_n} \cup \mathcal{C}_{x_l} \tag{320}$$

Where:

$$\mathcal{S}_{x_l, d'_n} = \left\{ y \in \mathcal{Y} : |\Delta(y, x_l)| > d'_n \right\} \tag{321}$$

$$\mathcal{M}_{x_l, d'_n} = \left\{ y \in \mathcal{Y} : 0 < |\Delta(y, x_l)| \leq d'_n \right\} \tag{322}$$

$$\mathcal{C}_{x_l} = \left\{ y \in \mathcal{Y} : |\Delta(y, x_l)| = 0 \right\} \tag{323}$$

Then, lemmas C.3 and C.4 and an argument analogous to the proof of proposition C.1 yield the desired results.

□

Proposition C.3. *Set* $\Psi^d_{x_l, \kappa, n} := L_\kappa(\eta_{\kappa,n}, x_l)$ *and define:*

$$P_{x_l | z_k, \kappa, n} = \frac{\sum_{i \in J_\kappa} \mathbf{1}\{X_i = x_l, Z_i = z_k\}}{\sum_{i=1}^{n} \mathbf{1}\{Z_i = z_k\}}. \tag{324}$$

Define the random vector:

$$\boldsymbol{\theta}^T_{\kappa,n} = (P_{x_0|z_0,\kappa,n}, ..., P_{x_1|z_0,\kappa,n}, P_{x_{L-1}|z_1,\kappa,n}, ..., P_{x_{L-1}|z_1,\kappa,n}, \Psi^d_{x_0,\kappa,n}, ..., \Psi^d_{x_{L-1},\kappa,n}) \tag{325}$$

With expectation $\boldsymbol{\mu}^T_\theta = (P_{x_0|z_0}, ..., P_{x_1|z_0}, P_{x_{L-1}|z_1}, ..., P_{x_{L-1}|z_1}, \Psi_{x_0}, ..., \Psi_{x_{L-1}})$ *and and variance-covariance matrix* $Var(\boldsymbol{\theta}^T_{\kappa,n} | S_1)$.

Then, a symmetric positive semi-definite matrix $\boldsymbol{\Omega}$ *exist such that:*

$$nVar(\boldsymbol{\theta}_{\kappa,n}^T|S_1) \xrightarrow{p} \boldsymbol{\Omega} \tag{326}$$

And:

$$\sqrt{n}\big(\boldsymbol{\theta}_{\kappa,n} - \boldsymbol{\mu}_\theta\big) \xrightarrow{d} \mathcal{N}(0, \boldsymbol{\Omega}) \tag{327}$$

Proof of proposition C.3.
An analogous argument to proposition 5.1 replacing $\mathbf{1}\{\eta_{n_1}^*(Y, X, Z; z_k, x_l) \neq k, X = x_l, Z = z_k\}$ with $\xi(Y, X, Z; z_k, x_l) = \mathbf{1}\{\eta_{n_1}^d(Y, X, Z; z_k, x_l) \neq k, X = x_l, Z = z_k\}$ and proposition C.1 with proposition C.2 yields asymptotic normality for non-degenerate directions, more specifically:

For any fixed $\boldsymbol{a} \in \mathbb{R}^{3L}$ such that there exists $c_a > 0$ that satisfies $\mathbb{P}\left[\boldsymbol{a}^T \boldsymbol{V}_{\kappa,n} \boldsymbol{a} > c_a\right] \longrightarrow 1$:

$$\sup_{t\in\mathbb{R}} \left| \mathbb{P}\left[\frac{\boldsymbol{a}^T \sqrt{\tilde{n}}\left(\boldsymbol{\theta}_{\kappa,n} - \boldsymbol{\mu}_\theta\right)}{\sqrt{\boldsymbol{a}^T \boldsymbol{V}_{\kappa,n} \boldsymbol{a}}} \leq t \right] - \Phi(t) \right| \xrightarrow{p} 0. \tag{328}$$

Where $n_2 \boldsymbol{V}_{\kappa,n}$ is the variance of the vector $\boldsymbol{\theta}_{\kappa,n}$ conditional on J_κ^C (i.e. the data in which the classifier is trained).

By lemma C.8, the entries of $\boldsymbol{V}_{\kappa,n}$ depend on conditional treatment probabilities and missclassification probabilities $\mathbb{P}[\eta_{\kappa,n}^d(Y, X) \neq k, X = x_l, Z = z_k]$. Then, lemmma C.9 implies that the sequence $\boldsymbol{V}_{S_1,n}$ converges to a deterministic limit $\boldsymbol{V}$. Thus:

$$\boldsymbol{a}^T \sqrt{\tilde{n}}\left(\boldsymbol{\theta}_{\kappa,n} - \boldsymbol{\mu}_\theta\right) \xrightarrow{d} \mathcal{N}(0, \boldsymbol{a}^T \boldsymbol{V} \boldsymbol{a}) \tag{329}$$

More over, for any degenerate direction $\boldsymbol{a}$ such that $\boldsymbol{a}^T \boldsymbol{V} \boldsymbol{a} = 0$, the sequence converges to a degenerate normal with variance zero.

Thus, the Cramér-Wold device implies that:

$$\sqrt{\tilde{n}}\big(\boldsymbol{\theta}_{\kappa,n} - \boldsymbol{\mu}_\theta\big) \xrightarrow{d} \mathcal{N}(0, \boldsymbol{\Omega}) \tag{330}$$

□

Proof of theorem C.1. Proposition C.3 implies that each of the terms $\sqrt{\tilde{n}}\big(\boldsymbol{\theta}_{\kappa,n} - \boldsymbol{\mu}_\theta\big)$ is assymptotically normal with variance $\boldsymbol{\Omega}$ for $\kappa = 1, ..., \tilde{K}$. Lemma C.10 implies

that each of the terms is asymptotically independent. Thus, the vector the vector $\sqrt{n}(\boldsymbol{\theta}_n - \boldsymbol{\theta})$ is asymptotically normal and its asymptotic variance is equal to $\boldsymbol{\Omega}$. □

Lemma C.9.

$$\lim_{n\longrightarrow\infty} \mathbb{P}[\eta^d_{\kappa,n}(y, x_l) \neq k, X = x_l, Z = z_k|S_1] = \mathbb{P}[\eta^*(y, x_l) \neq k, X = x_l, Z = z_k] \quad (331)$$

Lemma C.10. *(Influence function for K-folds cross-fitting)*
For any fixed classifier η, define the random variable $\xi_\eta(Y, X, Z; z_k, x_l) = \mathbf{1}\{\eta(Y, X) \neq k, X = x_l, Z = z_k\}$ and let $\mu_\eta(z_k, x_l)$ denote its mean.

Then:

$$L_\kappa(\eta_{\kappa,n}, x_l) - \Psi^d_{x_l} = \frac{1}{\tilde{n}} \sum_{i\in J_\kappa} \Big[\sum_{k=0,1} \frac{P_{z_k} - \mathbf{1}\{Z_i = z_k\}}{P^2_{z_k}} \mu_{\eta^d}(z_k, x_l) + \frac{1}{P_{z_k}} \xi_{\eta^d}(Y, X, Z; z_k, x_l) \Big] + o_p(n^{-1/2}) \quad (332)$$

Proof of lemma C.9.

$$\begin{aligned}
&\mathbb{P}[\eta^d_{\kappa,n}(Y, X) \neq k, X = x_l, Z = z_k|S_1] - \mathbb{P}[\eta^*(Y, X) \neq k, X = x_l, Z = z_k] && (333)\\
&\quad \leq \int_{\mathcal{Y}} \mathbf{1}\{\eta^d_{\kappa,n}(y, x_l) \neq \eta^*_{\kappa,n}(y, x_l)\} f_Y(y)dy \\
&\quad = \int_{\mathcal{S}_{x_l,d'_n}} \mathbf{1}\{\eta^d_{\kappa,n}(y, x_l) \neq \eta^*_{\kappa,n}(y, x_l)\} f_Y(y)dy && (334)\\
&\qquad + \int_{\mathcal{M}_{x_l,d'_n}} \mathbf{1}\{\eta^d_{\kappa,n}(y, x_l) \neq \eta^*_{\kappa,n}(y, x_l)\} f_Y(y)dy \\
&\qquad + \int_{\mathcal{C}_{x_l}} \mathbf{1}\{\eta^d_{\kappa,n}(y, x_l) \neq \eta^*_{\kappa,n}(y, x_l)\} f_Y(y)dy \\
&\quad \leq \int_{\mathcal{Y}} \mathbf{1}\{|\Delta^d_{\kappa,n}(Y, x_l) - \Delta(Y, x_l)| > d'_n\} f_Y(y) + \int_{\mathcal{M}_{x_l,d'_n}} f_Y(y)dy && (335)\\
&\quad \leq \mathbb{P}[||\Delta^d_{\kappa,n}(Y, x_l) - \Delta(Y, x_l)||_\infty > d'_n] + K {d'_n}^\alpha && (336)\\
&\quad = o({d'_n}^\alpha) && (337)
\end{aligned}$$

Where the first inequality follows from elemental properties. The second line par-

titions $\mathcal{Y}$ into the sets defined in the proof of proposition C.2. The third line notes that in the sets $\mathcal{S}_{x_l,d'_n}$ and $\mathcal{C}_{x_l}$ the classifier $\eta^d_{\kappa,n}$ can only disagree with $\eta^d_{\kappa,n}$ if $||\Delta^d_n(Y,x_l) - \Delta(Y,x_l)| > d'_n$, and the last uses the margin condition from assumption 5.1.3. □

Proof of lemma C.10.

Observe that $\mathbb{E}[\xi_\eta(Y,X,Z;z_k,x_l)] = L(\eta,x_l)$.

The estimator based on the semi-deterministic classifier trained on J^C_κ and evaluated in J_κ can be recast as:

$$L_\kappa(\eta^d_{\kappa,n},x_l) = \frac{1}{\tilde{n}}\sum_{i\in J_\kappa}\sum_{k=0,1}\frac{1}{P_{z_k,\kappa,n}}\xi_{\eta^d_{\kappa,n}}(Y,X,Z;z_k,x_l) \tag{338}$$

$$= \frac{1}{\tilde{n}}\sum_{i\in J_\kappa}\sum_{k=0,1}\Big[\frac{1}{P_{z_k}}\xi_{\eta^d}(Y,X,Z;z_k,x_l) + \Big(\frac{1}{P_{z_k,\kappa,n}} - \frac{1}{P_{z_k}}\Big)\xi_{\eta^d}(Y,X,Z;z_k,x_l) \tag{339}$$

$$+ \frac{1}{P_{z_k}}\Big(\xi_{\eta^d_{\kappa,n}}(Y,X,Z;z_k,x_l) - \xi_{\eta^d}(Y,X,Z;z_k,x_l)\Big)$$

$$+ \Big(\frac{1}{P_{z_k,\kappa,n}} - \frac{1}{P_{z_k}}\Big)\Big(\xi_{\eta^d_{\kappa,n}}(Y,X,Z;z_k,x_l) - \xi_{\eta^d}(Y,X,Z;z_k,x_l)\Big)\Big]$$

Now consider the random sum:

$$\frac{1}{\tilde{n}}\sum_{i\in J_\kappa}\sum_{k=0,1}\frac{1}{P_{z_k}}\Big(\xi_{\eta^d_{\kappa,n}}(Y,X,Z;z_k,x_l) - \xi_{\eta^d}(Y,X,Z;z_k,x_l)\Big) \tag{340}$$

Note that both $\xi_{\eta^d_{\kappa,n}}(Y,X,Z;z_k,x_l)$ and $\xi_{\eta^d}(Y,X,Z;z_k,x_l)\Big)$ are indicators, thus, their moments exist and are bounded. Moreover, all the terms in the sum are independent and identically distributed, and the probabilities P_{z_k} for $k = 0,1$ bounded away from zero, furthermore, part 1 of proposition C.2 implies that:

$$\mathbb{E}\Big[\sum_{k=0,1}\frac{1}{P_{z_k}}\Big(\xi_{\eta^d_{\kappa,n}}(Y,X,Z;z_k,x_l) - \xi_{\eta^d}(Y,X,Z;z_k,x_l)\Big)\Big|S_1\Big] = o_p(n^{-1/2}) \tag{341}$$

Moreover:

$$\mathbb{E}\Big\{\Big[\sum_{k=0,1}\frac{1}{P_{z_k}}\Big(\xi_{\eta^d_{\kappa,n}}(Y,X,Z;z_k,x_l)-\xi_{\eta^d}(Y,X,Z;z_k,x_l)\Big)\Big]^2\Big|S_1\Big\} \tag{342}$$

$$=\int_{\mathcal{Y}}\Big[\sum_{k=0,1}\frac{1}{P_{z_k}}\Big(\xi_{\eta^d_{\kappa,n}}(y,x_l,z_k;z_k,x_l)-\xi_{\eta^d}(y,x_l,z_k;z_k,x_l)\Big)\Big]^2 f_{Y,X,Z}(y,x_l,z_k)dy \tag{343}$$

$$\leq K\int_{\mathcal{Y}}\sum_{k=0,1}\mathbf{1}\{\eta^d_{\kappa,n}(y,x_l)\neq\eta^d(y,x_l)\}f_{Y,X,Z}(y,x_l,z_k)dy \tag{344}$$

$$=o_p(1) \tag{345}$$

Where the last equality follows from a similar reasoning to the proof of lemma C.9.

Then, 345 and 341 imply that the sum in 340 is $o_p(n^{-1/2})$.

On the other hand, assumption 5.1.1, Slutsky's theorem and the delta method imply that:

$$\frac{1}{\tilde{n}}\sum_{i\in J_\kappa}\sum_{k=0,1}\Big(\frac{1}{P_{z_k,\kappa,n}}-\frac{1}{P_{z_k}}\Big)\xi_{\eta^d}(Y,X,Z;z_k,x_l) \tag{346}$$

$$=\frac{1}{\tilde{n}}\sum_{i\in J_\kappa}\sum_{k=0,1}\frac{P_{z_k}-\mathbf{1}\{Z_i=z_k\}}{P^2_{z_k}}\mu_{\eta^d}(z_k,x_l)+o_p(n^{-1/2})$$

□

C.5 Proofs for subsection 5.4

Proof of theorem 5.2: Let n' be either $n/2$ or n depending on the estimator. Let $\boldsymbol{\Omega}_n$ denote $nVar(\boldsymbol{\theta}_n)$ for the estimator based on a semi-deterministic classifier, or $\frac{n}{2}'Var(\boldsymbol{\theta}_n|S_1)$ for the main estimator.

Theorems 5.1 and C.1 along with lemma C.8 imply that the sequence $n'\boldsymbol{\Sigma}_n^{-1/2}Var(\boldsymbol{\theta}_n)\boldsymbol{\Sigma}_n^{-1/2}$ converges in probability to $\boldsymbol{I}_m$ and the random sequence $\sqrt{n_2}\boldsymbol{\Omega}_n^{-1/2}\big(\boldsymbol{\theta}_n - \boldsymbol{\mu}_\theta\big)$ converges in distribution to a multivariate, non-degenerate normal with variance-covariance matrix $\boldsymbol{I}_m$.

Thus, assumption 3(a) from Cox and Shi (2023) is satisfied. □

C.6 Semi-deterministic classifiers

Here I propose a modification of the main overlap estimator, the modification restores a fixed limit distribution which makes the estimator compatible with a larger class of methods for inference on linear systems including Fang et al. (2023) and Goff and Mbakop (2025). Moreover, the modified estimator becomes asymptotically linear and thus allows for K-fold cross fitting. The modification is to use a sub-optimal estimated classifier that has the advantage of converging to a fixed function.

The key modification is to define a default pre-determined, arbitrary classification rule as well as a deterministic sequence d_n that dominates the rate of uniform convergence for KDEs r_n. Then, the classifier differs from the pre-determined rule only when the estimated difference $\Delta_{n_1}(y, x_l)$ is larger than d_n.

The modified classifier is assymptotically optimal in the sense that it converges to a classifier that only follows the pre-determined rule in the contact set, where, as proposition 5.1 shows, the loss is invariant to classification decisions.

Assumption C.1. *[**Deterministic decision rate:**] The sequence d_n satisfies:*

1. $\lim_{n\to\infty} \frac{d_n}{r_n} = \infty$.
2. $\lim_{n\to\infty} d_n = 0$.
3. *A constant $\tilde{\gamma} > 0$ exists such that $\lim_{n\to\infty} \frac{d_n}{n^{-\tilde{\gamma}}} = 0$ and $\alpha > \frac{d\tilde{\gamma}}{1-d\tilde{\gamma}}$.*

The first part of assumption C.1 simply requires that d_n dominates r_n so that the classifier only changes when KDE-based classifiers are unlikely to incur in classification mistakes. The second part requires that d_n tends to zero so that, asymptotically, the classifier only coincides with the pre-specified deterministic rule inside the contact set. The last condition is required to apply the margin condition and prove that the loss from classification mistakes near the boundary of the contact set is negligible. Note that this last condition is strictly stronger than its analogue in assumption 5.2.2.

For a sample of size n and some fixed integer $\tilde{K} \geq 2$, a $\tilde{K}$-fold partition randomly distributes all observations into $\tilde{K}$ sub-samples or folds denoted $(J_\kappa)_{\kappa=1}^{\tilde{K}}$ of size $\left\lfloor n/\tilde{K} \right\rfloor$ each. Let, J_κ^C denote the set of all observations except those in fold κ.

Let $\eta_{\kappa,n}^d(y, x_l)$ be the classifier trained in J_κ^C according to the following semi-deterministic rule:

$$\eta^d_{\kappa,n}(y,x_l) := \begin{cases} \mathbf{1}_{\{\Delta_{\kappa,n}(y,x_l)\geq 0\}} & \text{if } |\Delta_{\kappa,n}(y,x_l)| \geq d_n \\ \mathbf{1}_{\{y\in D\}} & \text{otherwise} \end{cases} \tag{347}$$

Where D is a fixed, pre-specified measurable subset of $\mathbb{R}^n$ and $\Delta_{\kappa,n}(y,x_l)$ is the difference between the kernel density estimators for $f_{Y,X|Z=z_1}$ and $f_{Y,X|Z=z_0}$ trained in J^C_κ. Note that $\eta^d_{\kappa,n}$ converges to the classifier η^d that coincides with $\mathbf{1}_{\{\Delta_{\kappa,n}(y,x_l)>0\}}$ everywhere except in $\mathcal{C}_{x_l}$ where it is equal to $\mathbf{1}_{\{y\in D\}}$.

Let $L_\kappa(\eta^d_{\kappa,n},x_l)$ denote the empirical loss of the classifier $\eta^d_{\kappa,n}$ evaluated in J_κ. Define the estimator $\Psi^d_{x_l,\kappa,n} := L_\kappa(\eta^d_{\kappa,n},x_l)$.

In appendix C I prove that the difference $L_\kappa(\eta^d_{\kappa,n},x_l) - \Psi_{x_l}$ admits the following decomposition:

$$\begin{aligned} L_\kappa(\eta_{\kappa,n},x_l) - \Psi_{x_l} =& \frac{1}{\tilde{n}} \sum_{i\in J_\kappa} \Big[\sum_{k=0,1} \frac{P_{z_k} - \mathbf{1}\{Z_i = z_k\}}{P^2_{z_k}} \mu_{\eta^d}(z_k,x_l) \\ &+ \frac{1}{P_{z_k}} \xi_{\eta^d}(Y,X,Z;z_k,x_l) \Big] + o_p(n^{-1/2}) \end{aligned} \tag{348}$$

Where $\mu_{\eta^d}(z_k,x_l)$ denotes the misclassification probability $\mathbb{P}[\eta^d(Y,X) \neq k, X = x_l, Z = z_k]$. That is, the estimator is asymptotically linear with influence function given by the sum in brackets. Thus, the estimator based on a semi-deterministic classifier is asymptotically normal and its asymptotic distribution does not depend on the trained classifier. This enables K-fold cross fitting.

Theorem C.1. *Set $\Psi^d_{x_l,\kappa,n} := L_\kappa(\eta_{\kappa,n},x_l)$ and define:*

$$P_{x_l|z_k,\kappa,n} = \frac{\sum_{i\in J_\kappa} \mathbf{1}\{X_i = x_l, Z_i = z_k\}}{\sum_{i=1}^n \mathbf{1}\{Z_i = z_k\}}. \tag{349}$$

Let $\boldsymbol{P}^T_{\mathcal{X}_1,n}$ and $\boldsymbol{\Psi}^T_n$ stack the estimators $P_{x_l|z_k,\kappa,n}$ associated with treatment values in $\mathcal{X}_k$ except the one with the highest index.

Define the random vector:

$$\boldsymbol{\theta}^T_{\kappa,n} = (\boldsymbol{P}^T_{\boldsymbol{\mathcal{X}_0},\boldsymbol{n}}, \boldsymbol{P}^T_{\boldsymbol{\mathcal{X}_1},\boldsymbol{n}}, \Psi^d_{x_0,\kappa,n}, ..., \Psi^d_{x_{L-1},\kappa,n}) \tag{350}$$

With expectation $\boldsymbol{\mu}_\theta^T = (\boldsymbol{P}_{\boldsymbol{\mathcal{X}_0}}^T, \boldsymbol{P}_{\boldsymbol{\mathcal{X}_1}}^T, P_{x_{L-1}|z_1}, \Psi_{x_0}, ..., \Psi_{x_{L-1}})$.

Furthermore, define:

$$\boldsymbol{\theta}_n^T = \frac{1}{\tilde{K}} \sum_{\kappa=1}^{\tilde{K}} \boldsymbol{\theta}_{\kappa,n}^T. \tag{351}$$

Then, a symmetric positive definite matrix $\boldsymbol{\Omega}$ *exist such that:*

$$nVar(\boldsymbol{\theta}_n^T) \xrightarrow{p} \boldsymbol{\Omega} \tag{352}$$

And:

$$\sqrt{n}\big(\boldsymbol{\theta}_n - \boldsymbol{\mu}_\theta\big) \xrightarrow{d} \mathcal{N}(0, \boldsymbol{\Omega}) \tag{353}$$

Furthermore, a consistent estimator $\boldsymbol{\Sigma}_n$ *for the matrix* $\boldsymbol{\Omega}$ *can be constructed using the formulas from lemma C.8 in appendix C.*

The proof of theorem C.1 is in appendix C.

Notice that the estimator from theorem C.1 has a fixed asymptotic variance. Hence, without imposing assumption 5.3, or including conditional treatment probabilities that are equal to zero, the limit would still be asymptotically normal, but potentially degenerate.

An alternative proof for the estimator based on the semi-deterministic classifier would be to use the asymptotically linear expression in equation (348) to verify assumption 1 from Cox and Shi (2023). Asymptotic linearity implies that other inference methods for linear systems such as Fang and Santos (2019) or Goff and Mbakop (2025) are valid too.

D Examples of shape restrictions encompassed by the framework

Multivalued ordered monotonicity: Ordered monotonicity captures the idea that treatment values are strictly ordered (e.g, by intensity) and that as the instrument increases, units may only switch to weakly higher treatment levels. That is, for any pair of indices $k' > k > 0$, response types such that $(t^Z)_k > (t^Z)_{k'}$ are ruled out. This restriction can be easily accommodated in the framework of assumption 2.2 by imposing that response types as the one described before occur with zero probability.

In their seminal paper about IV with heterogeneous treatment effects, Imbens and Angrist (1994) showed that if both the instrument and treatment are binary, and treatment is weakly monotonic in the instrument, the classical Wald estimand identifies a LATE for units that strictly increase their realized treatment when the instrument changes. This result has motivated the use of binary instruments and treatments in a wide range of empirical applications.

In a subsequent paper, Angrist and Imbens (1995) extended their analysis to multivalued ordered treatments. In this setting, no LATE is point identified, but the Wald estimand continues to identify a convex combination of LATEs among units that are induced to take strictly higher treatments.

Despite the interpretation provided by Angrist and Imbens (1995), it is common practice in applied work to binarize monotone multivalued treatments. That is, the researcher selects a threshold $\bar{x} \in \mathcal{X}$ and defines a binary treatment indicator $X^{Bin} = \mathbf{1}_{\{X \geq \bar{x}\}}$, thereby reducing the setting to the binary treatment case. A justification for this practice is that the resulting Wald estimand has a simpler interpretation.

However, this practice has two limitations, first, there are instances in which the binarized model is rejected by a specification test, even though the correctly specified multivalued model is not. Second, after binarization, the Wald estimand identifies a weighted sum of LATEs for certain response types, but there is no guarantee that the weights add up to one. As a result, the estimand cannot generally be interpreted as LATE or even as a convex combination of LATEs. An in-depth analysis of the second limitation is provided in Rose and Shem-Tov (2024).

To illustrate the first limitation, suppose that $\mathcal{X} = \{x_0, x_1, x_2\}$, $\mathcal{Y} = \{0, 1\}$ and $\mathcal{Z} = \{z_0, z_1\}$. Assume that half the population has the instrument-response type

(x_0, x_1), one fourth (x_1, x_2), and the last fourth (x_2, x_2)[19]. Suppose that potential outcomes are homogeneous so that $Y(x_0) = 0$, $Y(x_1) = 1$ and $Y(x_2) = 1$ for all $i \in \mathcal{I}$. Now suppose that a researcher sets $\bar{x} = x_2$. It is easy to see that units with instrument-response type (x_0, x_1) will not change their treatment status with respect to the artificially defined treatment variable X^{Bin} as Z increases, but, their potential outcomes will, violating the exclusion restriction in the binarized model.

Furthermore, Kitagawa (2015) shows that if instrument validity holds in the binary case, the following equation must be satisfied:

$$\mathbb{P}[Y = 1, X^{Bin} = 0|Z = 0] \geq \mathbb{P}[Y = 1, X^{Bin} = 0|Z = 1] \tag{354}$$

In the example, the left hand side is equal to $\frac{1}{4}$ while the right hand side is $\frac{1}{2}$, thus, instrument validity would be rejected for the binarized model, even if it holds in the true multivalued model with the right (multivalued ordered monotone) instrument-response restriction.

Furthermore, as studied by Rose and Shem-Tov (2024), even if equation 355 is not violated, the Wald estimand obtained using Z as an instrument for X^{Bin} may be difficult to interpret. To see this problem, suppose that the instrument-response types (x_0, x_1), (x_1, x_2), and (x_2, x_2) occur with equal probability $\frac{1}{3}$ in the population. In this case, the Wald estimand becomes:

$$W = \frac{\frac{1}{3}\mathbb{E}[Y(x_1) - Y(x_0)|i \in (x_0, x_1)] + \frac{1}{3}\mathbb{E}[Y(x_2) - Y(x_1)|i \in (x_1, x_2)]}{\frac{1}{3}} \tag{355}$$

While the numerator is a weighted sum of LATEs, the weights do not necessarily sum to one. Consequently, the estimand W cannot be interpreted as LATE or a convex combination of LATEs. Moreover, if instead of $\frac{1}{3}$, the denominator, i.e. the population probability associated with instrument-response type (x_1, x_2), was small, the magnitude of the Wald estimand could be arbitrarily large. Thus, even if binarizing a monotone multivalued treatment may simplify the model, it can produce estimands that are both biased and difficult to interpret.

These concerns provide a strong motivation for retaining a correctly specified monotone model when the treatment is multivalued.

[19]Recall that the instrument-response type $t = (x_l, x_{l'})$ represents units that realize treatment x_l when $Z = z_0$ and $x_{l'}$ when $Z = z_1$.

Generalizing monotonicity to settings with multiple instruments is nontrivial. Angrist and Imbens (1995) proposed one approach, which imposes binary monotonicity in one direction between every pair of treatment values. This definition was later criticized by Mogstad et al. (2021) who proposed an alternative extension. More recently, Goff (2020) introduced a tractable notion –vector monotonicity– which ensures identification of a series of LATEs.

Classical monotonicity and its three aforementioned generalizations can be expressed as restrictions that rule out specific response types, that is, they fall within assumption 2.2.

Unordered monotonicity and its refinements: Heckman and Pinto (2018) proposed an alternative notion of monotonicity for settings in which no natural ordering over treatment values exists. Intuitively, unordered monotonicity requires that, as the instrument changes from one value to another, a subset of treatments $X^I \subset \mathcal{X}$ are equally incentivized relative to the treatments in $\mathcal{X} \setminus \mathcal{X}^I$. This restriction not only implies that take-up of the incentivized treatments must weakly increase but also rules out unit-level switches within the set of promoted treatments $\mathcal{X}^I$, or its complement $\mathcal{X} \setminus \mathcal{X}^I$. This is another restriction on heterogeneity obtained from ruling out particular response types and thus, it can be accommodated within the framework of this paper.

Heckman and Pinto (2018) argue that the identifying power of unordered monotonicity can be strengthened by further restricting the set of admissible response types. For example, suppose that $x_0 \in \mathcal{X}^I$ represents a discounted car, and $x_1, x_2 \in \mathcal{X} \setminus \mathcal{X}^I$ represent, respectively, a similar full-priced car and a pick-up truck. One could impose the restriction that only individuals who would have chosen x_1 in the absence of the discount may switch to x_0 as a result of the discount, ruling out switches from x_2 to x_0. Unordered monotonicity augmented with this type of restriction remains within the framework of this paper.

Other response type restrictions: The identifying power of response type restrictions extends beyond ordered and unordered monotonicity. Goff (2024) shows that the ability of restrictions on treatment-response heterogeneity to point indetify some LATE without restricting the outcome space is underpined by the set of treatment-response types it allows. Leveraging this insight, Goff (2024) identifies a series of novel response-type restrictions that guarantee point identification of at least one LATE, and a procedure to find other such restrictions.

To advance the interpretability of IV models with multiple instruments and treatments, Lee and Salanié (2020) investigated identification when multiple instruments are available and each instrument targets (i.e. incentivizes) adoption of one particular treatment more than the rest. In particular, Lee and Salanié (2020) provide a general framework and then –to achieve point or partial identification– introduce the notions of strict targeting, and one-to-one targeting. Multiple instances of these two assumptions fit into the framework studied in this paper.

In a closely related paper, Bai et al. (2024c) study the case in which every treatment value is uniquely incentivized by a single instrument. For this particular targeting structure they show that multiple LATEs are point identified and derive the sharp testable implications of the model. However, the assumption that each instrument uniquely targets one –and only one– treatment (without affecting any other comparison) might be overly restrictive in multiple empirical settings. The framework developed in this paper can be used to assess the validity of the complete model proposed by Bai et al. (2024c), as well as more plausible relaxations.

Other linear restrictions on instrument-response types: Rather than ruling out specific response types, a less restrictive approach to IV identification is to place bounds on the prevalence or relative frequency of certain behaviors or instrument-response types. A notable example is the "more compliers than defiers" condition of De Chaisemartin (2017). Another example is Richardson and Robins (2010), who derive bounds on the outcome distribution under assumptions on the fraction of always takers in the population.

Similar assumptions are also used as intermediate steps in sensitivity analysis methods that assess the robustness of IV estimands to relaxations of monotonicity (Huber (2014), Noack (2021), Yap (2025)). The tests from this paper can be used to falsify these types of restrictions on their own or combined with response-type restrictions.

E Examples of submonotonicity

Ordered monotonicity: Ordered monotonicity captures the notion that an implicit ranking (a total order) of alternatives exists, and, as the instrument increases, adoption of higher ranked alternatives is weakly incentivized. This assumption is implicit in most random utility models in which a mean structural utility function is common to all members of the population and unit-specific unobservables are additively separable. Thus, holding unobserved heterogeneity fixed, as observables change all decision makers agree on which alternatives yield higher utility levels and thus are being promoted.

Because subomotonicity allows for general binary relations, ordered monotonicity is a particular case.

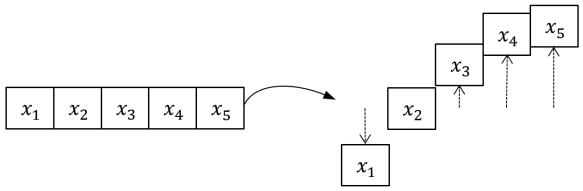


Figure 8: Ordered monotonicity

Notes: Visual representation of ordered monotonicity. The left of each figure represents alternatives before the intervention ($Z = z_0$). The right represents the effect of the intervention ($Z = z_1$). The higher an alternative is depicted on the left side, the more it is being incentivized. The black arrows represent the relative magnitude of the incentives.

Unordered monotonicity: Unordered monotonicity seeks to abstain from imposing an ex-ante ordering of alternatives and instead simply acknowledging that instruments incentivize adoption of some treatments relative to the rest. This idea is formalized by assuming that a subset of treatments $S \subset \mathcal{X}$ exists such that all alternatives in S are being promoted and no alternative in its complement $S' := \mathcal{X} \setminus S$ is.

While this assumption is general in spirit and in the multiple instruments setting unordered monotonicity is not implied nor implies ordered monotonicity, in the binary instrument case, it is a strictly more restrictive assumption. The reason is that unorderdered monotonicity not only requires that alternatives in S are being promoted, but also that they are being promoted with exactly the same intensity so that switches within the set S (and its complement) occur with zero probability.

The previous point is easy to see by noting that the binary relation associated to

unordered monotonicity is defined as follows:

$$x_l \geq^R x_{l'} \iff \begin{cases} x_l, x_{l'} \in S & or \\ x_l, x_{l'} \in S' & or \\ x_l \in S, x_{l'} \in S' \end{cases} \tag{356}$$

This restriction induces a partial order with two indifference classes S and S', so that if $x_l, x_{l'} \in S$ then $x_l \geq^R x_{l'}$ and $x_{l'} \geq^R x_l$. The name partial order is misleading in this setting –it suggests weakening a total order–. The weakening is achieved by imposing that different treatments belong to the same equivalence class, that is, adding more ordered pairs to the binary relation which imposes additional restrictions on heterogeneity.

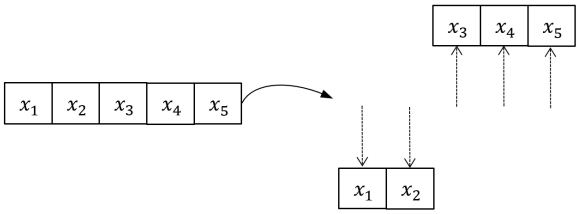


Figure 9: Unordered monotonicity

Notes: Visual representation of unordered monotonicity. The left of each figure represents alternatives before the intervention ($Z = z_0$). The right represents the effect of the intervention ($Z = z_1$). The higher an alternative is depicted, the more it is incentivized. The black arrows represent the relative magnitude of the incentives. The set $S = \{x_3, x_4, x_5\}$ represents the alternatives that are being incentivized relative to $S' = \{x_1, x_2\}$. Note that comparisons between alternatives within the same set are not altered.

Furthermore, note that if S has only one element s (i.e. $|S| = 1$), then, an instrument that encourages adoption of s (Lee and Salanié (2020)) or strictly targets s (Bai et al. (2024c)) satisfies a particular case of unordered monotonicity. Targeting or encouragement are not implied nor imply unordered monotonicity in the multiple instruments case.

The sets S and S' could be further partitioned into three sets instead of two. For example, let $S = S_1 \cap S_2$ and $S' = S'_1 \cap S'_2$ and let alternatives in S_1 and S'_1 represent compact cars and alternatives in S_2 and S'_2 SUVs. Suppose that S represents the set of all discounted vehicles (SUVs and compact) and S' the set of vehicles without a discount. In this setting, one plausible restriction is that the discount on compact cars

only affects decision makers who, absent the discount, would buy alternatives in S_1'; and the discount on SUVs only affects decision makers who would pick alternatives in S_2'. Thus, switches from S_1' to S_2 or S_2' to S_1 are assumed to occur with zero probability. This restriction is no longer represented by a partial order, but it is derived from adding ordered pairs to the partial order described in equation 356, thus, it is a submonotone restriction associated to the extension of a partial order.

Relaxations of unordered monotonicity: The assumption that all alternatives are equally discounted might be too strong, instead, one could assume that there are N discount bins and that comparisons within the same bin are not affected but comparisons across different bins are –always in favor of alternatives with larger discounts. For example, if $N = 3$ there could be a set of heavily discounted cars, a set of cars with payment facilities, and a set of cars that have to be payed upfront. This situation also corresponds to submonotonicity with respect to a partial order but now with three indifference classes.

Additional restrictions, as the one considered in the compact cars and SUVs examples could also be added across all three discount bins. The resulting restriction would remain within the submontonicity framework.

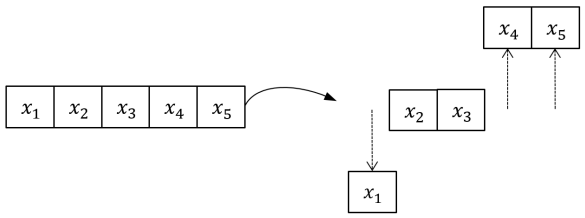


Figure 10: Submonotonicity with respect to a partial order with three indifference classes

Notes: Visual representation of submonotonicity with respect to a partial order with three indifference classes. The left of each figure represents alternatives before the intervention ($Z = z_0$). The right represents the effect of the intervention ($Z = z_1$). The higher an alternative is depicted, the more it is incentivized. The black arrows represent the relative magnitude of the incentives. The set $\{x_4, x_5\}$ represents the alternatives that are being incentivized the most (heavily discounted), the set $\{x_2, x_3\}$ the alternatives that for which payment facilities are offered (i.e. weakly incentivized). Alternative x_1 is not being incentivized in any way. Note that comparisons between alternatives within the same set are not altered and thus, their difference in height remains equal.

Relaxations of encouragement: Now consider an intervention in which an instrument encourages adoption of certain treatments in $S \subset \mathcal{X}$, but there is a concern that the instrument might also affect utility from other treatments. For example,

consider an an intervention aimed at increasing sports participation by providing information. It is clear that such an intervention could be used as an instrument for sports practice, but the intervention may also promote other activities that develop motor or social skills such as community service or playing board-games. Hence, the restriction that the instrument is not affecting comparisons between treatments other than practicing sports might not be plausible and a weaker assumption might be preferred.

One such weaker assumption could be that alternatives x_1 and x_2 which represent sports are certainly being promoted relative to all others (since they are the main target of the intervention), but pairwise comparisons between other treatments might also affected and the direction of this effect is heterogeneous across the population. Thus, restrictions on the direction of the flows between non-sports activities cannot be credibly imposed. This situation is encoded by submonotonicity with respect to a quasi-order, i.e. a binary relation that is transitive but not complete.

If x_5 and x_4 represent different sports and x_3, x_2 and x_1 other activities, the specific submonotonicity assumption would be $x_l \geq^R x_{l'}$ for $l = 5, 4$ and $l' = 3, 2, 1$ but neither $x_l \geq^R x_{l'}$ nor $x_{l'} \geq^R x_l$ for $l, l' = 5, 4$ or $l, l' = 3, 2, 1$. A graphical representation is given in figure ??.

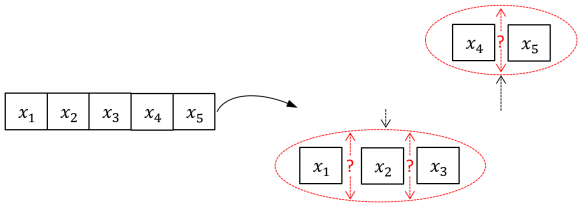


Figure 11: Submonotonicity with respect to a quasi-order - Sports example

Notes: Visual representation of submonotonicity with respect to a quasi-order. The left of each figure represents alternatives before the intervention ($Z = z_0$). The right represents the effect of the intervention ($Z = z_1$). The higher an alternative is depicted on the right side, the more it is incentivized. The set $\{x_4, x_5\}$ represents the alternatives that are being incentivized the most (sports), the set $\{x_1, x_2, x_3\}$ the alternatives that are not being incentivized. The black arrows represent the relative magnitude of the incentives but the red arrows represent relative uncertainty about this effect within the set of circled alternatives. That is, within the red elipses no restriction on how pairwise comparisons are affected is imposed.

Note that relaxing the completeness requirement allows for more heterogeneity. In particular, it allows decision makers to change their relative ranking of the same pair of alternatives in different directions. This situation cannot be captured by a random utility model in which the effect of Z is common to al decision makers, instead, it is

allowing the effect of Z to be determined by unobserved unit characteristics in the spirit of a random coefficients model.

Now consider an intervention aimed at preventing students from dropping college to take part-time jobs. Suppose this goal is achieved by providing additional classes aimed at developing professional qualifications but also by providing information on the importance of college completion.

It is clear that such an intervention would promote staying in college relative to taking part-time jobs, however, the classes could help students find full-time jobs, even without the degree. On the other hand, the information could reduce the willingness of students to accept any type of job. If the latter effect is stronger for full-time jobs (because they are more demanding and thus increase the risk of not finishing college), some students might choose to leave a full-time job and take a part-time one. Thus, it might be desirable to only impose the restriction that college is being incentivized relative to part-time jobs, but nothing else.

This restriction can also be accomodated by submonotonicity by assuming that college degrees x_4 and x_5 are being incentivized relative to half-time jobs x_3 and x_2 but no restrictions involving full-time jobs x_1 are imposed. Formally, $x_l \geq^R x_{l'}$ for $l = 1, 2$ and $l' = 3, 4$.

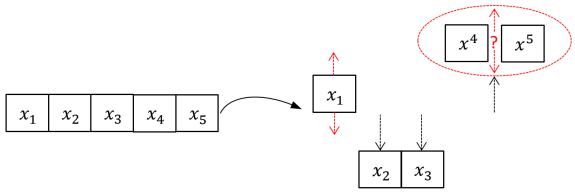


Figure 12: Submonotonicity with respect to a transitive quasi-order - College and full-time jobs example.

Notes: Visual representation of submonotonicity with respect to a quasi-order. The left of each figure represents alternatives before the intervention ($Z = 0$). The right represents the effect of the intervention ($Z = 1$). The higher an alternative is depicted, the more it is incentivized. The set $\{x_4, x_5\}$ represents the alternatives that are being incentivized the most (sports), the set $\{x_1, x_2, x_3\}$ the alternatives that are not being incentivized. The black arrows represent the relative magnitude of the incentives but the red arrows represent relative uncertainty about this effect relative to alternatives in the same class. Thus, within the red circle, no stance is taken on how pairwise comparisons are affected. Furthermore, no stance on whether x_1 is being promoted or not relative to x_4, x_5 or even x_2 and x_3 is taken.

Relaxing transitivity/encouraging experimentation: The submonotonicity framework does not require transitivity of the binary relation $\geq^R$ –i.e. it allows for cycles in $\geq^R$. This observation raises the question of how can such a restriction be

interpreted –specially in light of Goff (2024) who shows that some models associated to non-transitive binary relations can point identify LATEs for some specific response groups.

To answer this question, consider a three alternatives example and suppose that $x_1 >^R x_2$, $x_2 >^R x_3$ and $x_3 >^R x_1$ (where $x_l >^R x_{l'}$ indicates $x_l \geq^R x_{l'}$ but not $x_{l'} \geq^R x_l$).

This restriction implies that:

- Any decision maker who choses x_1 when $Z = 0$ will now choose x_1 or x_3.
- Any decision maker who choses x_2 when $Z = 0$ will now choose x_2 or x_1.
- Any decision maker who choses x_3 when $Z = 0$ will now choose x_3 or x_2.

Furthermore, if the assumption that there are no always takers at $\mathcal{X}$ is added (i.e. all types of always takers are ruled out), then:

- Any decision maker who choses x_1 when $Z = 0$ will now choose x_3.
- Any decision maker who choses x_2 when $Z = 0$ will now choose x_1.
- Any decision maker who choses x_3 when $Z = 0$ will now choose x_2.

In the first case, choices when $Z = 0$ restrict potential choices when $Z = 1$. In the second, there is a one to one correspondence between one and the other. In both cases, if decision makers change their choice, they pick an alternative different than the one they would have chosen absent the instrument. The result is a situation where every alternative is promoted among subjects who do not choose it at the initial value of the instrument; and discouraged among those that already prefer it over the rest. The previous discussion suggests interpreting the effect of the instrument as promoting experimentation or variety.

Interpreting more general non-transitive restrictions is not straightforward, but, in all cases, submonotonicity induces restrictions on heterogeneity that can be expressed as a correspondence of choices when $Z = 0$, and thus, can be seen as restricting heterogeneity in terms of status-quo choices.

□

Table 5: OHE summary statistics and lottery effects.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Female		Age >50		Hospital Utilization		Health	
Lost	0.589***	0.587***	0.326***	0.328***	-0.027***	-0.061**	0.006	-0.003
	(0.006)	(0.018)	(0.006)	(0.017)	(0.009)	(0.026)	(0.010)	(0.027)
Won	0.577***	0.575***	0.335***	0.352***	-0.010	-0.099***	0.062***	0.072***
	(0.006)	(0.018)	(0.006)	(0.017)	(0.009)	(0.027)	(0.009)	(0.027)
H_0 : Lost=Won								
p-value	0.177	0.201	0.282	0.0170	0.190	¡0.01	¡0.01	¡0.01
Observations	14,781	14,781	14,782	14,782	14,782	14,782	14,782	14,782
R-squared	0.583	0.589	0.330	0.336	0.001	0.047	0.004	0.024
Wave-HH Size FE	No	Yes	No	Yes	No	Yes	No	Yes